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UNIT 3: Introduction to Power Electronics and Electric Drive Train Systems

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Unit 3: Introduction to Power Electronics and Electric Drive Train Systems

An electric vehicle is, electrically, a chain of energy conversions: the grid supplies alternating current, the battery stores direct current, and the traction machine needs variable-frequency, variable-voltage alternating current. Power electronics connects these three worlds, and the electric drive turns the result into controlled mechanical motion. This unit therefore moves from converters (Sections 3.1 and 3.2), through the drive train and its mechanics (Sections 3.3 to 3.6), to the two dominant traction machines, the induction motor (Sections 3.7 to 3.11) and the permanent magnet synchronous machine (Sections 3.12 to 3.16).

Conventions Used Throughout This Unit

The machine and control equations depend on sign and scaling conventions, so one set is fixed here and kept throughout. Table 3.1 states them. Where a textbook uses another convention, the difference is noted at the point of use.

Table 3.1: Notation and conventions adopted in Unit 3

Item	Convention
Pole number	P is the number of poles and $p = P/2$ the number of pole *pairs*. All torque equations use p.
Angular speed	ω_m mechanical (rad/s), $\omega_r = p\,\omega_m$ rotor speed in electrical rad/s, ω_e electrical frequency of stator quantities (rad/s).
Current direction	Motor convention: positive current flows into the machine terminals. Battery current $I_b > 0$ means discharge.
Mechanical power	$P_m = T_e\omega_m$. $P_m > 0$ is motoring and $P_m < 0$ is generating.
Equivalent circuits	Steady-state phasors are per phase and rms.
Clarke and Park scaling	Amplitude-invariant (factor 2/3): a balanced set of peak I_m gives a space vector of length I_m. Power and torque therefore carry the factor 3/2.
Reference frame	The q axis leads the d axis by 90° electrical. d is aligned with rotor flux (induction motor) or the magnet north axis (PMSM).

3.1 Power Electronic Converters for Electric and Hybrid Vehicles

3.1.1 Introduction to Power Electronics

Power electronics is the efficient conversion and control of electrical power using semiconductor switches. A switch is ideally either fully on, with no voltage across it, or fully off, with no current through it, so it dissipates no power in either state. Output voltage is then controlled not by dropping voltage across a resistive element, but by the fraction of time the switch is on. Inductors and capacitors, which also store energy without dissipation, smooth the chopped waveform. This is why conversion efficiencies of 95–99% are routinely achieved at tens of kilowatts [1–3].

3.1.2 Importance of Power Electronic Converters in Electric Vehicles

Converters decide much of what a driver experiences: acceleration comes from fast torque control, range depends on conversion losses, and charging time depends on charger power [4,5]. Every watt lost in an inverter must also be removed by the cooling system. In summary, the converters must

- supply the machine with variable voltage and frequency from a fixed-chemistry DC source whose voltage varies with state of charge;
- permit energy to flow in reverse during braking;
- interface the vehicle to the grid at near-unity power factor;
- feed the low-voltage electrical system, the compressor and the heater from the high-voltage battery.

3.1.3 Power Electronic Conversion Stages in EVs

Figure 3.1 shows the complete power electronics architecture of a battery electric vehicle (BEV). The high-voltage battery is the central node. The traction path, drawn horizontally, runs through an optional bidirectional DC-DC converter and the traction inverter to the motor-generator, and every arrow on it is double-headed because braking reverses the flow. Around this path sit three supporting converters: the on-board charger, the off-board fast charger connection, and the isolated DC-DC converter for the 12 V network. Auxiliary inverters drive the air-conditioning compressor and similar loads.

Figure 3.2 arranges the same stages in the order of energy flow in a BEV. During AC charging, energy passes through rectification with power factor correction and then through an isolated DC-DC stage into the battery. During driving, it passes through the traction stage to the wheels. Note that the battery divides the system into a charging side that is often unidirectional and a traction side that must be bidirectional.

A hybrid electric vehicle (HEV) adds a second source. In the power-split arrangement of Figure 3.3, the engine drives a generator MG1, whose AC output is rectified by Inverter 1 on to a shared DC bus. Inverter 2 feeds the traction machine MG2. A boost converter links the bus

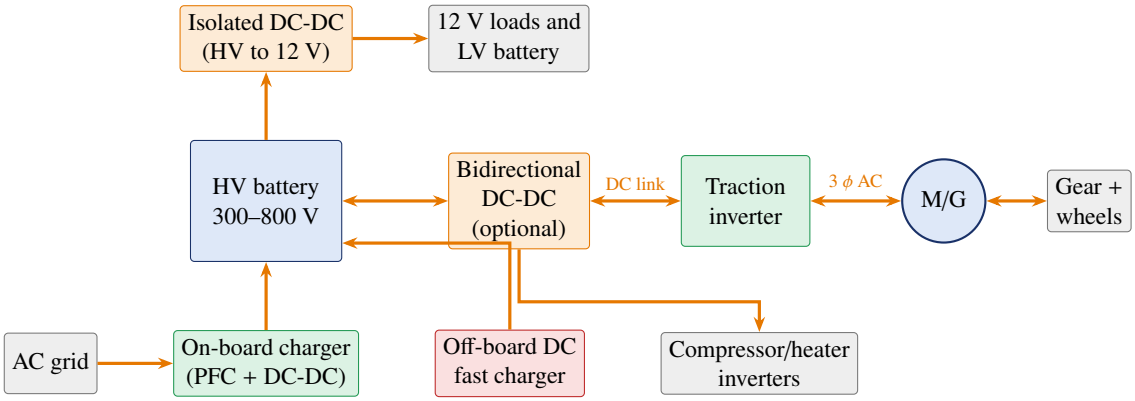


Figure 3.1: Complete EV power electronics architecture. Double-headed arrows are bidirectional paths.

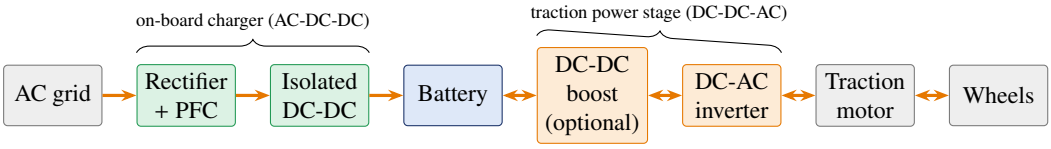


Figure 3.2: Power conversion stages in a battery electric vehicle.

to the lower-voltage battery, and part of the engine power reaches the wheels mechanically through a planetary gear. The converters therefore allow engine speed to be set independently of wheel speed, so that the engine stays near its best efficiency region.

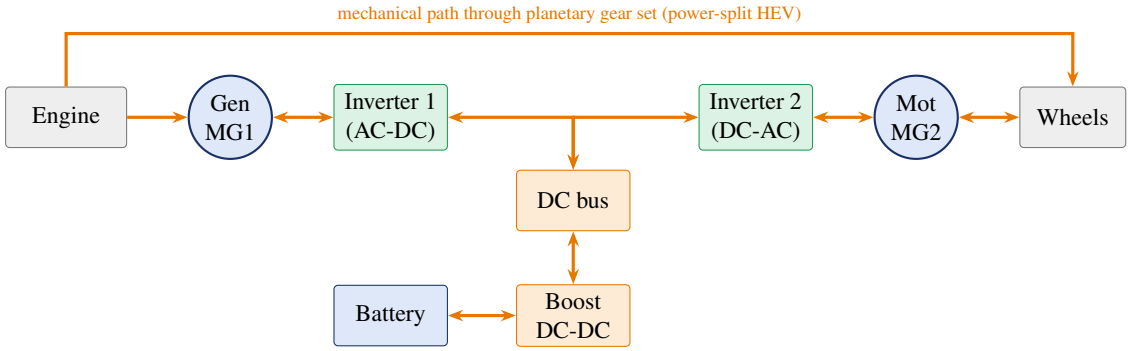


Figure 3.3: Power conversion stages in a power-split hybrid electric vehicle.

3.1.4 Classification of Converters

3.1.4.1 AC-DC Converters

A rectifier converts AC to DC. In the vehicle, it appears in the charger input, and the traction inverter itself acts as an active rectifier during regeneration.

3.1.4.2 DC-DC Converters

A DC-DC converter changes the level of a DC voltage. Examples include the traction boost stage, the HV-to-12 V converter and the DC stage of a charger.

3.1.4.3 DC-AC Converters

An inverter synthesises AC from DC. The traction inverter is the largest example.

3.1.4.4 DC-AC-DC Conversion

Isolation is not possible at DC, because a transformer needs changing flux. An isolated DC-DC converter therefore inverts the DC to high-frequency AC, passes it through a small transformer, and rectifies it again. Since transformer size falls as frequency rises, this DC-AC-DC structure is far smaller than a line-frequency transformer of the same rating.

3.1.4.5 Unidirectional and Bidirectional Converters

A unidirectional converter passes power one way and can use diodes in place of some switches, which is simpler and cheaper. A bidirectional converter uses active switches in every position, so current can reverse. Traction stages must be bidirectional, and chargers are increasingly so.

3.1.4.6 Isolated and Non-Isolated Converters

A non-isolated converter has a conductive path between input and output, and is small and efficient. An isolated converter has a transformer, which provides safety separation and a large voltage ratio, at some cost in size and loss. Table 3.2 compares the converters used in an EV.

Table 3.2: Power electronic converters used in EVs

Converter	Conversion	Power flow	Typical role
Traction inverter	DC-AC	Bidirectional	Drives motor, recovers braking energy
Traction boost stage	DC-DC	Bidirectional	Raises battery voltage to DC-link level
Auxiliary converter	DC-DC, isolated	Usually one way	Feeds 12 V or 48 V network
On-board charger	AC-DC-DC	One way or both	AC charging, optionally V2G
DC fast charger	AC-DC-DC	Mostly one way	Off-board high-power charging
Auxiliary inverter	DC-AC	One way	Compressor, pumps

The table shows that only the converters connected to the wheels are inherently bidirectional. The others are made bidirectional only when V2G or similar services justify the extra switches.

3.1.5 Basic DC-DC Converters

3.1.5.1 Governing Principles

Three facts support all the derivations below. The first is the inductor equation $v_L = L\,di_L/dt$, and the second is the capacitor equation $i_C = C\,dv_C/dt$. Thirdly, in periodic steady state, the inductor current and capacitor voltage return to their starting values each switching period

$T_s = 1/f_s$. Integrating over a period gives

$$\frac{1}{T_s}\int_0^{T_s} v_L\,dt = 0, \qquad \frac{1}{T_s}\int_0^{T_s} i_C\,dt = 0. \tag{3.1}$$

These are *volt-second balance* and *charge balance*. The duty ratio is the fraction of the period for which the switch conducts,

$$D = \frac{t_{\text{on}}}{T_s} = t_{\text{on}} f_s, \qquad 0 \le D \le 1, \tag{3.2}$$

where t_{on} is in seconds and f_s in hertz. D is dimensionless. We assume ideal switches, small ripple and continuous conduction unless stated otherwise.

3.1.5.2 Buck Converter

The buck converter in Figure 3.4 steps voltage down. While S is on, the source drives the inductor and $v_L = V_{\text{in}} - V_o$. While S is off, the inductor current freewheels through diode D and $v_L = -V_o$. Volt-second balance gives

$$(V_{\text{in}} - V_o)DT_s + (-V_o)(1-D)T_s = 0 \quad \Rightarrow \quad \frac{V_o}{V_{\text{in}}} = D. \tag{3.3}$$

The inductor current rises at $(V_{\text{in}} - V_o)/L$ and falls at V_o/L, so its peak-to-peak ripple is

$$\Delta i_L = \frac{V_o(1-D)}{L f_s}. \tag{3.4}$$

In a buck converter, the inductor current average equals the load current, $I_L = I_o$. By charge balance, the capacitor absorbs only the ripple, $i_C = i_L - I_o$. The charge stored in the positive half of this triangle is $\Delta Q = \Delta i_L T_s/8$, so the ripple voltage is

$$\Delta v_C = \frac{\Delta Q}{C} = \frac{\Delta i_L}{8 C f_s}. \tag{3.5}$$

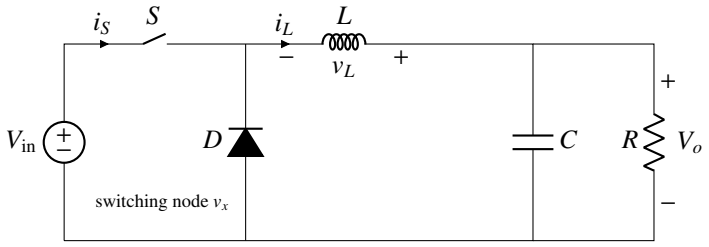


Figure 3.4: Buck converter circuit.

Figure 3.5 shows these waveforms. The switching-node voltage is a pulse train whose mean is V_o, the inductor current is a triangle about I_o, and the capacitor current is the same triangle with its mean removed.

In an EV, the buck stage is the basis of non-isolated low-voltage supplies, such as 48 V to

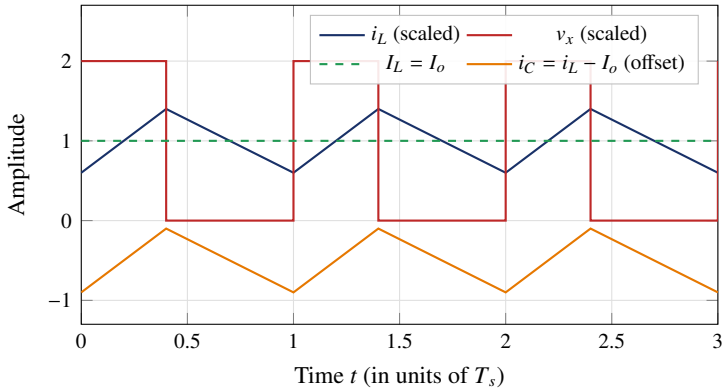


Figure 3.5: Buck converter waveforms in CCM at $D = 0.4$ (normalised).

12 V converters in mild hybrids, and of the regenerative direction of the bidirectional stage. Algorithm 1 gives its control sequence.

Algorithm 1 Buck converter operation with voltage regulation

Input: V_{in}, reference V_o^*, measured V_o, f_s
Output: Gate signal for S

Initialise $D \leftarrow V_o^*/V_{\text{in}}$ and PI integrator
for each switching period **do**
    $e \leftarrow V_o^* - V_o$; update PI to obtain D
    Limit D to $[D_{\min}, D_{\max}]$
    Turn S on for DT_s: i_L rises at $(V_{\text{in}} - V_o)/L$
    Turn S off for $(1-D)T_s$: i_L falls at V_o/L through D
    if i_L exceeds current limit **then**
        Terminate the on-time early
    end if
end for

Example 1: Buck converter voltage calculation

Problem. A 48 V to 14 V buck converter supplies the low-voltage bus of a mild hybrid. Find its duty ratio, inductor ripple, output voltage ripple and minimum inductance for CCM.

Given (assumed design values). $V_{\text{in}} = 48$ V, $V_o = 14$ V, $I_o = 50$ A, $f_s = 100$ kHz, $L = 33\ \mu$H, $C = 220\ \mu$F.

Required. D, Δi_L, Δv_C, L_{crit}.

Governing equations. (3.3), (3.4), (3.5) and (3.9).

Solution. $D = 14/48 = 0.292$.
$\Delta i_L = 14(1 - 0.292)/(33 \times 10^{-6} \times 10^5) = 3.01$ A.
$\Delta v_C = 3.01/(8 \times 220 \times 10^{-6} \times 10^5) = 17.1$ mV.
With $R = V_o/I_o = 0.28\ \Omega$, $L_{\text{crit}} = (1-D)R/(2f_s) = 0.99\ \mu$H.

Result. $D = 0.292$, $\Delta i_L = 3.01$ A, $\Delta v_C = 17.1$ mV, $L_{\text{crit}} = 0.99\ \mu$H.

Physical interpretation. Ripple is about 6% of the load current, and the chosen L is many times the critical value, so the converter stays in CCM down to light load.

Engineering significance. Equivalent series resistance of the real capacitor will normally dominate this very small ideal ripple, so capacitor selection is governed by ESR and ripple current rating.

3.1.5.3 Boost Converter

The boost converter in Figure 3.6 steps voltage up. While S is on, the inductor charges from the source, $v_L = V_{\text{in}}$. While it is off, the inductor current flows through the diode to the output, $v_L = V_{\text{in}} - V_o$. Volt-second balance gives

$$V_{\text{in}}D + (V_{\text{in}} - V_o)(1 - D) = 0 \quad \Rightarrow \quad \frac{V_o}{V_{\text{in}}} = \frac{1}{1 - D}. \tag{3.6}$$

For a lossless converter, $V_{\text{in}}I_L = V_oI_o$, so $I_L = I_o/(1 - D)$. The ripple quantities are

$$\Delta i_L = \frac{V_{\text{in}}D}{Lf_s}, \qquad \Delta v_C = \frac{I_oD}{Cf_s}. \tag{3.7}$$

The second expression follows because the capacitor alone supplies I_o during the on-time DT_s. Real boost converters cannot reach the ideal gain as $D \to 1$, because resistive drops grow with I_L, so practical EV stages stay below a gain of about three [2].

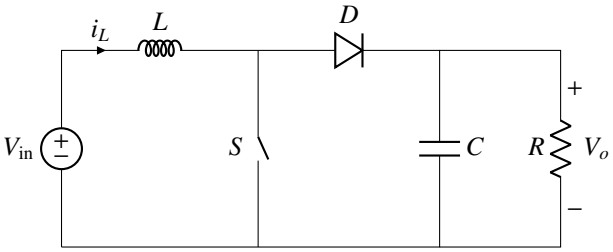


Figure 3.6: Boost converter circuit.

Example 2: Boost converter voltage calculation

Problem. A traction boost stage raises a 300 V battery to a 650 V DC link at 60 kW. Find the duty ratio, currents and ripples.

Given (assumed). $V_{\text{in}} = 300$ V, $V_o = 650$ V, $P = 60$ kW, $f_s = 20$ kHz, $L = 150\ \mu$H, $C = 500\ \mu$F; lossless.

Required. D, I_L, I_o, Δi_L, Δv_C.

Governing equations. (3.6), (3.7).

Solution. $D = 1 - 300/650 = 0.538$. $I_L = 60\,000/300 = 200$ A. $I_o = 60\,000/650 = 92.3$ A.
$\Delta i_L = 300 \times 0.538/(150 \times 10^{-6} \times 2 \times 10^4) = 53.8$ A.
$\Delta v_C = 92.3 \times 0.538/(500 \times 10^{-6} \times 2 \times 10^4) = 4.97$ V.

Result. $D = 0.538$, $I_L = 200$ A, $\Delta i_L = 53.8$ A (27%), $\Delta v_C = 4.97$ V.

Physical interpretation. The low-voltage side carries more than twice the output current, so the inductor and switches are sized by battery current.

Engineering significance. A higher DC link allows lower motor current for a given power, but the boost inductor adds mass. Designers weigh this against simply using a higher battery voltage.

Algorithm 2 summarises the operating sequence, which includes the overvoltage protection needed because the boost output rises if the load is disconnected.

Algorithm 2 Boost converter operation with cascaded control

Input: V_{in}, V_o^*, measured V_o, i_L
Output: Gate signal for S

```
Initialise D ← 1 − V_in/V_o*
for each control period do
    Voltage PI: i_L* ← PI_v(V_o* − V_o), limited to I_max
    Current PI: D ← PI_i(i_L* − i_L), limited to D_max
    S on for DT_s to store energy; off to transfer energy to output
    if V_o > V_OVP then
        Disable S
    end if
end for
```

3.1.5.4 Buck-Boost Converter

The buck-boost converter of Figure 3.7 stores energy in L during DT_s with $v_L = V_{\text{in}}$, then releases it to the output with $v_L = V_o$, where V_o is negative with respect to the reference polarity. Volt-second balance gives

$$V_{\text{in}} D + V_o(1 - D) = 0 \quad \Rightarrow \quad \frac{V_o}{V_{\text{in}}} = -\frac{D}{1 - D}. \tag{3.8}$$

The output magnitude is below the input for $D < 0.5$ and above for $D > 0.5$, and the polarity is inverted. In the vehicle this suits sources whose voltage overlaps the load voltage. Table 3.3 compares the three basic converters.

From the table, the boost converter has a smooth input current, which suits batteries and PFC stages, while the buck has a smooth output current, which suits loads. The filter on the pulsating side must be larger.

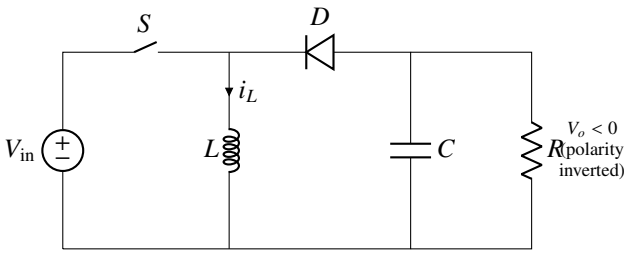


Figure 3.7: Buck-boost converter circuit.

Table 3.3: Comparison of buck, boost and buck-boost converters in CCM

Property	Buck	Boost	Buck-boost
Gain V_o/V_{in}	D	$1/(1-D)$	$-D/(1-D)$
Output magnitude	Lower	Higher	Lower or higher
Input current	Pulsating	Continuous	Pulsating
Output current	Continuous	Pulsating	Pulsating
Inductor ripple	$V_o(1-D)/(Lf_s)$	$V_{in}D/(Lf_s)$	$V_{in}D/(Lf_s)$
EV role	48/12 V supply	Traction boost, PFC	Fuel cell or wide-range interface

3.1.5.5 Continuous and Discontinuous Conduction

If the average inductor current falls below half the ripple, the current reaches zero before the period ends and stays there, as shown by the lower curve in Figure 3.8. This is discontinuous conduction mode (DCM). The boundary for a buck converter is $I_o = \Delta i_L/2$, which gives the critical inductance

$$L_{\text{crit}} = \frac{(1-D)R}{2f_s}. \tag{3.9}$$

In DCM, the gain depends on load as well as duty ratio. Let the current fall to zero after an interval D_2T_s. Volt-second balance gives $V_o/V_{in} = D/(D+D_2)$, and equating the mean inductor current to V_o/R gives, with $K = 2L/(RT_s)$,

$$\frac{V_o}{V_{in}} = \frac{2}{1+\sqrt{1+4K/D^2}}. \tag{3.10}$$

The corresponding boost converter result is $V_o/V_{in} = (1+\sqrt{1+4D^2/K})/2$ [2]. DCM reduces diode reverse-recovery loss, but its high peak currents limit it to low-power stages.

3.1.5.6 Bidirectional DC-DC Converter

Replacing the boost diode with a controlled switch yields the half-bridge bidirectional converter in Figure 3.9. With inductor current positive, S_2 is the active switch and S_1's diode conducts, which is a boost converter moving power from battery to DC link. With inductor current negative, S_1 is active and S_2's diode conducts, which is a buck converter moving power from DC

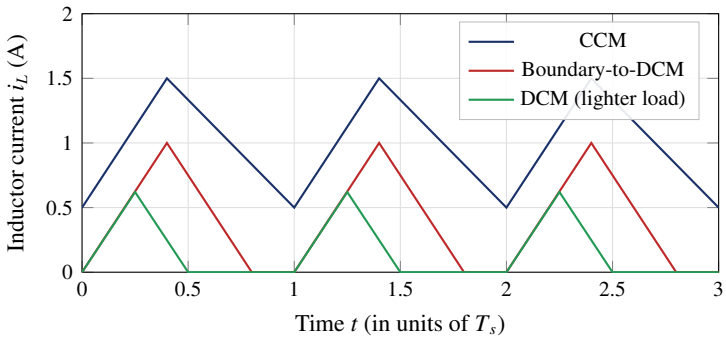


Figure 3.8: Inductor current in continuous, boundary and discontinuous conduction.

link to battery. If the duty ratio of the upper switch is D_1, complementary gating ensures

$$V_b = D_1 V_{\mathrm{dc}}, \tag{3.11}$$

whatever the current sign. The direction of power is therefore set by a small change in D_1, and current passes smoothly through zero. Algorithm 3 gives the control method.

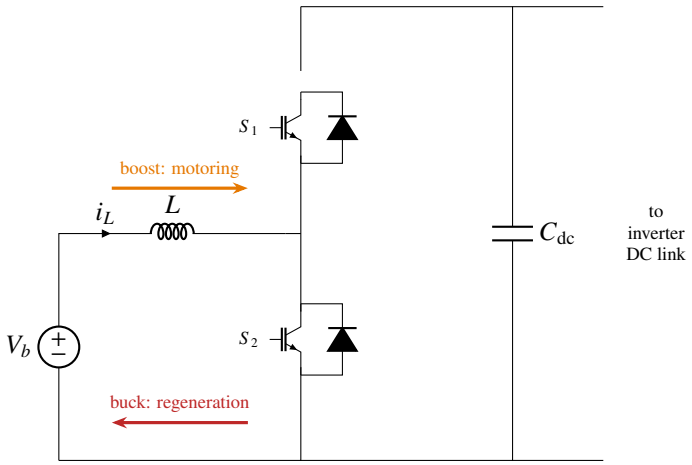


Figure 3.9: Bidirectional half-bridge DC-DC converter.

Example 3: Bidirectional DC-DC converter operation

Problem. A 350 V battery connects to a 600 V link through a bidirectional stage with efficiency 96%. Find switch duty ratios and battery current for 40 kW delivered to the link in motoring, and for 25 kW received from the link in regeneration.

Given. $V_b = 350$ V, $V_{\mathrm{dc}} = 600$ V, $\eta = 0.96$ (assumed).

Required. D_1, D_2, I_b in both modes.

Governing equations. (3.11), $P_b = V_b I_b$.

Solution. $D_1 = 350/600 = 0.583$, $D_2 = 1 - D_1 = 0.417$.
Motoring: $P_b = 40/0.96 = 41.67$ kW, $I_b = 41\,670/350 = +119.0$ A.

Algorithm 3 Bidirectional DC-DC converter control

Input: V_{dc}^{*}, V_{dc}, i_L, V_b, battery limits $I_{\mathrm{dis}}, I_{\mathrm{chg}}$
Output: Complementary gates for S_1, S_2

for each control period **do**
  $i_L^{*} \leftarrow \mathrm{PI}_v(V_{\mathrm{dc}}^{*} - V_{\mathrm{dc}})$
  Limit: $-I_{\mathrm{chg}} \le i_L^{*} \le I_{\mathrm{dis}}$
  $D_1 \leftarrow V_b/V_{\mathrm{dc}} - \mathrm{PI}_i(i_L^{*} - i_L)/V_{\mathrm{dc}}$
  if $i_L^{*} > 0$ **then**
    Mode $\leftarrow$ boost (motoring)
  else
    Mode $\leftarrow$ buck (regeneration)
  end if
  Apply D_1 to S_1, and $1 - D_1$ with dead time to S_2
end for

Regeneration: $P_b = -25 \times 0.96 = -24.0$ kW, $I_b = -24\,000/350 = -68.6$ A.

Result. Duty ratios are 0.583 and 0.417 in both modes. $I_b = +119.0$ A and -68.6 A.

Physical interpretation. Losses raise battery power in motoring and reduce it in regeneration, because they are always paid out of the energy flow.

Engineering significance. The ideal duty ratio is the same in both modes; only a small control correction, set by resistive drops, moves the current from positive to negative.

3.1.6 DC-AC Converters

3.1.6.1 Three-Phase Voltage Source Inverter

The three-phase voltage source inverter (VSI) in Figure 3.10 has three half-bridge legs across a stiff DC link. Each leg connects its phase to either rail, and the two switches in a leg are complementary, with a short dead time to avoid a shoot-through. Diodes (or the MOSFET body or channel) let current flow against the switch direction, which makes the inverter inherently bidirectional: regeneration needs no extra circuit.

Let $s_a, s_b, s_c \in \{0, 1\}$ denote whether the upper switch of each leg is on. Pole voltages with respect to the DC midpoint are $v_{a0} = (s_a - \frac{1}{2})V_{\mathrm{dc}}$ and so on. For a balanced star load the neutral voltage is the average of the pole voltages, so the phase voltage is

$$v_{an} = \frac{V_{\mathrm{dc}}}{3}\,(2s_a - s_b - s_c), \tag{3.12}$$

with similar expressions for b and c by rotation. In six-step operation, each leg is on for 180°, and Figure 3.11 shows the resulting quasi-square line voltage and stepped phase voltage. Fourier analysis yields the fundamental peak $2V_{\mathrm{dc}}/\pi$. With sinusoidal PWM, the amplitude

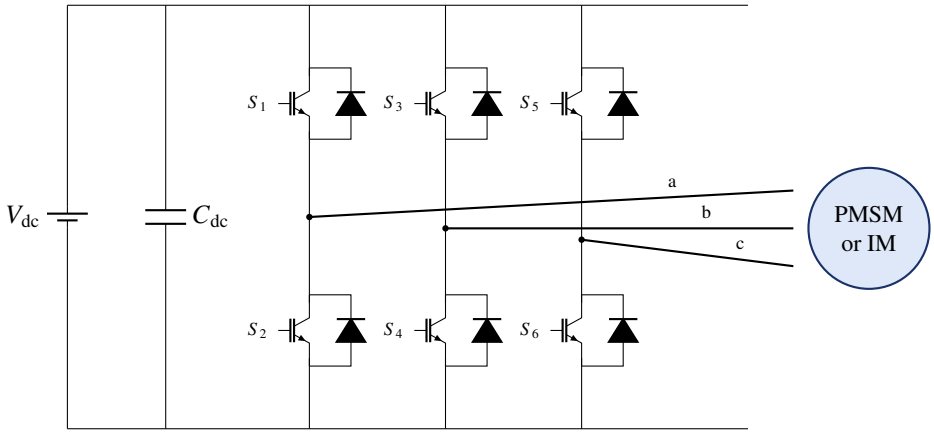


Figure 3.10: Three-phase voltage source inverter feeding a traction machine.

modulation index $m_a = \hat{v}_{ref}/\hat{v}_{tri}$ sets the fundamental in the linear range $m_a \leq 1$:

$$\hat{V}_{\phi 1} = m_a \frac{V_{dc}}{2}, \qquad V_{LL,rms} = \frac{\sqrt{3}}{2\sqrt{2}} m_a V_{dc}. \tag{3.13}$$

Space vector modulation (Section 3.16) raises the linear limit to $\hat{V}_{\phi 1} = V_{dc}/\sqrt{3}$, about 15% above SPWM, by exploiting the third-harmonic freedom of the star connection [6].

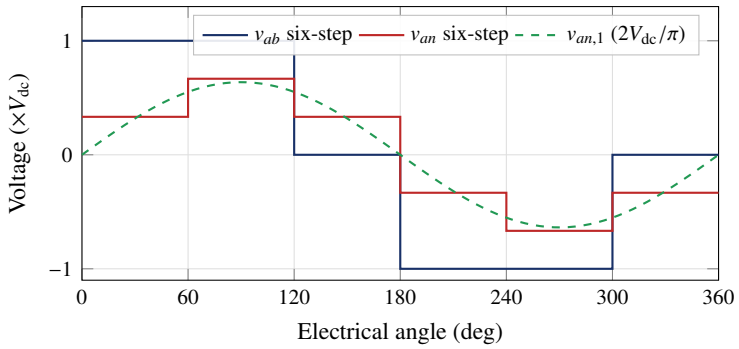


Figure 3.11: Six-step line and phase voltages of a VSI with the phase-voltage fundamental.

3.1.6.2 DC-Link Power Balance

Neglecting inverter losses, instantaneous power on the two sides is equal. For a balanced sinusoidal load, this power is constant,

$$V_{dc} I_{dc}\, \eta_{inv} = 3 V_\phi I_\phi \cos\varphi = \tfrac{3}{2} \hat{V}_\phi \hat{I}_\phi \cos\varphi, \tag{3.14}$$

where V_ϕ, I_ϕ are rms and $\hat{V}_\phi, \hat{I}_\phi$ peak values, and $\cos\varphi$ is the load displacement factor. Thus the average link current is DC; the pulsed switching current is supplied by the DC-link capacitor.

Example 4: Three-phase inverter output voltage

Problem. A 400 V inverter runs at $m_a = 0.9$ with SPWM and delivers 50 kW at 0.9 power

factor. The efficiency is 97%. Find output voltages, currents and loss. Also compare the SVPWM and six-step limits.

Given. $V_{\mathrm{dc}} = 400$ V, $m_a = 0.9$, $P_o = 50$ kW, $\cos\varphi = 0.9$, $\eta = 0.97$.

Required. $\hat{V}_{\phi 1}$, V_ϕ, V_{LL}, I_ϕ, I_{dc}, loss.

Governing equations. (3.13), (3.14).

Solution. $\hat{V}_{\phi 1} = 0.9 \times 200 = 180$ V, $V_\phi = 127.3$ V, $V_{LL} = \sqrt{3} \times 127.3 = 220.5$ V.
$I_\phi = 50\,000/(3 \times 127.3 \times 0.9) = 145.5$ A.
$P_{\mathrm{dc}} = 50/0.97 = 51.55$ kW, $I_{\mathrm{dc}} = 128.9$ A, loss $= 1.55$ kW.
Limits: SVPWM $400/\sqrt{3} = 230.9$ V peak; six-step $2 \times 400/\pi = 254.6$ V peak.

Result. $V_{LL} = 220.5$ V rms, $I_\phi = 145.5$ A rms, $I_{\mathrm{dc}} = 128.9$ A, loss 1.55 kW.

Physical interpretation. Phase current exceeds DC current because the phase voltage is lower than the link voltage; the inverter behaves as a step-down converter.

Engineering significance. The 1.55 kW loss sets the cooling load of the inverter at this operating point.

3.1.6.3 Current Source Inverter

The current source inverter (CSI) in Figure 3.12 is fed through a large inductor, so the DC side is a stiff current. Series diodes (or reverse-blocking devices) make the switches block both voltage polarities, and the inverter steers the DC current into the phases. A CSI gives naturally sinusoidal, low-dv/dt motor voltages with suitable output capacitors, and it tolerates short circuits. However, the DC inductor is heavy and regeneration requires reversal of DC voltage, so the VSI remains the standard choice for traction.

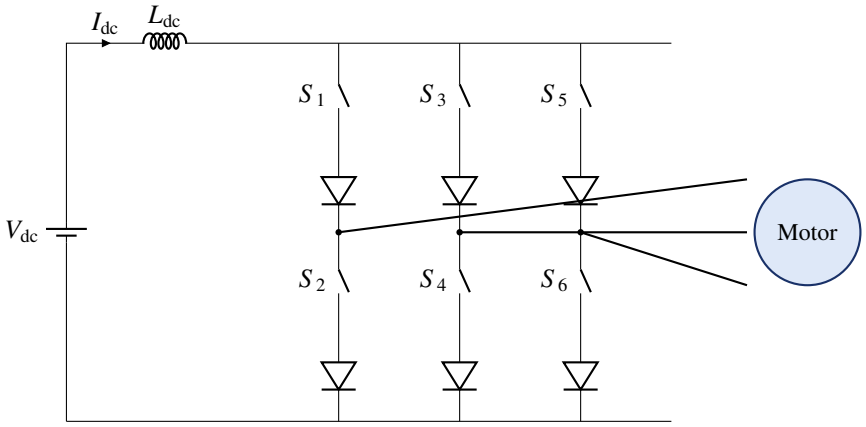


Figure 3.12: Current source inverter with series-blocking diodes.

3.1.6.4 Multilevel Inverters

A multilevel inverter builds its output from more than two voltage levels. The three-level neutral-point-clamped (NPC) leg of Figure 3.13, introduced by Nabae et al. [7], adds a capacitor

midpoint and clamping diodes so that the phase can connect to $+V_{dc}/2$, 0 or $-V_{dc}/2$. Each switch blocks only half the link voltage, the voltage steps are smaller, and the output has less harmonic distortion. These features become attractive with 800 V batteries, at the price of more devices and a need to balance the midpoint.

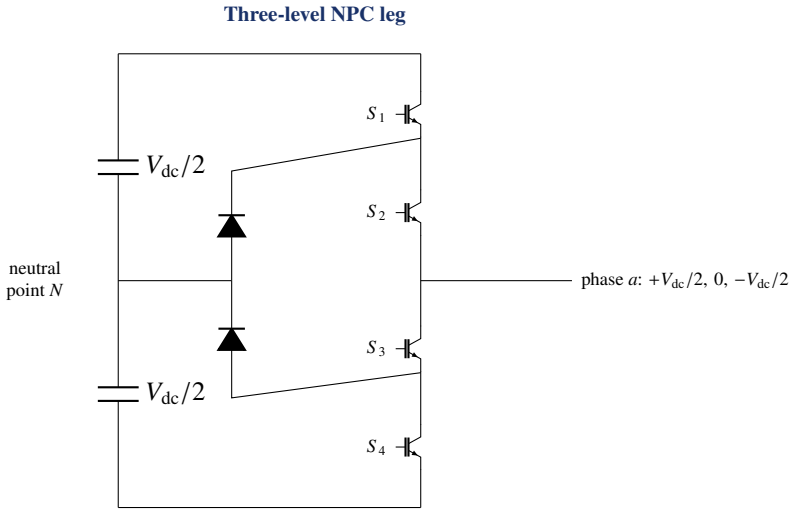


Figure 3.13: Multilevel inverter architecture: three-level NPC leg.

3.1.6.5 Active Front-End Converters

An active front end (AFE) is a PWM rectifier: a VSI connected to the grid through inductors. By shaping its AC voltage relative to grid voltage, it controls current magnitude and phase, so it draws sinusoidal current at any chosen power factor and can return power. It is therefore the grid-side stage of a bidirectional charger.

3.1.7 Power Electronic Interface Between Battery and Motor

Figure 3.14 shows the battery-to-motor path. The battery, modelled by V_b and R_b, connects through the inductor and half bridge to a regulated DC link, and the VSI converts this to three-phase AC. Without the boost stage, as in many BEVs, the battery connects directly to the link, and the link voltage then varies with state of charge and load. The controller sits below the two power stages and generates all gate signals.

3.1.8 Power Flow During Motoring and Regenerative Braking

Figure 3.15 shows the two power flow directions. In motoring, the battery discharges, $I_b > 0$, and each stage loses some power on the way to the wheels. In regeneration, the wheels drive the machine, the inverter acts as an active rectifier, and power returns with $I_b < 0$. The energy delivered to the battery is reduced by the same stage efficiencies. If stage efficiencies are η_i, the round-trip efficiency from battery to wheels and back is $(\prod \eta_i)^2$, which is why every percent of converter efficiency matters.

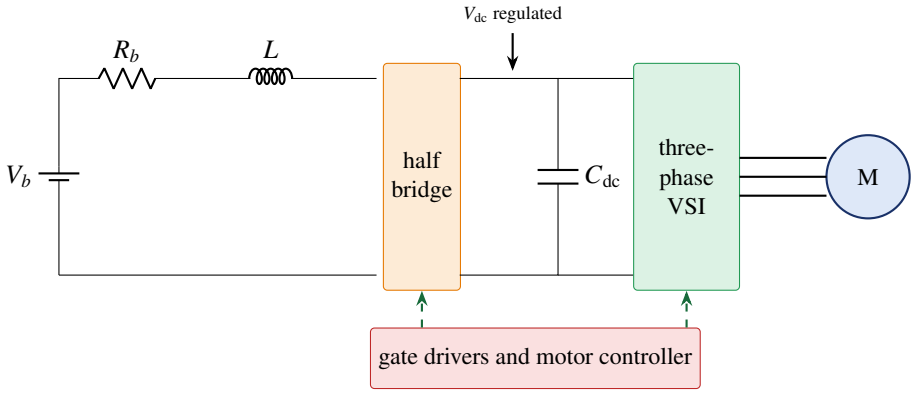


Figure 3.14: Battery-to-motor power conversion architecture.

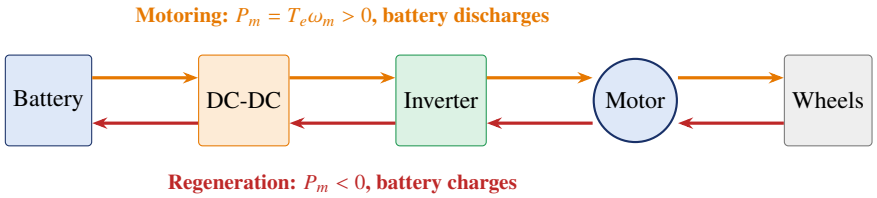


Figure 3.15: EV motoring and regenerative braking power flow.

3.1.9 Converter Efficiency and Losses

Efficiency is the ratio of output to input power,

$$\eta = \frac{P_o}{P_o + P_{\text{loss}}}, \qquad P_{\text{loss}} = P_{\text{cond}} + P_{\text{sw}} + P_{\text{mag}} + P_{\text{aux}}. \tag{3.15}$$

3.1.9.1 Conduction Losses

A MOSFET behaves as a resistance, and an IGBT or diode as a fixed drop plus a small resistance:

$$P_{\text{cond,MOS}} = I_{\text{rms}}^2 R_{\text{DS(on)}}, \qquad P_{\text{cond,IGBT}} = V_{\text{CE0}} I_{\text{avg}} + r_C I_{\text{rms}}^2. \tag{3.16}$$

3.1.9.2 Switching Losses

During each transition, voltage and current overlap. Assuming linear transitions with a switching voltage V and current I, the energy per cycle is $\frac{1}{2}VI(t_r + t_f)$, so

$$P_{\text{sw}} = \tfrac{1}{2} V I (t_r + t_f) f_s. \tag{3.17}$$

Device datasheets replace this with measured energies, $P_{\text{sw}} = (E_{\text{on}} + E_{\text{off}}) f_s$. Conduction loss is dominant at high current, while switching loss scales with frequency and voltage. Figure 3.16 gives the resulting typical shape of the efficiency curve for two illustrative loss models: efficiency falls at light load, where fixed loss dominates, and droops slightly at high load, where I^2R grows. The SiC curve stays higher at light load, the region in which vehicles spend most of their time.

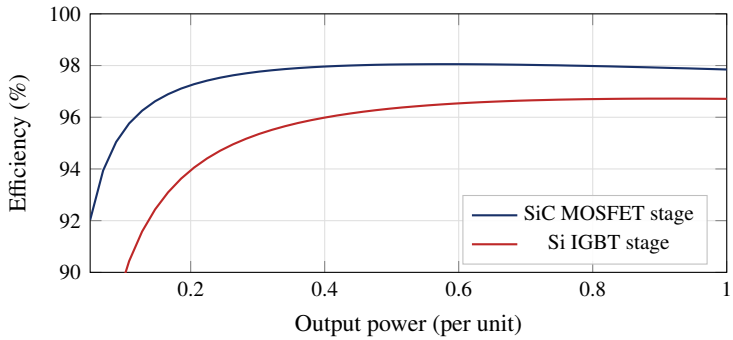


Figure 3.16: Illustrative efficiency against load for Si IGBT and SiC MOSFET stages (assumed loss models).

3.1.10 Semiconductor Devices Used in EV Power Electronics

- **Silicon MOSFET.** Fast and resistive, used at low voltage up to about 150 V, for example in 48 V systems.
- **Silicon IGBT.** Bipolar conduction gives a low drop at 600–1200 V and high current. Tail current limits switching frequency to around 10–20 kHz. IGBTs have been the mainstream traction device.
- **SiC MOSFET.** Silicon carbide has a wide bandgap and roughly ten times the critical field of silicon, so 1200 V devices have low on-resistance and switch quickly [8]. Lower losses, especially at part load, and higher temperature capability make SiC the favoured device for 800 V platforms, though at higher cost.
- **GaN HEMT.** Very fast and well-suited to on-board chargers and DC-DC converters at about 650 V.

3.1.11 Gate Driver Circuits

A gate driver turns a logic-level signal into the gate voltage and charge the device needs. For a MOSFET or IGBT, the driver charges the gate capacitance through a resistor, which sets di/dt and dv/dt. Upper-leg drivers need isolated supplies and isolated signal paths, because the source of the upper device swings at the full link voltage. Drivers also provide dead-time insertion, desaturation (short-circuit) detection, soft turn-off, undervoltage lockout and, for SiC, a negative turn-off voltage that resists false turn-on.

3.1.12 DC-Link Capacitor

The DC-link capacitor supplies the pulsed inverter current, keeps the link stiff and absorbs regenerative transients. Its current rating, not its capacitance, is usually the design constraint: for a two-level VSI the rms capacitor ripple current is of the order of half the load phase rms current at the worst-case modulation [6]. A simple sizing rule follows from the charge supplied in a period, $C \geq I_{\text{ripple}}/(2\pi f_s \Delta V)$ as an approximation. Film capacitors are preferred over

electrolytics for their long life, high ripple rating and low inductance. The link's parasitic inductance must be small, since it causes voltage overshoot when devices switch off.

3.1.13 Thermal Management of Power Converters

Heat flows from the semiconductor junction through a chain of thermal resistances to the coolant, and the steady-state junction temperature is

$$T_j = T_{\text{cool}} + P_{\text{loss}} R_{\text{th},jc}. \tag{3.18}$$

$R_{\text{th},jc}$ (in K/W) here is the total junction-to-coolant value. Traction modules typically use direct liquid cooling through pin-fin baseplates, on a water-glycol loop that often shares a circuit with the motor. Temperature cycling fatigues bond wires and solder, so lifetime depends on the amplitude of temperature swings as well as their peak value. Control software estimates T_j and derates current as limits are approached.

3.2 Electric Vehicle Charger Topologies

3.2.1 Introduction to EV Charger Topologies

Unit 2 treated charging from the standpoint of the battery and its standards. This section looks inside the charger. Every conductive charger has the same tasks: draw clean current from the grid, isolate the battery from the grid, and deliver controlled DC. Topologies differ in how they divide these tasks between stages [9, 10].

3.2.2 Onboard and Offboard Charging Systems

An *onboard* charger (OBC) travels with the vehicle, so its mass, volume and cost are tightly limited; ratings are commonly 3.3–22 kW. An *offboard* charger sits in the station and feeds the battery directly with DC, so it can be sized at hundreds of kilowatts with its own cooling. Table 3.4 compares them.

Table 3.4: Onboard and offboard chargers

Feature	Onboard	Offboard
Location	In the vehicle	In the charging station
Typical power	3.3–22 kW	30 kW to several hundred kW
Grid input	Single or three-phase AC	Three-phase AC, often medium-voltage fed
Constraint	Mass, volume, cost	Grid capacity, station cost
Cooling	Vehicle loop	Dedicated station system
Typical use	Home, workplace, overnight	Highway fast charging

The table explains why vehicles usually have both: the OBC serves daily needs at low cost, whereas the high power needed on long trips is impractical to carry on board.

3.2.3 Single-Stage and Two-Stage Chargers

A single-stage charger in Figure 3.17 performs power factor correction, isolation and battery regulation in one power stage. There are fewer components and potentially higher efficiency, but since the input power pulsates at twice line frequency and there is no large DC link, this ripple reaches the battery as low-frequency current. A two-stage charger in Figure 3.18 separates the tasks: the PFC stage shapes the grid current and regulates a DC link, and the DC-DC stage isolates and regulates the battery. The DC-link capacitor absorbs the double-line-frequency power, and each stage can be optimised separately. This is the dominant structure. Table 3.5 compares them.



Figure 3.17: Single-stage EV charger architecture.



Figure 3.18: Two-stage EV charger architecture.

Table 3.5: Single-stage and two-stage charging topologies

Feature	Single-stage	Two-stage
Power stages	One	PFC + DC-DC
Component count	Lower	Higher
100 Hz battery ripple	Present	Largely removed by DC link
Control	Coupled	Independent
Output voltage range	Narrower	Wide
Applications	Low-power, cost-driven	Mainstream OBC and fast charger

3.2.4 AC-DC Converter Topologies

3.2.4.1 Diode Bridge Rectifier

In Figure 3.19, the diode pair connected to the higher voltage conducts. With a capacitive filter, the diodes conduct only near the peak of the supply, drawing narrow current pulses that are rich in harmonics. With a resistive load, the average output of a single-phase bridge fed from rms voltage V_s is

$$V_{\mathrm{dc}} = \frac{1}{\pi}\int_0^{\pi} \sqrt{2}V_s \sin\theta\, d\theta = \frac{2\sqrt{2}}{\pi}V_s \approx 0.9V_s. \tag{3.19}$$

A three-phase bridge gives $V_{\mathrm{dc}} = (3\sqrt{2}/\pi)V_{LL} \approx 1.35V_{LL}$.

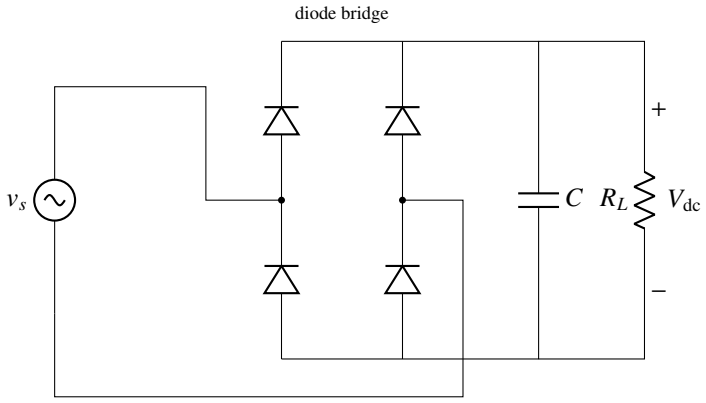


Figure 3.19: Single-phase diode bridge rectifier: an AC-DC rectifier topology.

3.2.4.2 Controlled Rectifier

Replacing diodes with thyristors delays conduction by a firing angle α. In continuous conduction, $V_{dc} = (2\sqrt{2}/\pi)V_s \cos\alpha$. The voltage is adjustable, but the displacement factor is $\cos\alpha$, so grid behaviour worsens as voltage falls. Thyristor rectifiers are hence restricted to older, low-cost installations.

3.2.4.3 Power Factor Correction

Power factor is defined in Unit 2 as $\mathrm{PF} = \cos\varphi_1 / \sqrt{1 + \mathrm{THD}_I^2}$. PFC aims for $\mathrm{PF} \approx 1$ by making the current sinusoidal and in phase with the voltage, so that the converter looks like a resistor to the grid. Grid-code limits on harmonic currents make this mandatory above low powers.

3.2.4.4 Boost PFC Converter

Figure 3.20 shows the classic solution: a diode bridge followed by a boost converter whose inductor current is forced to follow the rectified voltage shape. The output voltage controller sets the amplitude v_c, so the reference is $i_L^* = k\, v_c\, |v_s|$. From (3.6), the instantaneous duty ratio is

$$d(t) = 1 - \frac{|v_s(t)|}{V_o}, \qquad V_o > \sqrt{2}V_s. \tag{3.20}$$

The boost output must always exceed the peak grid voltage. Since input power is $p(t) = P(1 - \cos 2\omega t)$ for unity PF, the capacitor must buffer a ripple at 2ω, and the peak-to-peak link ripple is approximately

$$\Delta V_o = \frac{P}{\omega V_o C_o}. \tag{3.21}$$

ω is the grid angular frequency in rad/s. For this reason, two-stage chargers carry a sizeable link capacitor.

3.2.4.5 Totem-Pole PFC

The bridge diodes of Figure 3.20 drop voltage twice in the current path. The bridgeless totem-pole PFC of Figure 3.21 removes them [11]. One fast leg (GaN or SiC) performs high-frequency boost switching, while a slow leg is switched at line frequency to select the half

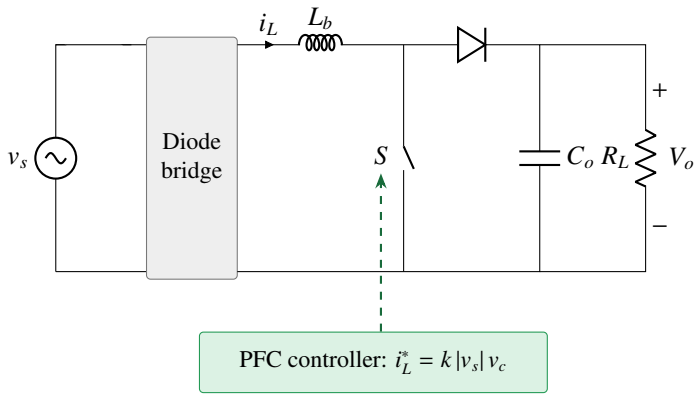


Figure 3.20: Boost PFC circuit with average-current control.

cycle. In the positive half cycle, Q_2 acts as boost switch and Q_1 as synchronous rectifier, and the roles swap in the negative half. Only two devices carry current at any time, so efficiency above 98% is achievable; because all positions are active, the circuit is also naturally bidirectional. Wide-bandgap devices are essential, as the poor reverse recovery of a silicon MOSFET body diode would otherwise cause severe loss.

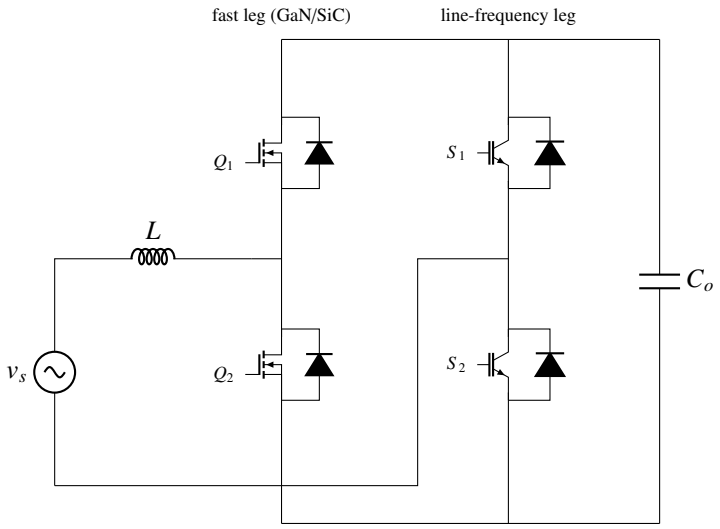


Figure 3.21: Totem-pole bridgeless PFC.

3.2.5 Isolated DC-DC Converter Topologies

Figure 3.22 shows the common structure: a high-frequency (HF) inverter chops the DC, a transformer provides isolation, and a rectifier and filter recover DC. The turns ratio gives a coarse voltage match and the switching pattern gives fine control. Table 3.6 later compares isolated and non-isolated types.

In the derivations below, $n = N_p/N_s$ unless stated otherwise, and CCM is assumed.

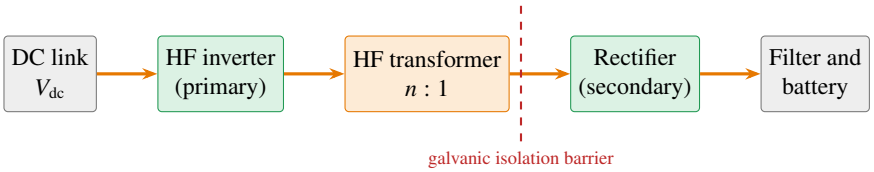


Figure 3.22: Isolated DC-DC converter architecture.

3.2.5.1 Flyback Converter

The flyback in Figure 3.23 is an isolated buck-boost whose transformer stores energy: during DT_s, energy is stored in the magnetising inductance L_m; during $(1-D)T_s$, it is released through the diode to the secondary. Volt-second balance gives

$$\frac{V_o}{V_{\text{in}}} = \frac{1}{n}\frac{D}{1-D}. \tag{3.22}$$

It has few parts, and suits auxiliary supplies under about 100 W to low kilowatts, but not traction-level power.

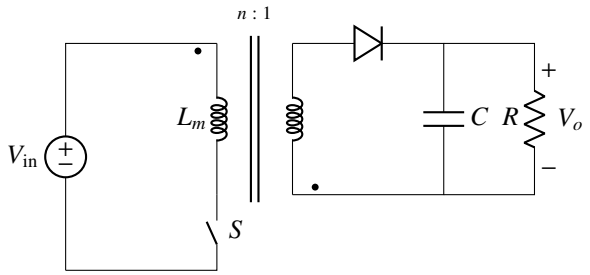


Figure 3.23: Flyback converter.

3.2.5.2 Forward Converter

The forward converter in Figure 3.24 transfers energy directly during the on-time, and the output inductor filters the result. With turns ratio $1:n$ as marked, $V_o = nDV_{\text{in}}$. The core must be reset each cycle through a reset winding or an active clamp, which limits D to about 0.5.

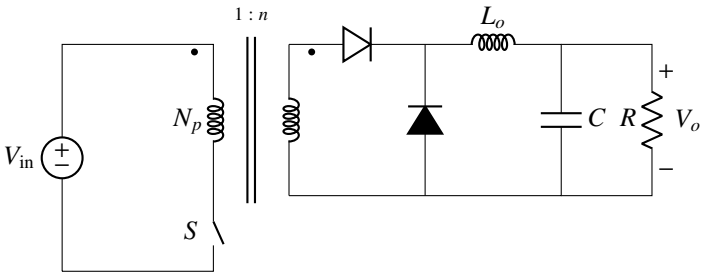


Figure 3.24: Forward converter (core reset winding not shown).

3.2.5.3 Push-Pull Converter

In Figure 3.25, two switches drive a centre-tapped primary alternately, exciting the core in both directions. Output voltage is $V_o = 2DV_{\text{in}}/n$ with $D < 0.5$ for each switch. It suits low-voltage

inputs, for example the 12 V to HV direction. Each switch must block $2V_{\text{in}}$, and flux imbalance needs care.

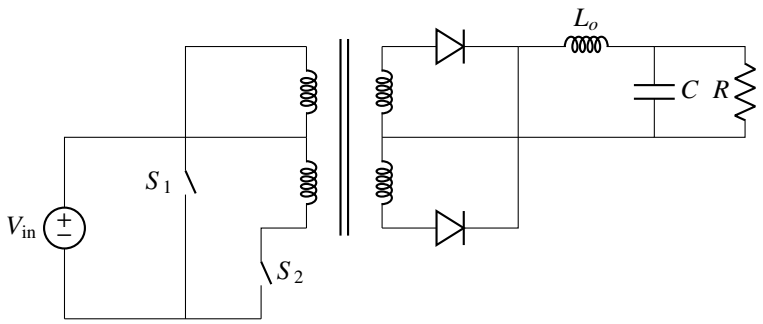


Figure 3.25: Push-pull converter with centre-tapped windings.

3.2.5.4 Half-Bridge Converter

In Figure 3.26, a capacitor divider holds the midpoint at $V_{\text{in}}/2$, and the switches apply $\pm V_{\text{in}}/2$ to the primary. Switch stress is only V_{in}, and $V_o = DV_{\text{in}}/n$, where $D < 0.5$ is the duty ratio of each switch. Half the voltage means twice the current compared with a full bridge at the same power.

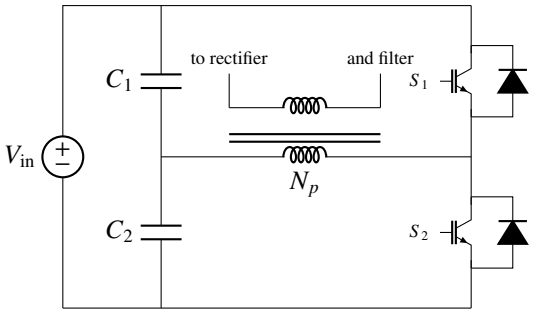


Figure 3.26: Half-bridge isolated converter.

3.2.5.5 Full-Bridge Converter

In Figure 3.27, diagonal switch pairs S_1, S_4 and S_2, S_3 apply $\pm V_{\text{in}}$ to the primary. Phase-shifting one leg relative to the other sets the effective duty, and thereby

$$V_o = \frac{2DV_{\text{in}}}{n}, \qquad 0 < D < 0.5. \tag{3.23}$$

Leakage inductance enables zero-voltage switching. This phase-shifted full bridge is the classic topology for kilowatt-class charger DC-DC stages.

3.2.5.6 LLC Resonant Converter

The LLC converter in Figure 3.28 places a resonant tank of C_r, L_r and the magnetising inductance L_m in the power path [12]. Rather than duty ratio, the control variable is frequency. Two

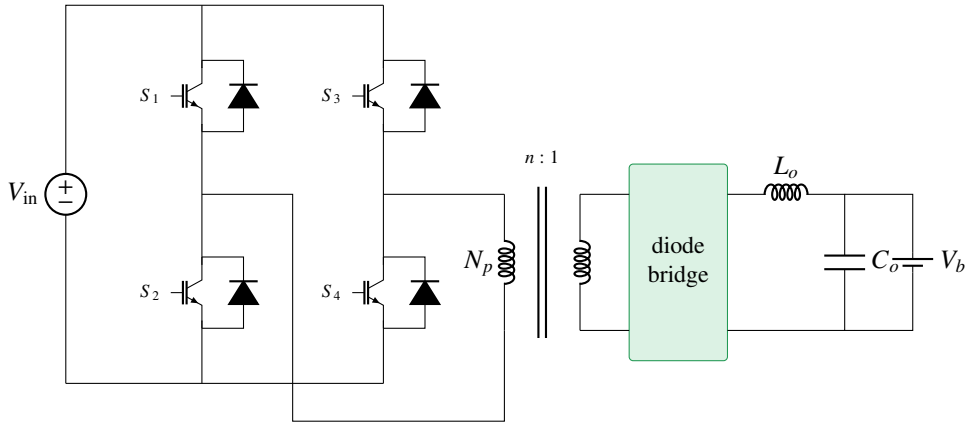


Figure 3.27: Full-bridge DC-DC converter.

resonant frequencies matter:

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}}, \qquad f_m = \frac{1}{2\pi\sqrt{(L_r + L_m)C_r}}. \tag{3.24}$$

The gain is unity near f_r and rises below it. Operation above f_m gives an inductive tank, so the primary switches turn on at zero voltage, and rectifier diodes turn off softly near f_r. Efficiency is thus very high near the design point. The trade-off is that a wide battery-voltage range demands a wide frequency range, which is one reason for pre-regulating the DC link from the PFC stage.

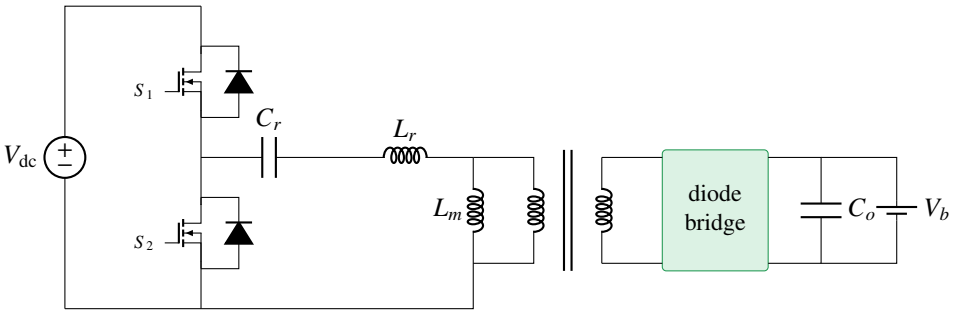


Figure 3.28: LLC resonant converter.

3.2.5.7 Dual Active Bridge

The dual active bridge (DAB) in Figure 3.29 has an active bridge on each side of the transformer, with an inductor L (usually the leakage) between them [13, 14]. Both bridges run at 50% duty, producing square waves with a phase shift ϕ. Inductor voltage is the difference of the square waves, so power flows from the leading bridge to the lagging bridge. With secondary voltage referred to the primary, $V_2' = nV_2$, the transferred power is

$$P = \frac{V_1 V_2' \phi\,(\pi - |\phi|)}{2\pi^2 f_s L}, \tag{3.25}$$

with ϕ in radians. Reversing ϕ reverses power without changing any hardware, so the DAB is the natural choice for bidirectional chargers.

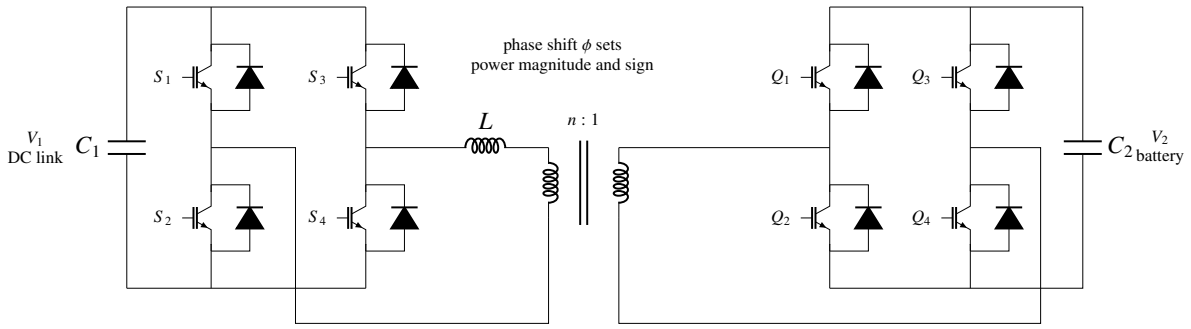


Figure 3.29: Dual active bridge converter.

Table 3.6: Isolated and non-isolated converters

Feature	Non-isolated	Isolated
Examples	Buck, boost, half bridge	Flyback, full bridge, LLC, DAB
Galvanic barrier	None	HF transformer
Voltage ratio	Limited	Set by turns ratio
Size and cost	Lower	Higher
Efficiency	Very high	High
Typical EV use	Traction stage	Charger, 12 V supply

As Table 3.6 shows, the choice is driven by safety: anything that links the grid to the battery or the HV bus to the chassis-referenced 12 V network normally needs isolation.

3.2.6 Bidirectional Charger Topologies

Figure 3.30 shows a typical bidirectional charger: an active front end, normally a totem-pole or three-phase AFE, followed by a DC link and a DAB or a symmetrical CLLC resonant stage. Both stages use active switches, so the structure runs in either direction. Three services follow.

- **Grid-to-Vehicle (G2V).** Ordinary charging with $P > 0$ into the battery.
- **Vehicle-to-Grid (V2G).** The AFE operates as a grid-tied inverter and the vehicle supplies energy or reactive support under an aggregator's control.
- **Vehicle-to-Home (V2H).** The charger forms a local supply for household loads, including during an outage, which needs grid-forming control and islanding protection.

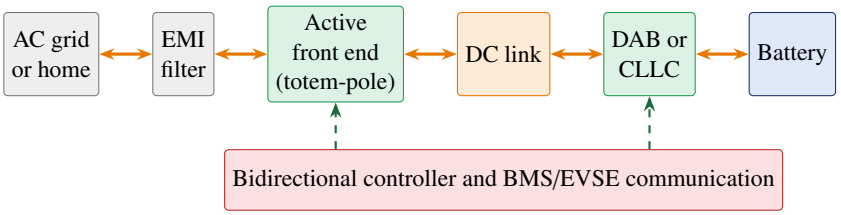


Figure 3.30: Bidirectional charger architecture.

Figure 3.31 summarises these flows. The upper arrows represent G2V, while the lower arrows show the return path. V2G and V2H add battery cycling, so operators must weigh revenue against degradation, a matter treated in Unit 2. Table 3.7 compares the two classes of charger.

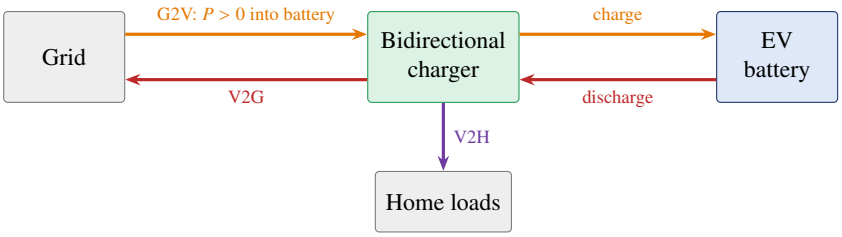


Figure 3.31: G2V, V2G and V2H power flow.

Table 3.7: Unidirectional and bidirectional chargers

Feature	Unidirectional	Bidirectional
Grid stage	Diode bridge + boost, or Vienna rectifier	Totem-pole or AFE
DC-DC stage	LLC or phase-shifted full bridge	DAB or CLLC
Secondary devices	Diodes	Active switches
Reactive support	Limited	Available
Grid services	None	V2G, V2H, V2L
Cost	Lower	Higher

3.2.7 Power Factor and Harmonic Considerations

Distorted current heats transformers and neutral conductors and disturbs other equipment. Modern PFC stages reach PF above 0.99 and current THD below 5% near rated power. Performance degrades at light load, where ripple current is large relative to the fundamental. EMI filters are also needed, since switching at tens to hundreds of kilohertz generates high-frequency conducted noise.

3.2.8 Galvanic Isolation and High-Frequency Transformers

Galvanic isolation means that no conductive path exists between circuits. It prevents a single insulation fault from connecting the grid to touchable parts. The transformer core area follows from Faraday's law. For a square wave,

$$N_p A_c = \frac{V_p}{4 f_s B_{\max}}. \tag{3.26}$$

Here V_p is the square wave amplitude (V), A_c the core area (m^2), and $B_{\max}$ the flux density (T). Raising f_s from 50 Hz to 100 kHz therefore shrinks the $N_p A_c$ product by a factor of 2000, though in practice core loss, winding proximity effect and insulation requirements limit the gain. Ferrite cores and litz wire are the common choices.

3.2.9 Charging Efficiency

Overall efficiency is the product of stage efficiencies,

$$\eta_{\text{chg}} = \eta_{\text{EMI}}\,\eta_{\text{PFC}}\,\eta_{\text{DCDC}}. \tag{3.27}$$

Typical modern OBC efficiency is in the region of 93–96%. Efficiency matters both for cost and because each kilowatt of loss must be removed from the vehicle.

Example 5: EV charger power calculation

Problem. An 11 kW OBC delivers rated power to a 60 kWh battery from a 415 V three-phase supply. Find the input power, line current, loss and the energy and time to charge from 20% to 80%. Tapering is ignored.

Given (assumed). $\eta_{\text{PFC}} = 0.98$, $\eta_{\text{DCDC}} = 0.97$, PF $= 0.99$, $P_b = 11$ kW.

Required. P_{in}, I_L, P_{loss}, t, E_{grid}.

Governing equations. (3.27), $P = \sqrt{3}V_{LL}I_L\text{PF}$, $t = \Delta E/P_b$.

Solution. $\eta_{\text{chg}} = 0.98 \times 0.97 = 0.9506$.
$P_{\text{in}} = 11/0.9506 = 11.57$ kW, $P_{\text{loss}} = 0.57$ kW.
$I_L = 11\,570/(\sqrt{3} \times 415 \times 0.99) = 16.26$ A.
$\Delta E = 0.6 \times 60 = 36$ kWh, so $t = 36/11 = 3.27$ h. $E_{\text{grid}} = 36/0.9506 = 37.9$ kWh.

Result. $P_{\text{in}} = 11.57$ kW, $I_L = 16.26$ A, $t = 3.27$ h, $E_{\text{grid}} = 37.9$ kWh.

Physical interpretation. About 1.9 kWh is lost as heat in the charger for this session.

Engineering significance. The current fits a standard 16 A or 20 A three-phase supply circuit, given a suitable margin under the applicable wiring rules.

3.3 Electric Drive Train Overview

3.3.1 Introduction to Electric Drive Trains

The drive train is everything that carries energy from the battery to the road. It splits naturally into an electrical part and a mechanical part, joined at the motor. An EV drive train is much simpler than that of a conventional car, since a traction motor produces torque from zero speed over a wide speed range, so there is no clutch and usually only one gear ratio [15].

3.3.2 EV Powertrain Architecture and Components

Figure 3.32 shows the complete architecture. Heavy double arrows form the energy path, and dashed arrows are signals. The driver's pedal request reaches the vehicle control unit, which

issues a torque command to the motor controller after considering limits from the battery management system. The components are:

- **Battery system.** Stores energy and sets the DC voltage, as covered in Unit 2.
- **Power electronic converter.** Controls voltage, current and frequency (Section 3.1).
- **Electric traction motor.** Converts electrical to mechanical power and back.
- **Transmission system.** A reduction gear trades speed for torque, so a small high-speed motor can provide high wheel torque.
- **Differential.** Divides torque equally between wheels of an axle, while permitting different wheel speeds in a turn.
- **Drive shaft.** Carries torque to the wheel hubs through constant-velocity joints.
- **Wheels and tyres.** Convert torque to tractive force through the tyre-road contact. Adhesion, not motor torque, is often the limit.
- **Motor controller.** Executes the fast current loops and generates inverter switching commands.
- **Vehicle control unit (VCU).** Interprets driver demand, arbitrates limits, blends regenerative and friction braking and coordinates all subsystems.

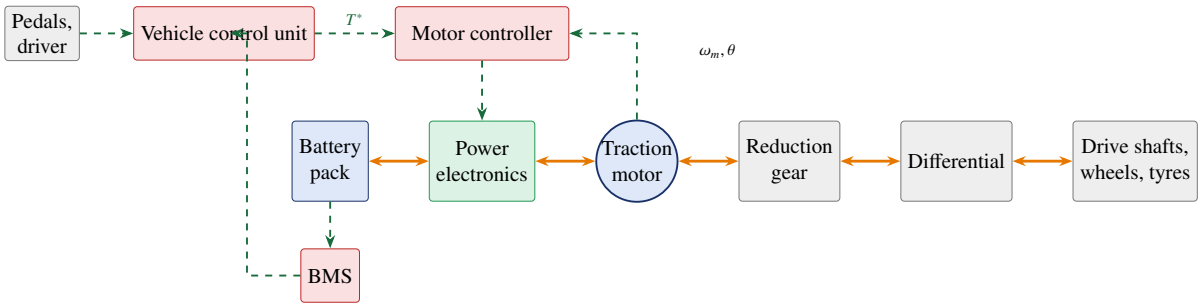


Figure 3.32: Complete EV drivetrain architecture.

3.3.3 Electrical and Mechanical Power Transmission

Figure 3.33 marks the power at each interface. Electrical transmission uses voltage and current, and mechanical transmission uses torque and speed. Multiplying the stage efficiencies gives the drivetrain efficiency,

$$\eta_{\mathrm{dt}} = \frac{F_t v}{V_b I_b} = \eta_{\mathrm{inv}}\, \eta_m\, \eta_g. \tag{3.28}$$

Illustrative values $\eta_{\mathrm{inv}} = 0.97$, $\eta_m = 0.94$ and $\eta_g = 0.97$ give $\eta_{\mathrm{dt}} = 0.88$. This figure varies strongly with operating point.

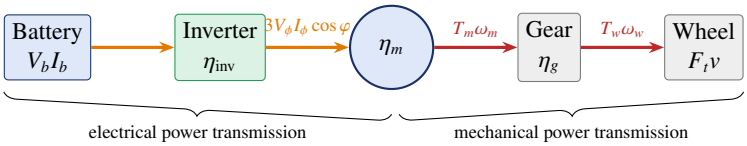


Figure 3.33: Electrical and mechanical power flow.

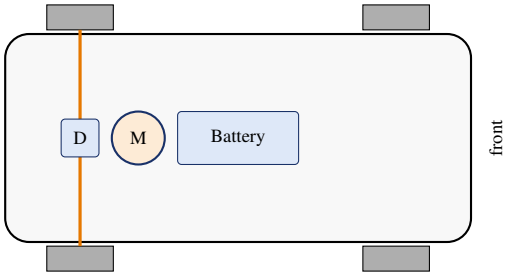
3.3.4 Drivetrain Configurations

3.3.4.1 Single-Motor Drive

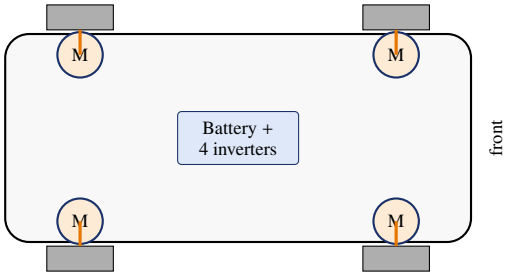
In Figure 3.34a, one motor drives one axle through a reduction gear and differential. It is the simplest and lowest-cost layout.

3.3.4.2 Dual-Motor and Multi-Motor Drive

Two motors can be placed one per axle, or one per wheel. Figure 3.34b shows the multi-motor limit, with in-wheel or near-wheel motors and no mechanical differential, so that the software performs the differential's role and can apply torque vectoring. In-wheel motors raise unsprung mass, which harms ride quality.



(a) Single-motor drivetrain (rear axle).



(b) Multi-motor drivetrain with individually driven wheels.

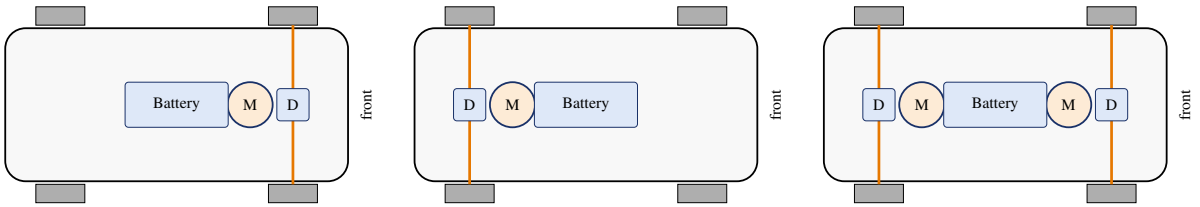
Figure 3.34: Single-motor and multi-motor drivetrains. D denotes the differential and M the motor.

3.3.4.3 Front-, Rear- and All-Wheel Drive

In the front-wheel-drive (FWD) layout of Figure 3.35a, the drive unit sits under the bonnet. The front tyres both steer and drive, and load transfer under acceleration reduces their grip. The rear-wheel-drive (RWD) layout of Figure 3.35b benefits from that load transfer and is preferred for performance. The all-wheel-drive (AWD) layout of Figure 3.35c usually fits a motor to each axle, without a propeller shaft. This dual-motor arrangement improves traction and allows one motor to be shut down at cruise to save energy, particularly if an induction machine or disconnect clutch keeps its drag low.

3.3.4.4 Integrated Electric Axle

An e-axle, in Figure 3.36, combines inverter, motor, gear and differential in one housing. The short connections eliminate external three-phase cables, the parts share cooling, and the assembly drops into a skateboard platform as a single module.



(a) FWD. **(b)** RWD. **(c)** Dual-motor AWD.

Figure 3.35: Front-wheel-drive, rear-wheel-drive and all-wheel-drive architectures.

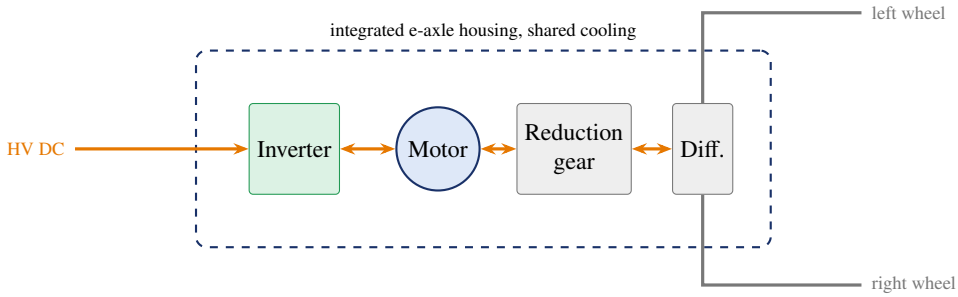


Figure 3.36: Integrated electric axle (e-axle).

3.3.4.5 Single-Speed and Multi-Speed Transmission

With wide-speed motors, a single fixed ratio is usually sufficient, and it is light, cheap and quiet. A multi-speed gearbox lets a smaller motor meet both gradeability and top-speed requirements, and can shift the operating point into a higher-efficiency region. The cost is added mass, shift-quality effort and mechanical loss, so it mainly suits heavy vehicles and high-performance models. Table 3.8 compares the configurations.

Table 3.8: EV drivetrain configurations

Configuration	Cost	Traction	Comment
FWD single motor	Low	Moderate	Compact, common in city and mainstream cars
RWD single motor	Low	Good	Better acceleration grip
Dual-motor AWD	Medium	Very good	Torque split between axles
In-wheel multi-motor	High	Very good	Torque vectoring, high unsprung mass
E-axle	Low–medium	As layout	High integration; applied to any axle
Multi-speed	Medium	Good	Useful for buses and trucks

The table shows that traction performance is largely bought through more motors, whereas integration (the e-axle) reduces cost independently of wheel layout.

3.3.5 Torque Distribution

In a dual-motor vehicle, total torque demand T^* is split as $T_f^* = kT^*$ and $T_r^* = (1-k)T^*$. The split factor k follows axle load for maximum traction, or efficiency maps for minimum loss.

Under hard acceleration, the load moves rearward, so k falls. Yaw control adds a left-right difference when individual-wheel drives exist.

3.3.6 Regenerative Braking in the Drivetrain

Figure 3.37 traces the path during deceleration: wheels turn the gear and machine, and the inverter rectifies the generated power into the battery. Regeneration is limited by battery charge acceptance, machine torque and the fact that torque available to the wheels fades at very low speed. The VCU therefore blends friction brakes, and these remain the primary safety system.

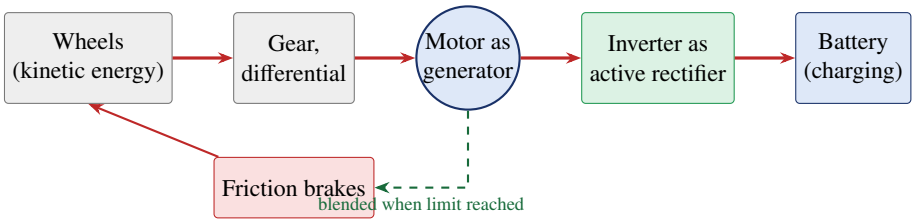


Figure 3.37: Regenerative braking drivetrain.

3.3.7 EV Motor Control Architecture

Figure 3.38 expands the control path. Pedal position gives a torque request; the VCU limits it with BMS data; and the controller converts torque to current references, closes current loops, and produces PWM signals. Feedback of phase current and rotor position closes the inner loop. Sections 3.10 and 3.16 describe these blocks in detail.

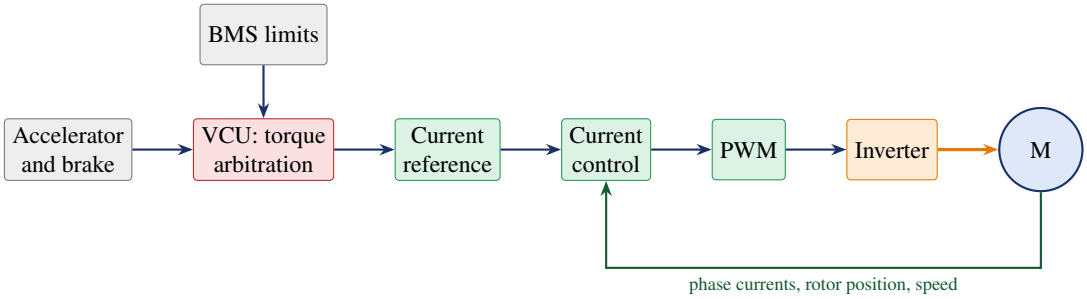


Figure 3.38: EV motor control architecture.

3.3.8 Powertrain Operation During a Drive

- **Acceleration.** Maximum current produces high torque. Battery power rises with speed until the power limit is reached.
- **Cruising.** Torque just balances road load. The motor often runs at part load, in which case efficiency depends on low iron and switching losses.
- **Deceleration.** Lifting the pedal commands negative torque, so the vehicle coasts down under a mild regenerative load.

- **Regenerative braking.** Pressing the brake commands larger negative torque, and the friction brakes supplement it as required.

3.3.9 EV Powertrain Performance

Performance is expressed by acceleration, gradeability, top speed and range. Each maps to the torque-speed envelope. Low-speed torque with gear ratio sets acceleration and gradeability, power sets high-speed acceleration, the base speed and maximum speed set the ratio choice, and drivetrain efficiency sets range. Example 6 computes the power at a cruising point.

Example 6: EV powertrain mechanical power calculation

Problem. A car climbs a 3% grade at 100 km/h. Find tractive force, wheel power, motor torque and battery power.

Given (assumed). $m = 1500$ kg, $C_{rr} = 0.012$, $C_d = 0.3$, $A = 2.2$ m^2, $\rho = 1.2$ kg/m^3, $r_w = 0.31$ m, $G = 9$, $\eta_g = 0.92$ (transmission), electrical efficiency 0.94 (inverter and motor).

Required. F_t, P_w, T_m, ω_m, P_b.

Governing equations. (3.44), (3.36), (3.37).

Solution. $v = 27.78$ m/s, grade angle $\theta = \arctan 0.03$.
$F_r = C_{rr} mg\cos\theta = 176.5$ N, $F_a = \frac{1}{2}\rho C_d A v^2 = 305.6$ N, $F_g = mg\sin\theta = 441.3$ N.
$F_t = 923.3$ N, $P_w = F_t v = 25.65$ kW.
$\omega_w = v/r_w = 89.6$ rad/s, $\omega_m = 9\omega_w = 806$ rad/s.
$T_m = F_t r_w/(G\eta_g) = 286.2/(9 \times 0.92) = 34.6$ N m.
$P_m = 25.65/0.92 = 27.88$ kW, $P_b = 27.88/0.94 = 29.66$ kW.

Result. $F_t = 923$ N, $P_w = 25.65$ kW, $T_m = 34.6$ N m at 806 rad/s, $P_b = 29.7$ kW.

Physical interpretation. Gradient and drag dominate; rolling resistance is the smallest term here.

Engineering significance. Only a modest fraction of the motor's torque capability is needed for steady driving, so it runs well below rated torque for most of its life.

3.4 Systems with Linear Motion and Rotating Systems

3.4.1 Linear Motion Systems

3.4.1.1 Introduction and Fundamental Quantities

A body in linear motion is described by its displacement x (m), velocity $v = dx/dt$ (m/s) and acceleration $a = dv/dt$ (m/s^2). Mass m (kg) measures its resistance to change of velocity, and

force F (N) is the cause of that change. Figure 3.39 shows a mass acted on by a driving force against a load force and friction.

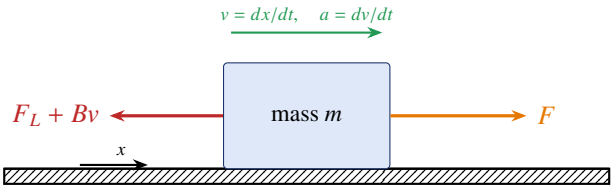


Figure 3.39: Linear motion system.

3.4.1.2 Newton's Law

Newton's second law says that net force equals the rate of change of momentum, $F = d(mv)/dt$. With constant mass, this reduces to

$$F = m\frac{dv}{dt} = ma. \tag{3.29}$$

The unit of force follows: $1\ \text{N} = 1\ \text{kg m/s}^2$.

3.4.1.3 Power and Kinetic Energy

Work done over a small displacement is $dW = F\,dx$. Dividing by dt gives mechanical power,

$$P = \frac{dW}{dt} = F\frac{dx}{dt} = Fv. \tag{3.30}$$

Power is in watts. Integrating the work needed to accelerate a mass from rest,

$$E_k = \int F\,dx = \int m\frac{dv}{dt}v\,dt = \int_0^v mv\,dv = \tfrac{1}{2}mv^2. \tag{3.31}$$

Energy is in joules. Kinetic energy grows with v^2, so braking from 100 km/h has about 2.8 times the recoverable energy available at 60 km/h.

3.4.1.4 Friction and Load Dynamics

Friction opposes motion and has three common components: static friction at rest, roughly constant Coulomb friction when moving, and viscous friction proportional to speed, $F_f = Bv$. Including a load force F_L, the motion equation is

$$F = m\frac{dv}{dt} + Bv + F_L. \tag{3.32}$$

The force-speed characteristic of the load is the plot of $F_L + Bv$ against v, and steady motion occurs where the drive force equals it.

3.4.1.5 Applications in Electric Transportation

A vehicle is a linear system on a road with F as tractive force. Linear models are also used for lifts, conveyor belts and linear motors in transit systems.

3.4.2 Rotating Systems

3.4.2.1 Introduction and Fundamental Quantities

Rotational motion has direct analogues of the linear quantities: angular displacement θ (rad), angular velocity $\omega = d\theta/dt$ (rad/s), angular acceleration $\alpha = d\omega/dt$ (rad/s^2), torque $T = Fr$ (N m), and moment of inertia $J = \int r^2\, dm$ (kg m^2). A speed in revolutions per minute converts by $\omega = 2\pi n/60$. Figure 3.40 shows a rotating inertia with a driving torque and a load torque.

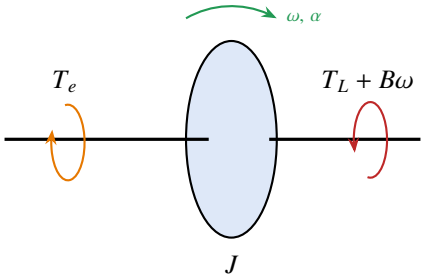


Figure 3.40: Rotating mechanical system.

3.4.2.2 Equation of Motion

Apply Newton's law to a mass element at radius r: its tangential acceleration is $r\alpha$, so $dF = r\alpha\, dm$ and the elemental torque is $dT = r\, dF = r^2\alpha\, dm$. Integrating gives $T_{\text{acc}} = J\alpha$. The motor torque T_e must supply this acceleration torque, the viscous frictional torque $B\omega$, and the load torque T_L:

$$T_e = J\frac{d\omega}{dt} + B\omega + T_L. \tag{3.33}$$

- $J\, d\omega/dt$ is the *acceleration torque*, which is zero at constant speed.
- $B\omega$ is the *frictional torque*, with B in N m s/rad.
- T_L is the *load torque*, the useful or imposed resistance.
- T_e is the *motor (electromagnetic) torque*.

Figure 3.41 shows this motor-load system. Motor and load inertias add, $J = J_m + J_L$, when they share a rigid shaft. If $T_e > T_L + B\omega$, the drive accelerates; if it is less, the drive decelerates.

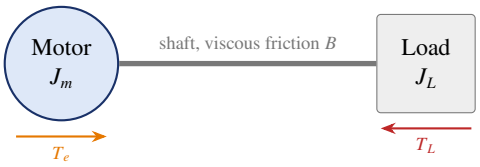


Figure 3.41: Rotating motor-load system.

3.4.2.3 Rotational Power and Energy

Work is $dW = T\,d\theta$, so that

$$P = T\omega, \qquad E_k = \int T\,d\theta = \int J\omega\,d\omega = \tfrac{1}{2}J\omega^2. \tag{3.34}$$

3.4.3 Gear Ratio and Mechanical Transmission

A gear pair with tooth counts N_1 (motor side) and N_2 (load side) has ratio $G = N_2/N_1$, as shown in Figure 3.42. Tooth speed at the mesh is common, so speeds are in inverse ratio. An efficiency η_g accounts for mesh and bearing loss while driving forward:

$$\omega_L = \frac{\omega_m}{G}, \qquad T_L = \eta_g G T_m. \tag{3.35}$$

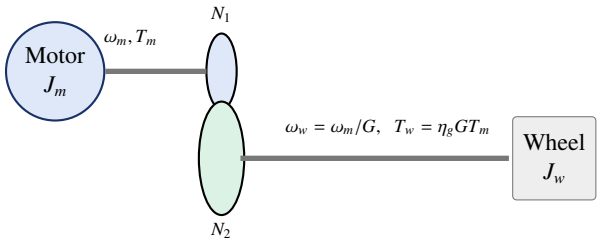


Figure 3.42: Gearbox and drivetrain arrangement.

3.4.4 Relationship Between Linear and Rotational Motion

Figure 3.43 connects motor, gear, wheel and road. A wheel of radius r_w rolling without slip has

$$v = \omega_w r_w, \qquad \omega_w = \frac{\omega_m}{G} \quad \Rightarrow \quad v = \frac{\omega_m r_w}{G}. \tag{3.36}$$

The wheel torque produces a tractive force, and hence

$$F_t = \frac{T_w}{r_w} = \frac{\eta_g G T_m}{r_w}. \tag{3.37}$$

Equations (3.36) and (3.37) show the trade: raising G gives more force but less speed for the same motor. Power is preserved apart from loss, $F_t v = \eta_g T_m \omega_m$.

3.4.4.1 Reflected Inertia and Load Torque

To write one motion equation at the motor shaft, load quantities are *reflected* through the gear. Equating kinetic energy, a vehicle of mass m gives $\frac{1}{2}mv^2 = \frac{1}{2}J_{\text{eq}}\omega_m^2$. Using (3.36),

$$J_{\text{v}} = m\left(\frac{r_w}{G}\right)^2, \qquad J_{\text{eq}} = J_m + \frac{J_w}{G^2} + m\left(\frac{r_w}{G}\right)^2. \tag{3.38}$$

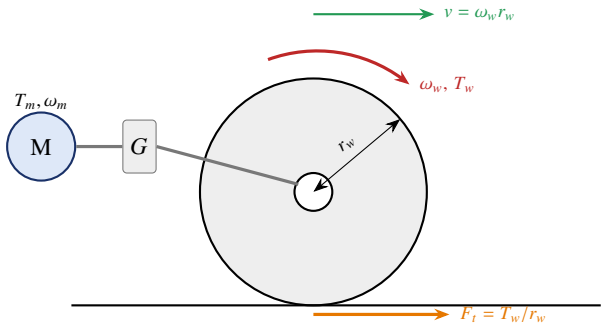


Figure 3.43: Vehicle wheel and motor torque relationship.

A load torque is reflected by equating power, $T'_L \omega_m \eta_g = T_L \omega_L$, so that

$$T'_L = \frac{T_L}{\eta_g G} \quad \text{(motoring)}, \tag{3.39}$$

and η_g moves to the numerator when power flows back from the wheels. Table 3.9 sets out the analogy.

Table 3.9: Linear and rotational motion

Linear quantity	Symbol, unit	Rotational analogue	Symbol, unit
Displacement	x, m	Angle	θ, rad
Velocity	v, m/s	Angular velocity	ω, rad/s
Mass	m, kg	Moment of inertia	J, kg m^2
Force	F, N	Torque	T, N m
Newton's law	$F = ma$	Motion equation	$T = J\alpha$
Power	$P = Fv$	Power	$P = T\omega$
Kinetic energy	$\frac{1}{2}mv^2$	Kinetic energy	$\frac{1}{2}J\omega^2$

Every rotational result can thus be read from its linear counterpart by substitution, which reduces the amount of independent material to be learned.

Example 7: Linear motion and rotational motion relationship

Problem. A vehicle has wheels of radius 0.3 m and a gear ratio of 9. Find wheel and motor speed at 72 km/h.

Given. $v = 72$ km/h, $r_w = 0.3$ m, $G = 9$.

Required. ω_w, ω_m, n_m.

Governing equations. (3.36).

Solution. $v = 72/3.6 = 20$ m/s. $\omega_w = 20/0.3 = 66.7$ rad/s. $\omega_m = 9 \times 66.7 = 600$ rad/s. $n_m = 600 \times 60/(2\pi) = 5730$ r/min.

Result. $\omega_w = 66.7$ rad/s, $\omega_m = 600$ rad/s, $n_m = 5730$ r/min.

Physical interpretation. The motor turns nine times faster than the wheel.

Engineering significance. Choosing a motor maximum speed of, say, 15 000 r/min at this ratio sets a vehicle top speed of about 188 km/h.

Example 8: Gear ratio and motor torque calculation

Problem. The wheels need 1800 N m. Find the motor torque and the total inertia at the motor shaft.

Given (assumed). $G = 9$, $\eta_g = 0.95$, $m = 1500$ kg, $r_w = 0.3$ m, $J_m = 0.05$ kg m^2, total wheel inertia $J_w = 1.2$ kg m^2.

Required. T_m, J_{eq}, and the acceleration of the vehicle with motor torque 200 N m and load neglected.

Governing equations. (3.35), (3.38), (3.33).

Solution. $T_m = 1800/(9 \times 0.95) = 210.5$ N m.
$J_{\text{v}} = 1500(0.3/9)^2 = 1.667$ kg m^2. $J_w/G^2 = 0.0148$ kg m^2.
$J_{\text{eq}} = 0.05 + 0.0148 + 1.667 = 1.73$ kg m^2.
At 200 N m, $d\omega_m/dt = 200/1.73 = 115.5$ rad/s^2. Then $a = (r_w/G)\, d\omega_m/dt = 3.85$ m/s^2.

Result. $T_m = 210.5$ N m, $J_{\text{eq}} = 1.73$ kg m^2, $a = 3.85$ m/s^2.

Physical interpretation. Reflected vehicle mass is far larger than rotor inertia in this example.

Engineering significance. Rotor inertia is reduced by G^2 on the wheel side but it appears in full at the motor, so control-loop tuning must use J_{eq}, not J_m alone. Gear efficiency is ignored in this inertia estimate.

3.5 Types of Loads

3.5.1 Introduction to Mechanical Loads

A drive is selected by matching the torque-speed capability of the motor to the torque-speed demand of the load. A load is therefore classified by how its torque T_L varies with speed ω, and since $P_L = T_L\omega$, the power curve follows directly.

3.5.2 Constant Torque Loads

For a constant torque load,

$$T_L = T_0, \qquad P_L = T_0\omega. \tag{3.40}$$

As shown in Figure 3.44, torque is independent of speed, so power is proportional to speed. Conveyors, hoists and rolling friction are examples. In a **hoist or crane**, the load torque equals the weight force times drum radius, $T_L = mgr$, and acts in the same direction whether raising or lowering: the load is *active*, and it can drive the motor when lowering. A **conveyor** carries a load which mostly appears as friction, nearly constant over its speed range, though breakaway torque at start is higher. A **rolling mill** has a high, suddenly applied torque when metal enters the rolls, and is commonly treated as a constant torque load with an impact component, with a flywheel or strong field control for support.

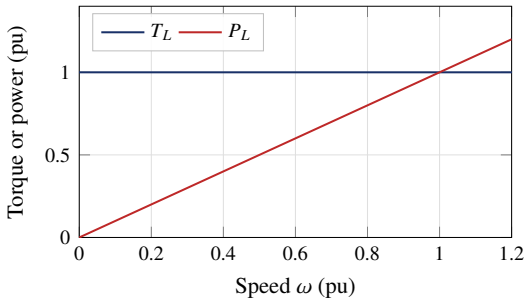


Figure 3.44: Constant torque load curve.

3.5.3 Variable Torque Loads

A variable torque load has $T_L = k\omega^n$, with $n \neq 0$. Figure 3.45 shows the cases $n = 1$, typical of viscous friction and a generator feeding a fixed resistor, and $n = 2$, the *quadratic torque* load, for which $P \propto \omega^3$.

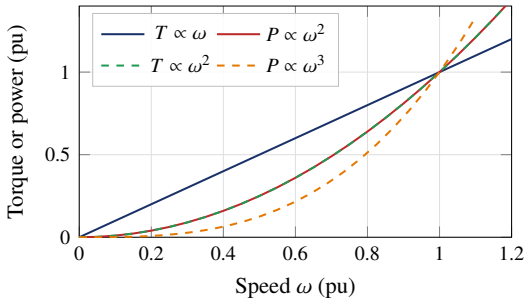


Figure 3.45: Variable torque load curves.

3.5.3.1 Fan-Type and Pump-Type Loads

Fan torque arises from air momentum: mass flow rate is proportional to speed and velocity change is proportional to speed, so torque is quadratic. Including a small friction torque T_f,

$$T_{\text{fan}} = T_f + k_f\omega^2, \qquad P_{\text{fan}} \approx k_f\omega^3. \tag{3.41}$$

A centrifugal pump obeys similar affinity laws, with a larger static term, since it must first reach a head and overcome seal friction:

$$T_{\text{pump}} = T_s + k_p\omega^2. \tag{3.42}$$

Figure 3.46 compares them. These laws govern EV cooling fans and coolant pumps: a modest reduction in speed saves a large amount of energy.

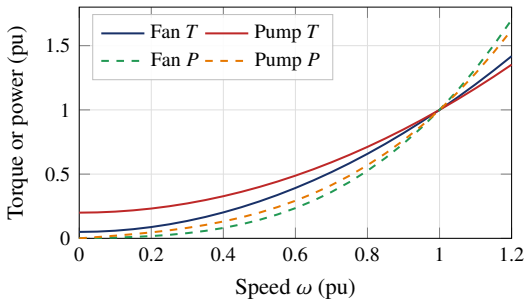


Figure 3.46: Fan and pump characteristics (per unit).

3.5.4 Constant Power Loads

In a constant power load, $T_L\omega = P_0$, so

$$T_L = \frac{P_0}{\omega}. \tag{3.43}$$

Torque falls hyperbolically with speed, as in Figure 3.47. Machine tools and winders behave in this way, and the motor limits of a vehicle at high speed have this form.

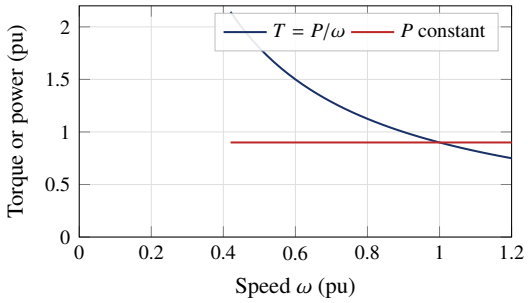


Figure 3.47: Constant power characteristic.

3.5.5 Traction Loads and Vehicle Road-Load Characteristics

The force needed to drive a vehicle up a slope of angle θ is the sum of rolling, aerodynamic, gradient and inertial terms:

$$F_t = C_{rr} mg\cos\theta + \tfrac{1}{2}\rho C_d A v^2 + mg\sin\theta + \delta m\frac{dv}{dt}, \tag{3.44}$$

where $\delta \geq 1$ allows for rotating inertia. Reflecting this to the motor by $T_L = F_t r_w/(\eta_g G)$ gives

$$T_L(\omega_m) = \frac{r_w}{\eta_g G}\left[C_{rr} mg\cos\theta + mg\sin\theta + \tfrac{1}{2}\rho C_d A\left(\frac{r_w\omega_m}{G}\right)^2\right]. \tag{3.45}$$

This is a constant torque term plus a quadratic term, so a vehicle is in effect a combined constant and fan-type load. Figure 3.48 plots a representative set of road-load curves against a

motor envelope. The motor's constant torque region gives acceleration and gradeability, and the constant power region gives speed range. At high speed, the crossing of the road-load and envelope curves sets the top speed.

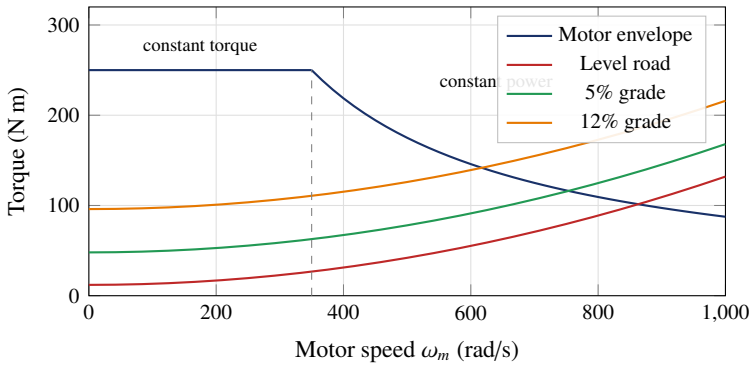


Figure 3.48: EV traction torque-speed characteristic with road load curves (illustrative).

3.5.6 Motor-Load Matching

The operating point of a drive is where motor and load curves intersect, as in Figure 3.49. Between the two curves, the excess torque $T_e - T_L$ accelerates the system through (3.33). The point is stable if a small speed rise makes the load torque exceed the motor torque,

$$\left.\frac{dT_L}{d\omega}\right|_{\text{op}} > \left.\frac{dT_e}{d\omega}\right|_{\text{op}}. \tag{3.46}$$

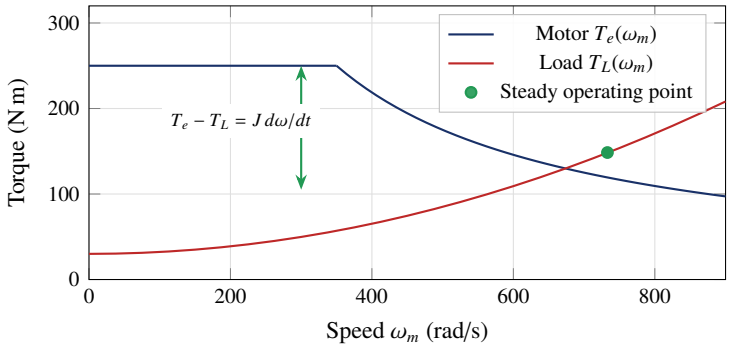


Figure 3.49: Motor and load torque-speed matching.

Table 3.10 summarises the classes of load.

The EV entry shows that traction combines several classes: the constant term dominates at low speed or on a gradient, and the quadratic term dominates at motorway speed. Traction motors must therefore be specified for both regions.

Example 9: Constant torque and variable torque load calculation

Problem. Two loads each take 100 N m at 1500 r/min. One is constant torque and the other is

Table 3.10: Load types: torque and power relationships

Load type	Torque	Power	Applications
Constant torque	T_0	$\propto \omega$	Conveyor, hoist, rolling resistance
Linear variable	$\propto \omega$	$\propto \omega^2$	Viscous friction
Quadratic	$\propto \omega^2$	$\propto \omega^3$	Fans, pumps, aerodynamic drag
Constant power	$\propto 1/\omega$	P_0	Winders, machine tools
Traction	$a + b\omega^2$	$a\omega + b\omega^3$	EV on road

fan-type. Find torque and power at 1500 and 1000 r/min.

Given. $T_{\text{rated}} = 100$ N m, $n_1 = 1500$ r/min, $n_2 = 1000$ r/min. Static fan torque neglected.

Required. T and P for each load at each speed.

Governing equations. (3.40), (3.41), $P = T\omega$.

Solution. $\omega_1 = 157.1$ rad/s, $\omega_2 = 104.7$ rad/s. At 1500 r/min, both give $P = 15.71$ kW.
Constant torque at 1000 r/min: $T = 100$ N m, $P = 10.47$ kW.
Fan at 1000 r/min: $T = 100(1000/1500)^2 = 44.4$ N m, $P = 4.65$ kW.

Result. Reducing speed by one third saves 33% of the constant torque power, but 70% of the fan power.

Physical interpretation. Fan power falls with the cube of speed.

Engineering significance. Variable-speed control of EV coolant pumps and blowers is strongly justified, since auxiliary load directly affects range.

3.6 Four Quadrant Operation

3.6.1 Introduction

A drive that can produce torque in either direction at speed in either direction has four possible combinations. A vehicle uses all of them: driving forward, braking, reversing and braking in reverse. Because the power electronics of Section 3.1 is bidirectional, an electric drive provides these modes without any mechanical reversing gear.

3.6.2 Speed-Torque Plane and Sign Convention

Figure 3.50 shows the plane with torque on the horizontal axis and speed on the vertical axis. The conventions are:

- Speed ω_m is positive in the forward direction.
- Torque T_e is positive when it acts to accelerate the rotor in the forward direction.

- Mechanical power out of the machine is

$$P_m = T_e \omega_m. \tag{3.47}$$

If torque and speed have the same sign, $P_m > 0$ and the machine is motoring. If they have opposite signs, $P_m < 0$ and the machine absorbs mechanical power, which is generating or braking.

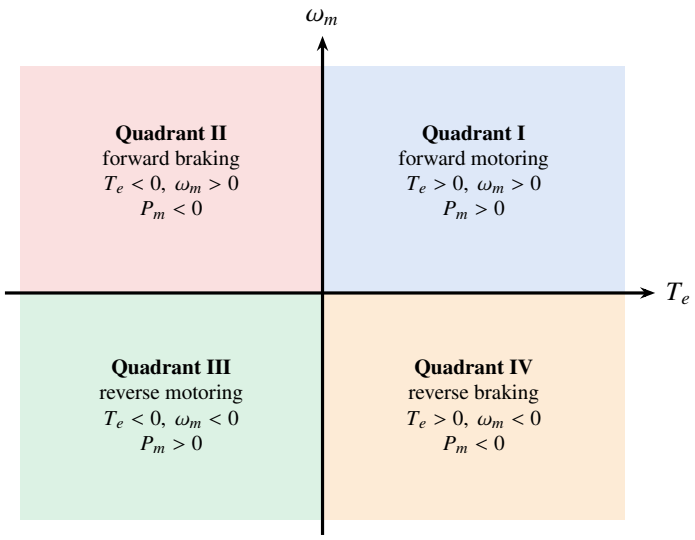
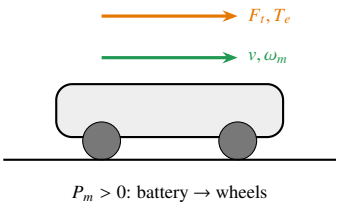


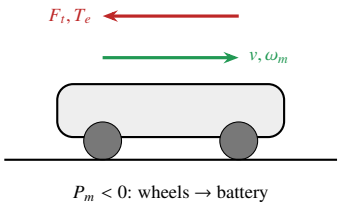
Figure 3.50: Four-quadrant torque-speed plane.

3.6.3 The Four Quadrants

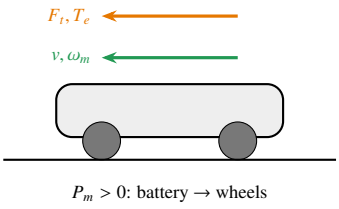
Figure 3.51 illustrates each mode in terms of a vehicle.



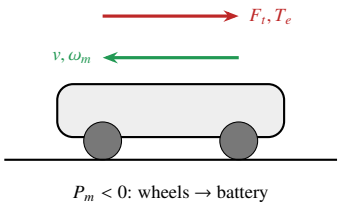
(a) Quadrant I: forward motoring.



(b) Quadrant II: forward regenerative braking.



(c) Quadrant III: reverse motoring.



(d) Quadrant IV: reverse regenerative braking.

Figure 3.51: Operating modes in the four quadrants. Green arrows give velocity, and upper arrows the force from the drive.

3.6.3.1 Quadrant I: Forward Motoring

Speed and torque are positive, as in Figure 3.51a. Energy flows from battery to wheels. This is ordinary driving and hill climbing.

3.6.3.2 Quadrant II: Forward Braking

The vehicle moves forward while torque acts backward (Figure 3.51b). Kinetic energy passes through the machine into the battery. Lifting off and braking belong here, as does descending a long hill at steady speed.

3.6.3.3 Quadrant III: Reverse Motoring

Both signs are negative (Figure 3.51c), and power is positive again. This is reversing, in which the inverter simply changes the phase sequence.

3.6.3.4 Quadrant IV: Reverse Braking

The vehicle moves backward with forward torque (Figure 3.51d). Examples are braking while reversing, and holding a car that is rolling backward on a slope until drive torque stops it.

Table 3.11 draws these results together.

Table 3.11: Four-quadrant drive operation

Q	ω_m	T_e	P_m	Mode	Energy flow	EV application
I	+	+	+	Forward motoring	Battery to wheels	Driving
II	+	−	−	Forward braking	Wheels to battery	Regenerative braking
III	−	−	+	Reverse motoring	Battery to wheels	Reverse gear
IV	−	+	−	Reverse braking	Wheels to battery	Braking in reverse, roll-back on a hill

Note from the table that only the *product* of the signs determines the energy direction: quadrants II and IV both return energy, whatever the direction of travel.

3.6.4 Electrical and Mechanical Power Flow

The electrical power at the battery differs from P_m by losses, whose direction never changes:

$$P_b = \begin{cases} P_m/\eta, & P_m > 0, \\ \eta P_m, & P_m < 0. \end{cases} \tag{3.48}$$

Figure 3.52 shows the bidirectional path. A reversal of battery current sign happens without any contactor operation, as the inverter diodes and switches conduct whichever way current flows.

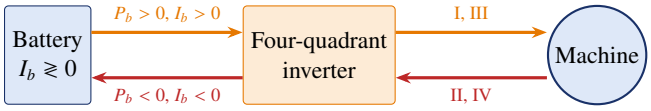


Figure 3.52: Bidirectional power flow between battery and motor.

3.6.5 Reversible DC and AC Drives

A DC machine reverses speed by reversing armature voltage, and reverses torque by reversing armature current, for which a four-quadrant H-bridge chopper provides both. In an AC drive, torque reverses through the sign of the q-axis current, and speed direction reverses through the phase sequence. Field-oriented control handles both smoothly through zero speed.

3.6.6 Four-Quadrant Operation in EV Traction

Figure 3.53 sketches the operating map of a traction drive. Forward motoring has the largest envelope, whereas the regeneration envelope is often smaller, since it is limited by battery charging current at high state of charge and by low-temperature restrictions. Reverse is usually restricted by software to low speed. Figure 3.54 shows the supervisory decision made in every control cycle.

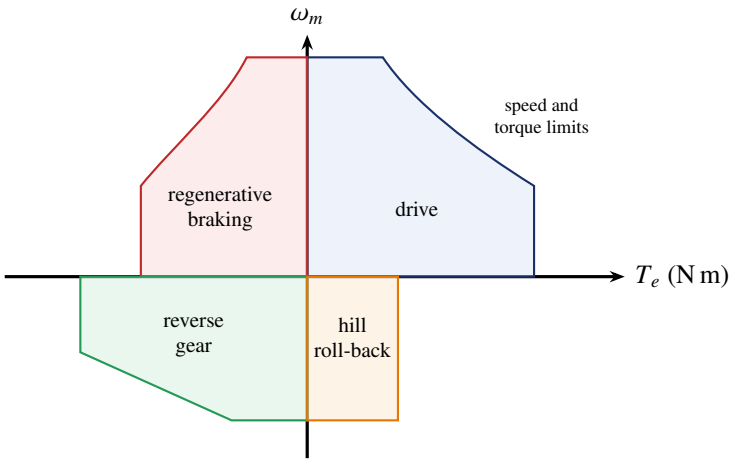


Figure 3.53: EV four-quadrant drive operation with typical operating areas.

Algorithm 4 shows the logic: positive power is limited by discharge current, negative power by charge current, and any torque shortfall in braking is passed to the friction brakes.

3.6.7 Regenerative Braking and Energy Recovery

Energy recovered in a stop from v_1 to v_2 is

$$E_{\text{rec}} = \eta_{\text{regen}} \left[\tfrac{1}{2} m (v_1^2 - v_2^2) - E_{\text{road}} \right], \tag{3.49}$$

where E_{road} is energy spent against drag and rolling resistance during the stop, and η_{regen} is the wheel-to-battery efficiency. Recovery matters most in stop-start traffic. At very low speed the

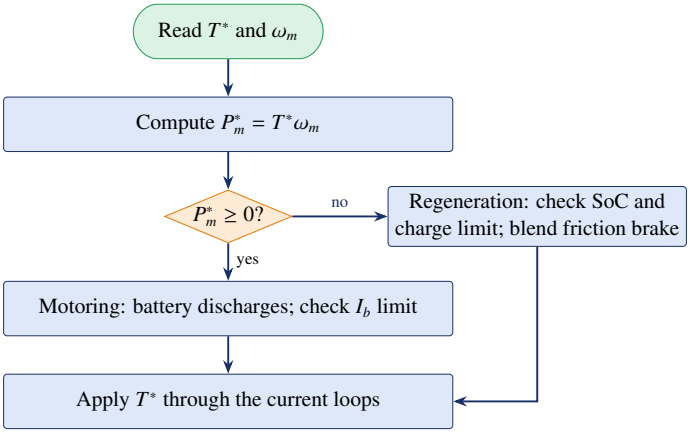


Figure 3.54: Four-quadrant supervisory logic.

Algorithm 4 Four-quadrant drive operation

Input: Torque demand T^*, speed ω_m, SoC, limits $I_{\text{dis}}, I_{\text{chg}}$
Output: Limited torque T^*_{lim}, brake blending command

Compute $P_m^* \leftarrow T^*\omega_m$
if $P_m^* \geq 0$ **then** ▷ Quadrants I, III
    $T^*_{\text{lim}} \leftarrow$ limit T^* so that $P_m^*/\eta \leq V_b I_{\text{dis}}$
else ▷ Quadrants II, IV
    $T^*_{\text{lim}} \leftarrow$ limit T^* so that $\eta \left|P_m^*\right| \leq V_b I_{\text{chg}}$
    if SoC high or battery cold **then**
        Reduce regeneration further
    end if
    $T_{\text{fric}} \leftarrow T^* - T^*_{\text{lim}}$ ▷ Friction brake share
end if
Pass T^*_{lim} to the current controllers

machine's induced voltage is small and the mechanical power is low, so recovery declines and friction brakes complete the stop.

Example 10: Four-quadrant operation and power sign calculation

Problem. A motor runs at $|T_e| = 150$ N m and $|\omega_m| = 400$ rad/s. Find P_m and the battery power for each quadrant, with one-way efficiency 0.9.

Given. Values above; $\eta = 0.9$ (assumed).

Required. P_m, P_b, mode.

Governing equations. (3.47), (3.48).

Solution. Magnitude: $150 \times 400 = 60$ kW.
Q-I: $T_e = +150$, $\omega_m = +400$: $P_m = +60$ kW, $P_b = +66.7$ kW.
Q-II: $T_e = -150$, $\omega_m = +400$: $P_m = -60$ kW, $P_b = -54$ kW.
Q-III: $T_e = -150$, $\omega_m = -400$: $P_m = +60$ kW, $P_b = +66.7$ kW.
Q-IV: $T_e = +150$, $\omega_m = -400$: $P_m = -60$ kW, $P_b = -54$ kW.

Result. Motoring draws 66.7 kW; braking returns 54 kW.

Physical interpretation. The 12.7 kW asymmetry is the two-way loss.

Engineering significance. Only $54/66.7 = 81\%$ of the battery energy spent reaches the battery again, even under perfect conditions, so anticipation and coasting remain useful.

3.7 Induction Motor for EV Powertrain

3.7.1 Introduction to Induction Motors

The three-phase induction motor (IM) has no magnets and no brushes. Its rotor current is induced from the stator, which is why it is also called an asynchronous machine. Rugged construction, a lack of rare-earth material, and the absence of magnet-induced drag when unexcited all make it a serious traction candidate, particularly for the secondary axle of a dual-motor vehicle [15, 20].

3.7.2 Construction of Three-Phase Induction Motor

Figure 3.55 shows a cross-section. The stator and rotor are laminated silicon-steel cylinders separated by a small air gap, and the laminations limit eddy-current loss from the alternating flux.

3.7.2.1 Stator Construction

The stator has slots that carry a distributed three-phase winding. Each phase occupies a group of slots, displaced by 120° electrical from the other phases as in Figure 3.56. With sinusoidal

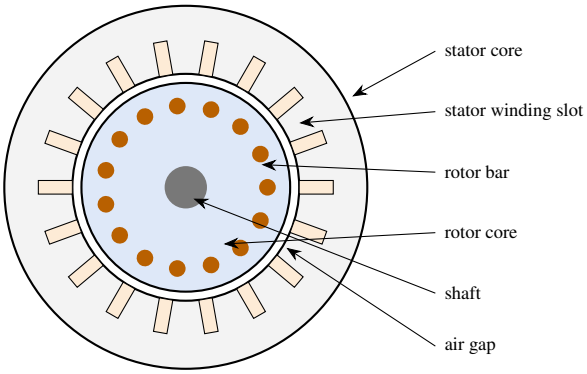


Figure 3.55: Induction motor construction (cross-section).

winding distribution, each phase produces a nearly sinusoidal MMF around the gap.

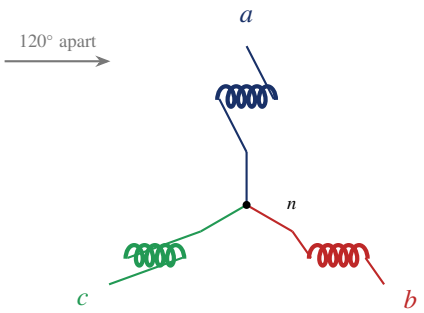


Figure 3.56: Three-phase stator winding (star connection).

3.7.2.2 Rotor Construction

Squirrel-cage rotor. Conducting bars lie in rotor slots and are short-circuited by end rings, as in Figure 3.57. Aluminium is cheap and easily cast, while copper has lower resistance and hence lower rotor loss, which is valuable in traction machines. **Wound rotor.** A three-phase winding is brought out to slip rings, so that external resistance or converters can be connected. Table 3.12 compares them. The wound rotor's brushes and maintenance rule it out for EV traction.

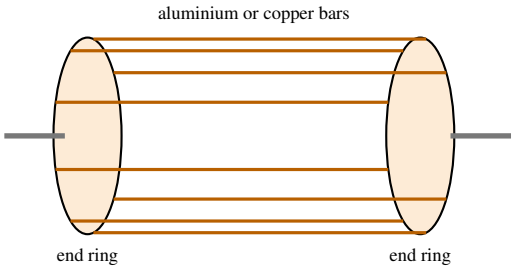


Figure 3.57: Squirrel-cage rotor with iron removed.

Table 3.12: Squirrel-cage and wound-rotor induction motors

Feature	Squirrel cage	Wound rotor
Rotor winding	Bars and end rings	Three-phase insulated winding
External access	None	Slip rings and brushes
Starting torque control	Through the inverter	External resistance
Maintenance	Very low	Brush and ring upkeep
Maximum speed	High	Limited
Cost	Lower	Higher
EV suitability	Well suited	Not used

3.7.3 Rotating Magnetic Field

Let balanced currents $i_a = I_m \cos \omega_e t$, $i_b = I_m \cos(\omega_e t - 120^\circ)$ and $i_c = I_m \cos(\omega_e t + 120^\circ)$ flow in windings whose axes are at 0°, 120° and 240°. At gap angle θ, the MMF of each phase is proportional to its current times the cosine of the angle from its axis:

$$F(\theta, t) = N[i_a \cos \theta + i_b \cos(\theta - 120^\circ) + i_c \cos(\theta - 240^\circ)]. \tag{3.50}$$

Using $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$, the three $\cos(\omega_e t + \theta + \ldots)$ terms sum to zero and the others add:

$$F(\theta, t) = \tfrac{3}{2} N I_m \cos(\theta - \omega_e t). \tag{3.51}$$

This is a travelling wave of constant amplitude, $3/2$ of a single phase peak, which rotates at ω_e electrical rad/s. Figure 3.58 shows the resultant MMF vector (orange) at three instants: it keeps its length and rotates with time. Exchanging two phases reverses the direction, which gives reverse operation.

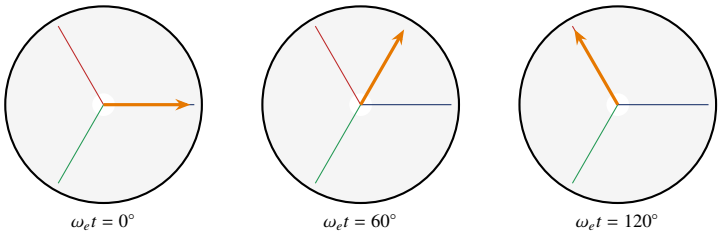


Figure 3.58: Rotating magnetic field: resultant MMF at three instants.

3.7.4 Synchronous Speed, Slip and Rotor Frequency

In a machine with p pole pairs, one electrical cycle covers $1/p$ of a revolution. The field therefore rotates at the synchronous speed

$$\omega_s = \frac{\omega_e}{p} \ \text{(rad/s)}, \qquad n_s = \frac{60 f_e}{p} = \frac{120 f_e}{P} \ \text{(r/min)}. \tag{3.52}$$

The rotor turns at ω_m. Slip is the relative speed as a fraction of synchronous speed:

$$s = \frac{\omega_s - \omega_m}{\omega_s} = \frac{n_s - n_r}{n_s}. \tag{3.53}$$

So $\omega_m = (1 - s)\omega_s$. Rotor conductors see the field moving past at $s\omega_s$, so the frequency of rotor currents is

$$f_r = s f_e. \tag{3.54}$$

Rotor currents are therefore low in frequency at normal load, typically a few hertz.

Example 11: Induction motor synchronous speed calculation

Problem. Find the synchronous speed of a 4-pole motor at 50 Hz and on an inverter at 200 Hz.

Given. $P = 4$, so $p = 2$; $f_e = 50$ and 200 Hz.

Required. n_s, ω_s.

Governing equations. (3.52).

Solution. $n_s = 120 \times 50/4 = 1500$ r/min, $\omega_s = 157.1$ rad/s. At 200 Hz, $n_s = 6000$ r/min and $\omega_s = 628.3$ rad/s.

Result. 1500 r/min and 6000 r/min.

Physical interpretation. Speed follows frequency, so the inverter acts as a variable gearbox.

Engineering significance. EV motors run at several hundred hertz, which increases iron loss and calls for thin laminations.

Example 12: Induction motor slip and rotor frequency calculation

Problem. The motor of Example 11 runs at 5820 r/min on the 200 Hz supply. Find the slip, the rotor frequency and the slip speed.

Given. $n_s = 6000$ r/min, $n_r = 5820$ r/min, $f_e = 200$ Hz.

Required. s, f_r, slip speed.

Governing equations. (3.53), (3.54).

Solution. $s = (6000 - 5820)/6000 = 0.03$. $f_r = 0.03 \times 200 = 6$ Hz. Slip speed = 180 r/min = 18.85 mechanical rad/s.

Result. $s = 3\%$, $f_r = 6$ Hz, slip speed 180 r/min.

Physical interpretation. Low-frequency rotor currents mean that rotor leakage reactance and skin effect are small at this load.

Engineering significance. Control keeps slip small, as rotor copper loss equals slip times air-gap power, derived in Section 3.8.

3.7.5 Torque Production

The rotating field cuts the rotor bars and induces voltages in them. As the cage is short-circuited, currents flow and these interact with the field to give force on each bar. By Lenz's law, the force acts to reduce the relative motion, so it drags the rotor along behind the field. If the rotor reached synchronous speed, the relative motion would vanish, as would the induced current and the torque; hence a motor always runs with some slip. Above synchronous speed, slip is negative, and the machine generates: this is how an induction machine regenerates.

Torque is produced by the interaction of the field with the in-phase component of rotor current,

$$T_e = k\,\Phi_m I_r \cos\varphi_r, \tag{3.55}$$

where φ_r is the rotor power-factor angle and k a machine constant. Near zero slip, rotor current is proportional to slip and largely resistive, so torque rises linearly with slip. At large slip, rotor reactance dominates and torque falls. Figure 3.59 shows this complete torque-speed characteristic, including motoring, generating and plugging regions, computed from the example parameters of Section 3.8.

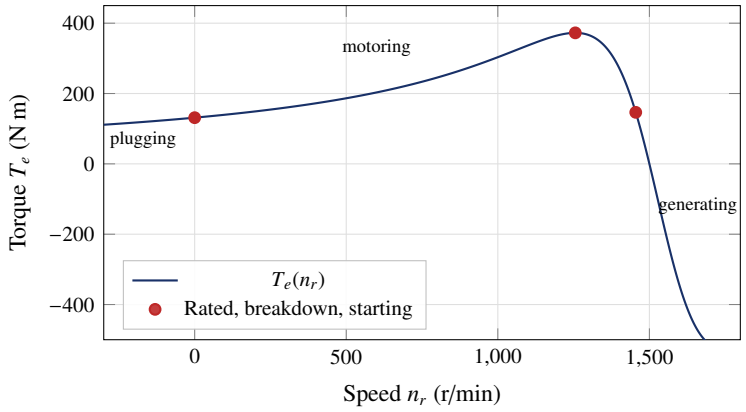


Figure 3.59: Induction motor torque-speed characteristic at 50 Hz (machine of Example 13).

3.7.6 Per-Phase Equivalent Circuit

Figure 3.60 gives the machine's equivalent circuit, which is derived in Section 3.8. It looks like a transformer whose secondary is short-circuited, with the mechanical load represented by the slip-dependent resistance.

3.7.7 EV Traction Operating Requirements and Regions

A traction machine must supply high torque at low speed and high power at high speed. The drive achieves this by varying voltage and frequency together.

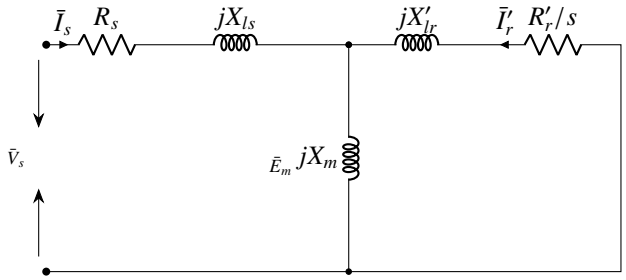


Figure 3.60: Induction motor per-phase equivalent circuit (core loss neglected).

3.7.7.1 Constant Torque Region

Below base speed, the inverter holds air-gap flux at its rated value by keeping V/f approximately constant, so that full torque is available at any speed. Power rises with speed.

3.7.7.2 Field Weakening and the Constant Power Region

At base speed, the inverter reaches its voltage limit. Above that, frequency increases at fixed voltage, so flux falls as $1/f_e$. With current held at its limit, torque falls as $1/\omega$ and power stays roughly constant. This is the field-weakening or constant power region. Finally, since breakdown torque falls as $1/\omega^2$, a third region exists at very high speed, where power falls because the machine reaches its pull-out limit. Figure 3.61 shows these effects using the example machine.

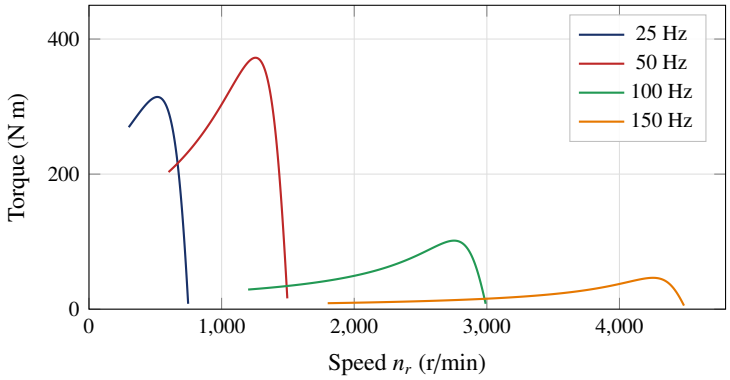


Figure 3.61: Torque-speed curves of the example machine with constant V/f to 50 Hz and constant voltage above.

3.7.8 Induction Motor EV Traction Drive

Figure 3.62 shows the drive. Battery and VSI supply the machine, and the controller closes loops on current and speed, using a speed sensor. Both V/f and vector control are possible, although EVs use vector control, which is treated in Sections 3.9 and 3.10.

3.7.9 Advantages and Limitations in EV Applications

Advantages:

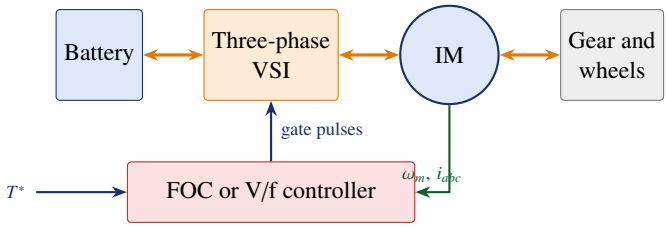


Figure 3.62: Induction motor EV traction drive.

- No rare-earth magnets, with lower material cost and supply risk.
- Robust rotor that tolerates high temperature and high speed.
- Zero spinning drag when de-excited, which is useful on the secondary axle.
- Safe fault behaviour: no back EMF exists when excitation is removed.

Limitations:

- Rotor copper loss produces heat in a rotor that is hard to cool.
- Magnetising current lowers power factor and raises inverter current.
- Lower power density and efficiency than a PMSM, especially at part load.
- The controller depends on rotor parameters that vary with temperature.

3.8 Steady-State Analysis of Induction Drive

3.8.1 Per-Phase Equivalent Circuit Derivation

Assume a balanced machine, sinusoidal fields, no saturation, and steady state. Let the air-gap field induce an rms voltage $\bar{E}_s$ per stator phase at frequency f_e. In the rotor, this is $s\bar{E}_r$ at frequency sf_e, and rotor leakage reactance at that frequency is sX_{lr}. Rotor current is thus

$$\bar{I}_r = \frac{s\bar{E}_r}{R_r + jsX_{lr}} = \frac{\bar{E}_r}{R_r/s + jX_{lr}}. \tag{3.56}$$

Dividing by s moves the circuit to stator frequency, without changing the current magnitude. Referring rotor quantities to the stator by the effective turns ratio $a = N_s/N_r$ gives $\bar{E}'_r = a\bar{E}_r = \bar{E}_s$, $\bar{I}'_r = \bar{I}_r/a$, $R'_r = a^2R_r$ and $X'_{lr} = a^2X_{lr}$. This gives Figure 3.60, in which:

- R_s is the stator resistance per phase (Ω).
- $X_{ls} = \omega_e L_{ls}$ is the stator leakage reactance, due to flux that links only the stator.
- $X_m = \omega_e L_m$ is the magnetizing reactance that carries the air-gap flux.
- R'_r is the rotor resistance referred to the stator.

- X'_{lr} is the rotor leakage reactance referred to the stator.
- R'_r/s is the slip-dependent rotor resistance, which accounts for both rotor loss and output power.

Writing $R'_r/s = R'_r + R'_r(1-s)/s$ separates the rotor resistance from a fictitious resistance that represents mechanical output.

3.8.2 Power Flow

Air-gap power is all the power crossing to the rotor, which is the power dissipated in R'_r/s:

$$P_{ag} = 3\left|\bar{I}'_r\right|^2 \frac{R'_r}{s}. \tag{3.57}$$

Rotor copper loss is the actual dissipation:

$$P_{rc} = 3\left|\bar{I}'_r\right|^2 R'_r = sP_{ag}. \tag{3.58}$$

The remainder is developed mechanical power:

$$P_{\text{mech}} = P_{ag} - P_{rc} = (1-s)P_{ag}. \tag{3.59}$$

Equation (3.58) is the key result: rotor loss is proportional to slip, so high slip means low rotor efficiency, $\eta_r \le 1-s$. Shaft power subtracts friction, windage and stray loss, $P_{\text{shaft}} = P_{\text{mech}} - P_{\text{rot}}$. Electromagnetic torque is mechanical power divided by mechanical speed. Since $\omega_m = (1-s)\omega_s$,

$$T_e = \frac{P_{\text{mech}}}{\omega_m} = \frac{P_{ag}}{\omega_s} = \frac{3}{\omega_s}\left|\bar{I}'_r\right|^2 \frac{R'_r}{s}. \tag{3.60}$$

Air-gap power therefore serves as a measure of torque, sometimes quoted in "synchronous watts". Input power and efficiency are

$$P_{\text{in}} = 3V_s I_s \cos\varphi, \qquad \eta = \frac{P_{\text{shaft}}}{P_{\text{in}}}, \tag{3.61}$$

and $\cos\varphi$ is the input power factor, which falls at light load because the magnetising current then dominates.

3.8.3 Thevenin Equivalent and Torque-Slip Relationship

To express torque in terms of supply voltage, the stator side is replaced by its Thevenin equivalent at the air-gap node, as in Figure 3.63:

$$\bar{V}_{\text{th}} = \bar{V}_s \frac{jX_m}{R_s + j(X_{ls} + X_m)}, \qquad R_{\text{th}} + jX_{\text{th}} = \frac{jX_m(R_s + jX_{ls})}{R_s + j(X_{ls} + X_m)}. \tag{3.62}$$

Then $\left|\bar{I}_r'\right| = V_{\text{th}} / \sqrt{(R_{\text{th}} + R_r'/s)^2 + (X_{\text{th}} + X_{lr}')^2}$, and using (3.60),

$$T_e = \frac{3V_{\text{th}}^2 \, (R_r'/s)}{\omega_s \left[(R_{\text{th}} + R_r'/s)^2 + (X_{\text{th}} + X_{lr}')^2\right]}. \tag{3.63}$$

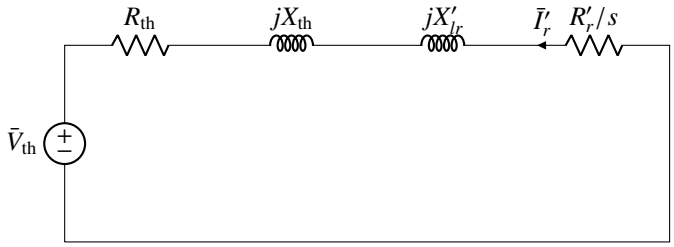


Figure 3.63: Thevenin equivalent seen from the rotor.

3.8.4 Starting, Maximum and Breakdown Torque

Starting torque follows by putting $s = 1$ in (3.63). Maximum torque occurs when power delivered to R_r'/s is maximum. By the maximum power transfer theorem, this requires R_r'/s to equal the magnitude of the remaining impedance:

$$s_{\max} = \frac{R_r'}{\sqrt{R_{\text{th}}^2 + (X_{\text{th}} + X_{lr}')^2}}. \tag{3.64}$$

Substitution gives the maximum, or *breakdown*, torque

$$T_{\max} = \frac{3V_{\text{th}}^2}{2\omega_s \left[R_{\text{th}} + \sqrt{R_{\text{th}}^2 + (X_{\text{th}} + X_{lr}')^2}\right]}. \tag{3.65}$$

Notably, $T_{\max}$ does not depend on R_r'; only the slip at which it occurs does. If R_{th} is neglected, then dividing (3.63) by (3.65) gives the Kloss approximation

$$\frac{T_e}{T_{\max}} \approx \frac{2}{s/s_{\max} + s_{\max}/s}. \tag{3.66}$$

For $s \ll s_{\max}$, this is the linear relation $T_e \approx 2T_{\max} s/s_{\max}$. Figure 3.64 shows the torque-slip curve with this linear approximation; Figure 3.59 shows the same data against speed. Operation to the left of the peak (low slip) is stable. To the right, the torque falls as slip rises. With $\omega_s \propto f_e$ and all reactances $\propto f_e$, (3.65) with $R_{\text{th}} \approx 0$ gives $T_{\max} \propto (V/f_e)^2$, which is why constant V/f preserves pull-out torque and field weakening reduces it.

Algorithm 5 gives the complete procedure.

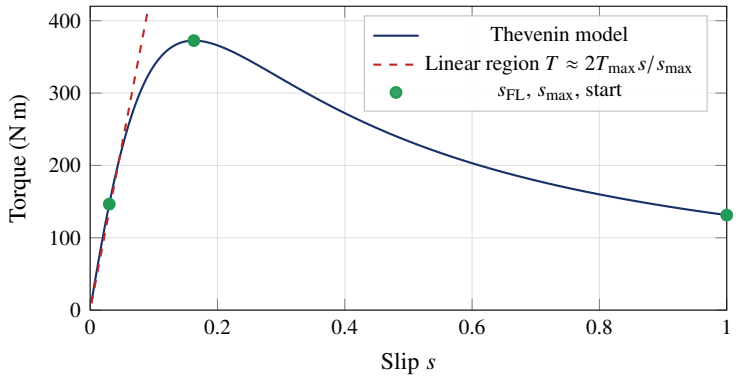


Figure 3.64: Torque-slip characteristic of the machine in Example 13.

Algorithm 5 Induction motor steady-state analysis

Input: V_s, f_e, p, R_s, R'_r, X_{ls}, X'_{lr}, X_m, P_{rot}, slip s
Output: I_s, PF, T_e, P_{shaft}, η, $T_{\max}$, $s_{\max}$

$\omega_s \leftarrow 2\pi f_e/p$
$Z_r \leftarrow R'_r/s + jX'_{lr}$; $Z_p \leftarrow jX_m Z_r/(jX_m + Z_r)$
$Z \leftarrow R_s + jX_{ls} + Z_p$; $\bar{I}_s \leftarrow \bar{V}_s/Z$
$\bar{E} \leftarrow \bar{I}_s Z_p$; $\bar{I}'_r \leftarrow \bar{E}/Z_r$
$P_{ag} \leftarrow 3\left|\bar{I}'_r\right|^2 R'_r/s$; $P_{rc} \leftarrow sP_{ag}$; $P_{\mathrm{mech}} \leftarrow (1-s)P_{ag}$
$T_e \leftarrow P_{ag}/\omega_s$; $P_{\mathrm{shaft}} \leftarrow P_{\mathrm{mech}} - P_{\mathrm{rot}}$
$P_{\mathrm{in}} \leftarrow 3\,\mathrm{Re}(\bar{V}_s \bar{I}_s^*)$; $\eta \leftarrow P_{\mathrm{shaft}}/P_{\mathrm{in}}$
Compute $V_{\mathrm{th}}, R_{\mathrm{th}}, X_{\mathrm{th}}$ from (3.62)
Compute $s_{\max}$ and $T_{\max}$ from (3.64) and (3.65)
if $s > s_{\max}$ **then**
    Flag operating point as beyond breakdown
end if

Example 13: Induction motor equivalent circuit analysis

Problem. A 400 V (line), 50 Hz, 4-pole, star-connected motor runs at slip 3%. Find the stator and rotor currents, power factor, power flow and efficiency.

Given (assumed). $R_s = 0.2\ \Omega$, $R'_r = 0.18\ \Omega$, $X_{ls} = 0.5\ \Omega$, $X'_{lr} = 0.6\ \Omega$, $X_m = 18\ \Omega$, $P_{\text{rot}} = 800$ W; core loss neglected.

Required. I_s, PF, I'_r, P_{ag}, P_{rc}, P_{mech}, η.

Governing equations. (3.57)–(3.59), (3.61).

Solution. $V_s = 400/\sqrt{3} = 230.9$ V. $Z_r = 6 + j0.6\ \Omega$.
$Z_p = j18(6 + j0.6)/(6 + j18.6) = 5.09 + j2.22\ \Omega$. $Z = 5.29 + j2.72\ \Omega$.
$\bar{I}_s = 230.9/Z = 38.8\angle -27.2^\circ$ A, so PF = 0.889 lagging.
$\left|\bar{E}\right| = 38.8 \times \left|Z_p\right| = 215.6$ V, $I'_r = 215.6/\left|Z_r\right| = 35.75$ A.
$P_{ag} = 3(35.75)^2(6) = 23.01$ kW, $P_{rc} = 0.03P_{ag} = 690$ W, $P_{\text{mech}} = 22.32$ kW.
Stator copper loss $= 3(38.8)^2(0.2) = 904$ W. $P_{\text{in}} = 23.91$ kW.
$P_{\text{shaft}} = 22.32 - 0.80 = 21.52$ kW, $\eta = 21.52/23.91 = 90.0\%$.

Result. $I_s = 38.8$ A, PF = 0.889, $P_{\text{shaft}} = 21.5$ kW, $\eta = 90.0\%$.

Physical interpretation. Stator and rotor copper loss are similar in magnitude; the rotor loss, however, is deposited in the rotor.

Engineering significance. The input check holds: $P_{\text{in}} = P_{ag}$+ stator loss $= 23.01 + 0.90 = 23.91$ kW.

Example 14: Induction motor electromagnetic torque calculation

Problem. For the motor of Example 13, find the electromagnetic and shaft torques at $s = 0.03$, and verify the result using the Thevenin model.

Given. Data of Example 13; $\omega_s = 157.08$ rad/s.

Required. T_e, T_{shaft}, Thevenin check.

Governing equations. (3.60), (3.62), (3.63).

Solution. $T_e = 23\,010/157.08 = 146.5$ N m.
$\omega_m = 0.97 \times 157.08 = 152.4$ rad/s, so $T_{\text{shaft}} = 21\,519/152.4 = 141.2$ N m.
Thevenin: $V_{\text{th}} = 224.7$ V, $R_{\text{th}} = 0.189\ \Omega$, $X_{\text{th}} = 0.489\ \Omega$.
Eq. (3.63): $T_e = 3(224.7)^2(6)/[157.08((6.19)^2 + (1.09)^2)] = 146.5$ N m.

Result. $T_e = 146.5$ N m, $T_{\text{shaft}} = 141.2$ N m.

Physical interpretation. The difference of 5.2 N m is the torque lost to friction and windage.

Engineering significance. The Thevenin model gives exactly the same result, and it is faster when a whole torque curve is required.

Example 15: Induction motor maximum torque calculation

Problem. For the same motor, find s_{max}, T_{max}, the speed at breakdown, the starting torque and the overload capability.

Given. Thevenin values of Example 14.

Required. s_{max}, T_{max}, n at T_{max}, T_{st}.

Governing equations. (3.64), (3.65), (3.63) with $s = 1$.

Solution. $\sqrt{0.189^2 + 1.0885^2} = 1.105\ \Omega$, so $s_{max} = 0.18/1.105 = 0.163$.
$T_{max} = 3(224.7)^2/[2(157.08)(0.189 + 1.105)] = 372.5$ N m.
Speed $= 1500(1 - 0.163) = 1256$ r/min.
$T_{st} = 3(224.7)^2(0.18)/[157.08(0.369^2 + 1.0885^2)] = 131.3$ N m.
Overload ratio $= 372.5/146.5 = 2.54$.

Result. $s_{max} = 0.163$, $T_{max} = 372.5$ N m at 1256 r/min, $T_{st} = 131.3$ N m.

Physical interpretation. Starting torque on a fixed-frequency supply is less than the running torque, as the rotor is highly inductive at $s = 1$.

Engineering significance. An inverter avoids this by starting at low frequency, so that slip stays small and full torque is available from standstill.

3.9 Direct and Indirect Vector Control

3.9.1 Introduction to Vector Control

A separately excited DC motor is easy to control because its flux and torque are set independently: field current gives flux, armature current gives torque, and the two are perpendicular by commutator action. In an induction motor, one set of stator currents produces both flux and torque, so a sudden change in voltage disturbs both. Vector control, based on the principle of field orientation due to Blaschke [21], restores the DC-machine behaviour by viewing the stator current in a frame that rotates with the rotor flux.

3.9.2 Scalar Control Versus Vector Control

The simplest IM drive is the open-loop inverter of Figure 3.65, in which a speed command sets frequency and a V/f profile sets voltage. Closed-loop scalar control, as in Figure 3.66, adds a slip regulator: the speed error gives a slip command, which is added to the measured speed

to give the stator frequency ω_e^*, and the voltage follows from $V = V_0 + K\omega_e$. The boost V_0 compensates stator resistance drop at low speed. Only magnitudes are controlled, so transients in flux and torque are slow and coupled. Table 3.13 compares the methods.



Figure 3.65: Open-loop induction motor drive.

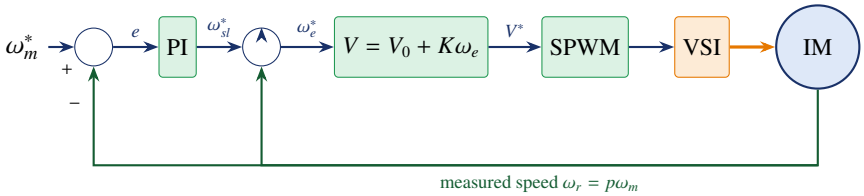


Figure 3.66: Scalar V/f control with slip regulation.

Table 3.13: Scalar control and vector control

Feature	Scalar V/f	Vector control
Controlled quantities	Magnitude and frequency	Magnitude and angle of current vector
Torque response	Slow, tens to hundreds of ms	Fast, a few ms
Flux-torque coupling	Coupled	Decoupled
Low-speed torque	Limited	Full torque at zero speed
Sensors	Voltage and optional speed	Currents and speed or flux
Computation	Light	Reference-frame transforms in real time
Typical use	Pumps, fans	Traction, servo drives

For traction, the fast torque response in the table is the decisive advantage, because driveability and regenerative blending require accurate torque.

3.9.3 Need for Decoupled Flux and Torque Control

Three fluxes appear in the machine, and any can be chosen for orientation.

- **Air-gap flux** $\vec{\lambda}_m = L_m(\vec{i}_s + \vec{i}'_r)$ links both stator and rotor.
- **Stator flux** $\vec{\lambda}_s = \vec{\lambda}_m + L_{ls}\vec{i}_s$ adds stator leakage.
- **Rotor flux** $\vec{\lambda}_r = \vec{\lambda}_m + L'_{lr}\vec{i}'_r$ adds rotor leakage.

Rotor flux orientation gives complete decoupling and is the choice in this unit. Stator and air-gap orientation retain a coupling term which must be compensated [16, 17]. Torque is produced by the component of stator current perpendicular to rotor flux. Vector control therefore keeps the flux component constant and varies the perpendicular component to change torque immediately.

3.9.4 Reference Frames and Space Vectors

3.9.4.1 Space Vector Representation

For a three-phase set x_a, x_b, x_c with $x_a + x_b + x_c = 0$, the space vector is

$$\vec{x} = \frac{2}{3}\left(x_a + \underline{a}\, x_b + \underline{a}^2 x_c\right), \qquad \underline{a} = e^{j2\pi/3}. \tag{3.67}$$

The factor $2/3$ is the amplitude-invariant scaling of Table 3.1. For balanced currents of peak I_m, $\vec{i} = I_m e^{j\omega_e t}$. This compact form is a mathematical picture of the rotating MMF of (3.51).

3.9.4.2 Stationary and Rotating Frames

Figure 3.67 shows a stationary $\alpha\beta$ frame, where α is fixed to phase a, and a dq frame rotating at speed ω. When a frame turns at the synchronous speed ω_e, called the *synchronous reference frame*, balanced sinusoidal quantities appear as constants, so ordinary PI controllers can regulate them to zero error.

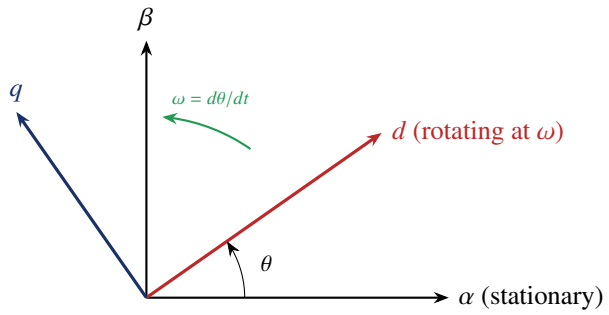


Figure 3.67: Stationary and rotating reference frames.

3.9.4.3 Clarke Transformation

Taking the real and imaginary parts of (3.67) gives the Clarke transformation [22], illustrated in Figure 3.68:

$$\begin{bmatrix} x_\alpha \\ x_\beta \\ x_0 \end{bmatrix} = \frac{2}{3}\begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}\begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix}. \tag{3.68}$$

With zero sum, $x_\alpha = x_a$ and $x_\beta = (x_b - x_c)/\sqrt{3}$; thus only two current sensors are needed. The inverse is

$$x_a = x_\alpha + x_0, \quad x_b = -\tfrac{1}{2}x_\alpha + \tfrac{\sqrt{3}}{2}x_\beta + x_0, \quad x_c = -\tfrac{1}{2}x_\alpha - \tfrac{\sqrt{3}}{2}x_\beta + x_0. \tag{3.69}$$

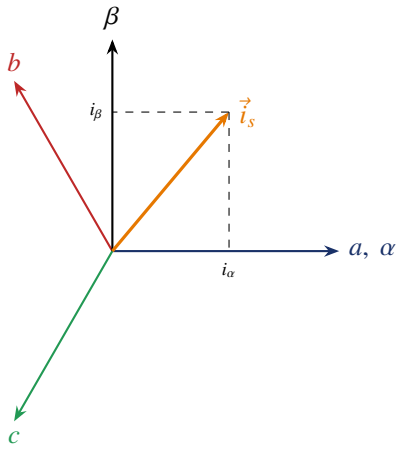


Figure 3.68: Clarke transformation: projection of the space vector on the $\alpha\beta$ axes.

3.9.4.4 Park Transformation

The Park transformation [23] rotates this vector by $-\theta$: $x_d + jx_q = (x_\alpha + jx_\beta)e^{-j\theta}$. In component form, as in Figure 3.69,

$$\begin{bmatrix} x_d \\ x_q \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x_\alpha \\ x_\beta \end{bmatrix}. \tag{3.70}$$

The inverse Park transformation is

$$\begin{bmatrix} x_\alpha \\ x_\beta \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x_d \\ x_q \end{bmatrix}. \tag{3.71}$$

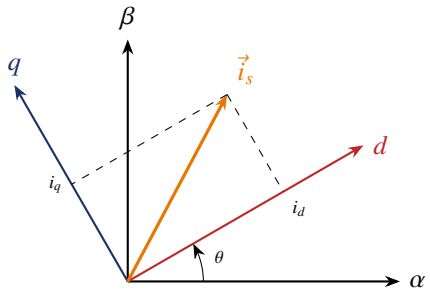


Figure 3.69: Park transformation: the same vector resolved on rotating dq axes.

3.9.4.5 Power Under the Chosen Scaling

Because amplitude-invariant scaling shrinks vectors relative to a power-invariant transform, three-phase instantaneous power is

$$p = v_a i_a + v_b i_b + v_c i_c = \tfrac{3}{2}\left(v_d i_d + v_q i_q\right). \tag{3.72}$$

With power-invariant scaling (factor $\sqrt{2/3}$ in place of $2/3$), the $3/2$ disappears from power and torque, and magnitudes are $\sqrt{3/2}$ larger. The two should not be mixed. Note that some texts,

such as Krause [19], align q with phase a at $\theta = 0$; the physics is the same, but the signs of individual terms shift.

Algorithms 6 and 7 list the two transformations as they appear in firmware.

Algorithm 6 Clarke transformation

Input: Sampled phase currents i_a, i_b (balanced: $i_c = -i_a - i_b$)
Output: i_α, i_β
1: Remove ADC offsets from i_a, i_b
2: $i_\alpha \leftarrow i_a$
3: $i_\beta \leftarrow (i_a + 2i_b)/\sqrt{3}$
4: **if** $|i_a + i_b + i_c| > \epsilon$ with a third sensor **then**
5: Flag a current-sensor fault
6: **end if**

Algorithm 7 Park transformation

Input: i_α, i_β, angle θ
Output: i_d, i_q
1: Wrap θ to $[0, 2\pi)$; compute $c \leftarrow \cos\theta$, $s \leftarrow \sin\theta$
2: $i_d \leftarrow c i_\alpha + s i_\beta$
3: $i_q \leftarrow -s i_\alpha + c i_\beta$

Example 16: Clarke and Park transformation numerical example

Problem. The phase currents are $i_a = 80$ A, $i_b = -20$ A, $i_c = -60$ A at $\theta = 30°$. Find $i_\alpha, i_\beta, i_d, i_q$ and verify by inverse transformation.

Given. As above, with $i_a + i_b + i_c = 0$.

Required. Stationary and synchronous components.

Governing equations. (3.68), (3.70), (3.71), (3.69).

Solution. $i_\alpha = 80$ A. $i_\beta = (-20 + 60)/\sqrt{3} = 23.09$ A.
$i_d = 80\cos 30° + 23.09 \sin 30° = 80.83$ A.
$i_q = -80 \sin 30° + 23.09 \cos 30° = -20.0$ A.
Check: $i_\alpha = 80.83(0.866) + 20(0.5) = 80$ A; $i_\beta = 80.83(0.5) - 20(0.866) = 23.09$ A; $i_b = -40 + 20 = -20$ A.

Result. $i_\alpha = 80$ A, $i_\beta = 23.09$ A, $i_d = 80.83$ A, $i_q = -20.0$ A; vector length 83.3 A.

Physical interpretation. The vector lies at 16.1° in the $\alpha\beta$ frame, which is 13.9° behind the d axis at 30°, so i_q is negative.

Engineering significance. In a drive, negative i_q at positive speed would indicate regenerative braking torque.

3.9.4.6 d-Axis and q-Axis Interpretation

Figure 3.70 shows the outcome: with d along the rotor flux, the stator current has a flux-producing part i_{ds} and a torque-producing part i_{qs}. This is the direct analogue of field and armature current in a DC machine.

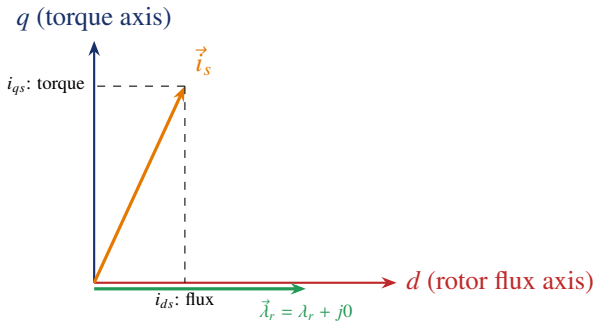


Figure 3.70: d-q axis representation of stator current under rotor flux orientation.

3.9.5 Rotor Flux Orientation

Figure 3.71 shows the d axis locked to the rotor flux vector, whose angle is θ_e. The stator current vector sits at an angle δ from it. Hence $i_{ds} = \left|\vec{i}_s\right| \cos\delta$ and $i_{qs} = \left|\vec{i}_s\right| \sin\delta$. The essential relations, derived in Section 3.10, are

$$\tau_r \frac{d\lambda_r}{dt} + \lambda_r = L_m i_{ds}, \qquad \tau_r = \frac{L_r}{R_r}, \tag{3.73}$$

$$T_e = \frac{3}{2} p \frac{L_m}{L_r} \lambda_r i_{qs}, \tag{3.74}$$

$$\omega_{sl} = \frac{R_r}{L_r} \frac{L_m}{\lambda_r} i_{qs}. \tag{3.75}$$

Parameters here are stator-referred. Figure 3.72 presents the result as a block diagram: flux responds slowly to i_{ds} through the rotor time constant, whereas torque responds to i_{qs} as fast as the current loop. Keeping λ_r constant therefore yields a linear torque actuator.

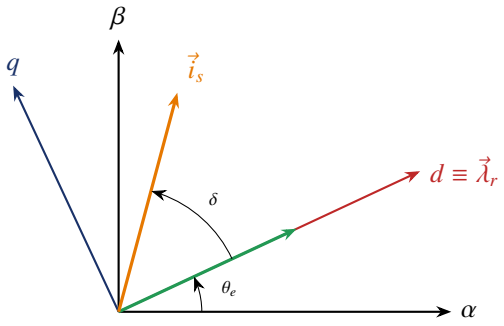


Figure 3.71: Rotor flux orientation.

3.9.6 Direct Vector Control

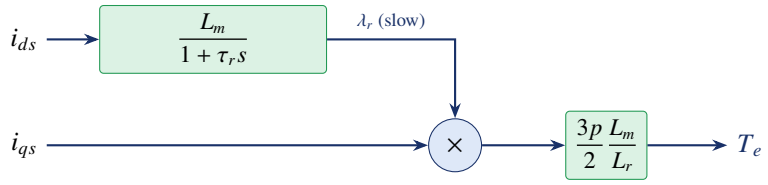


Figure 3.72: Flux and torque decoupling under rotor flux orientation.

3.9.6.1 Definition and Flux Estimation

In direct vector control (DVC), the flux angle θ_e is obtained directly from a measurement or estimate of flux. Hall sensors or search coils in the gap are possible but fragile, so in practice an estimator is employed, as in Figure 3.73. The voltage model integrates the stator voltage equation in the $\alpha\beta$ frame, then corrects for leakage:

$$\lambda_{\alpha s} = \int (v_{\alpha s} - R_s i_{\alpha s})\, dt, \qquad \lambda_{\beta s} = \int (v_{\beta s} - R_s i_{\beta s})\, dt, \tag{3.76}$$

$$\lambda_{\alpha r} = \frac{L_r}{L_m}(\lambda_{\alpha s} - \sigma L_s i_{\alpha s}), \qquad \lambda_{\beta r} = \frac{L_r}{L_m}(\lambda_{\beta s} - \sigma L_s i_{\beta s}), \tag{3.77}$$

$$\theta_e = \operatorname{atan2}(\lambda_{\beta r}, \lambda_{\alpha r}), \qquad \hat{\lambda}_r = \sqrt{\lambda_{\alpha r}^2 + \lambda_{\beta r}^2}. \tag{3.78}$$

$\sigma = 1 - L_m^2/(L_s L_r)$ is the leakage coefficient, developed in Section 3.11. The pure integration drifts and fails at low speed, where the back EMF is small and R_s errors dominate, so real estimators replace it with a low-pass filter or blend with a current model.

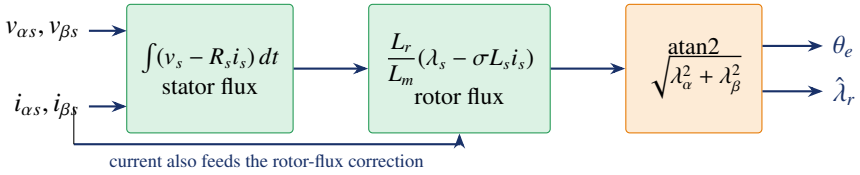


Figure 3.73: Rotor flux estimation from terminal quantities.

3.9.6.2 Direct Vector Control Architecture

Figure 3.74 shows the complete scheme. Inputs are the speed and flux commands ω_m^* and λ_r^*; outputs are the inverter gate signals. The speed PI generates i_{qs}^* (scaled from T_e^*), and the flux PI generates i_{ds}^*. Measured currents pass through Clarke and Park transforms, using θ_e from the estimator, to yield i_{ds} and i_{qs}. The current PIs produce v_d^* and v_q^*, and inverse Park with θ_e brings them to the $\alpha\beta$ frame for the SVPWM stage. The flux feedback $\hat{\lambda}_r$ closes the outer flux loop.

3.9.6.3 Advantages, Limitations and Sensitivity

Algorithm 8 shows that no slip calculation is needed, and rotor resistance is not required in the voltage model, so DVC is immune to rotor temperature rise. Its weakness is sensitivity to R_s and integrator error at low speed, which limits accuracy near standstill and thus at vehicle launch. σL_s error also affects the angle.

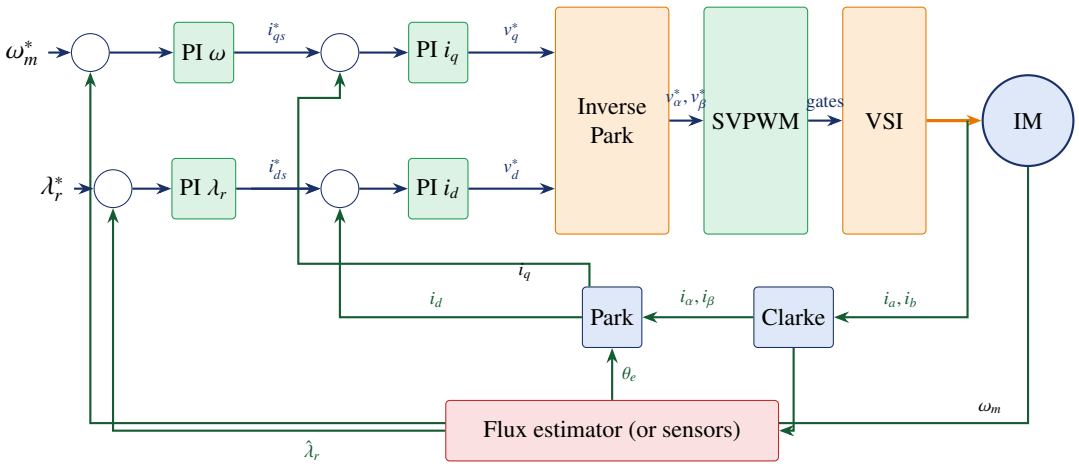


Figure 3.74: Direct vector control architecture.

Algorithm 8 Direct vector control of an induction motor

Input: ω_m^*, λ_r^*, measured i_a, i_b, V_{dc}, ω_m; parameters R_s, L_s, L_r, L_m
Output: Gate signals $S_1 \ldots S_6$

Initialise integrators and flux estimate
for each control period T_c **do**
  $(i_{\alpha s}, i_{\beta s}) \leftarrow$ Clarke(i_a, i_b); reconstruct $v_{\alpha s}, v_{\beta s}$ from duties and V_{dc}
  Update stator flux, then rotor flux, with (3.76), (3.77)
  $\theta_e, \hat{\lambda}_r \leftarrow$ (3.78)
  $(i_{ds}, i_{qs}) \leftarrow$ Park$(i_{\alpha s}, i_{\beta s}, \theta_e)$
  $T_e^* \leftarrow \mathrm{PI}_\omega(\omega_m^* - \omega_m)$; $i_{qs}^* \leftarrow 2L_r T_e^* / (3pL_m \hat{\lambda}_r)$
  $i_{ds}^* \leftarrow \mathrm{PI}_\lambda(\lambda_r^* - \hat{\lambda}_r)$
  Limit: $\sqrt{i_{ds}^{*2} + i_{qs}^{*2}} \le I_{\max}$
  $v_{ds}^*, v_{qs}^* \leftarrow$ current PIs with decoupling
  $v_\alpha^*, v_\beta^* \leftarrow$ inverse Park; compute SVPWM duties
end for

3.9.7 Indirect Vector Control

3.9.7.1 Definition

Indirect vector control (IVC) does not estimate flux. It instead forces orientation by computing the slip that must exist for the rotor flux to lie on the d axis, then adding it to measured rotor speed. It is also called *feed-forward* or *slip-frequency* control.

3.9.7.2 Governing Relationships

Under rotor flux orientation in steady flux, (3.73) gives the d-axis reference, and (3.74) gives the q-axis reference:

$$i_{ds}^{*} = \frac{\lambda_r^{*}}{L_m}, \tag{3.79}$$

$$i_{qs}^{*} = \frac{2}{3p}\frac{L_r}{L_m}\frac{T_e^{*}}{\lambda_r^{*}}. \tag{3.80}$$

The slip frequency follows from (3.75). In steady flux, it can be written using the current references alone:

$$\omega_{sl}^{*} = \frac{R_r}{L_r}\frac{i_{qs}^{*}}{i_{ds}^{*}} = \frac{1}{\tau_r}\frac{i_{qs}^{*}}{i_{ds}^{*}}. \tag{3.81}$$

Then the synchronous frequency and rotor flux angle are

$$\omega_e = \omega_r + \omega_{sl}^{*} = p\omega_m + \omega_{sl}^{*}, \qquad \theta_e = \int \omega_e \, dt. \tag{3.82}$$

3.9.7.3 Indirect Vector Control Architecture

In Figure 3.75, the upper part is the same as in DVC, but the flux reference is applied through (3.79) without a flux loop. The lower part computes slip from the current references, adds $p\omega_m$ and integrates to obtain θ_e. Only the speed sensor is needed, and it works down to and through zero speed.

Algorithm 9 Indirect vector control of an induction motor

Input: ω_m^{*}, λ_r^{*}, measured i_a, i_b, ω_m; parameters L_m, L_r, R_r, p
Output: Gate signals $S_1 \ldots S_6$

1: Initialise $\theta_e \leftarrow 0$; magnetise with $i_{ds}^{*} = \lambda_r^{*}/L_m$ for about $3\tau_r$
2: **for** each control period T_c **do**
3: $\quad T_e^{*} \leftarrow \mathrm{PI}_{\omega}(\omega_m^{*} - \omega_m)$
4: $\quad i_{ds}^{*} \leftarrow \lambda_r^{*}/L_m$; $i_{qs}^{*} \leftarrow 2L_r T_e^{*}/(3pL_m\lambda_r^{*})$
5: $\quad \omega_{sl}^{*} \leftarrow (R_r/L_r)\, i_{qs}^{*}/i_{ds}^{*}$
6: $\quad \omega_e \leftarrow p\omega_m + \omega_{sl}^{*}$; $\theta_e \leftarrow \theta_e + \omega_e T_c$
7: $\quad (i_{ds}, i_{qs}) \leftarrow \mathrm{Park}(\mathrm{Clarke}(i_a, i_b), \theta_e)$
8: $\quad v_{ds}^{*}, v_{qs}^{*} \leftarrow$ current PIs with decoupling
9: $\quad$ Apply inverse Park and SVPWM
10: **end for**

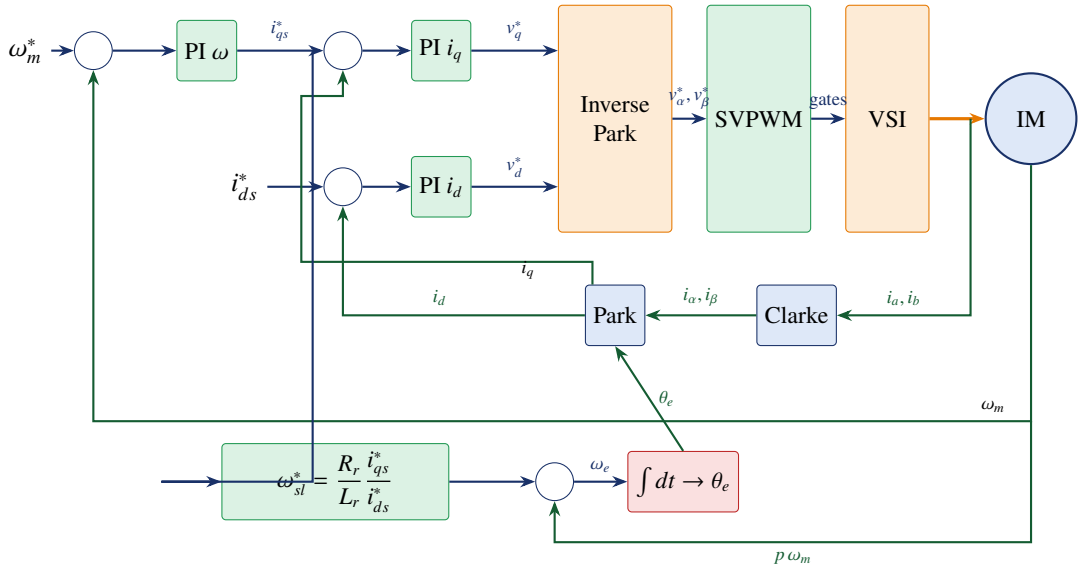


Figure 3.75: Indirect vector control architecture.

3.9.7.4 Advantages, Limitations and Rotor Time Constant Dependence

Algorithm 9 is simple and has good low-speed behaviour, which explains its popularity for traction. Its weakness lies in (3.81), which depends on $\tau_r = L_r/R_r$. As rotor bars warm by 100 K, R_r can rise by about 40% for aluminium or copper, whereas saturation alters L_r. An error in τ_r gives a wrong slip, the frame then misaligns with the actual flux, and both flux and torque deviate from their commands. On-line estimation or a thermal model is used to update R_r. Table 3.14 compares the two.

Table 3.14: Direct and indirect vector control

Feature	Direct	Indirect
Source of θ_e	Flux estimate or sensor	Slip + measured speed
Flux loop	Yes	Usually open loop
Sensitive parameter	R_s, σL_s	τ_r (R_r, L_r)
Zero speed	Difficult with voltage model	Works well
High speed	Very good	Good if τ_r tracked
Sensor	Optional (sensorless variants)	Speed or position needed

From the table, each method is strong where the other is weak. Commercial drives often combine them, using the current model at low speed and the voltage model at high speed.

Example 17: Indirect vector control slip frequency calculation

Problem. A traction IM with rotor flux 0.2 Wb has $i_{ds} = 50$ A and $i_{qs} = 262.5$ A, at $\omega_m = 300$ rad/s. Find the slip frequency and flux-frame speed, and the effect of a 30% rise in R_r if the controller is not updated.

Given (assumed). $p = 2$, $L_m = 4$ mH, $L_r = 4.2$ mH, $R_r = 0.012\ \Omega$.

Required. τ_r, ω_{sl}, ω_e, slip, frequency error.

Governing equations. (3.81), (3.82).

Solution. $\tau_r = 0.0042/0.012 = 0.35$ s.
$\omega_{sl}^* = (1/0.35)(262.5/50) = 15.0$ rad/s ($f_{sl} = 2.39$ Hz).
$\omega_r = 2 \times 300 = 600$ rad/s, $\omega_e = 615$ rad/s (97.9 Hz).
$s = 15/615 = 0.0244$.
With R_r up 30%, the correct slip is 19.5 rad/s, whereas the controller still applies 15 rad/s.

Result. $\omega_{sl}^* = 15$ rad/s, $\omega_e = 615$ rad/s, $s = 2.44\%$, and an error of 4.5 rad/s in slip when R_r is not tracked.

Physical interpretation. Rotor flux adjusts to the wrong slip, and the actual flux angle no longer matches the frame.

Engineering significance. The electrical frequency error is small relative to ω_e but large relative to slip, so torque accuracy suffers, particularly at high torque.

3.10 Field Oriented Control of Induction Motor

3.10.1 Introduction and Principle of FOC

Field oriented control (FOC) is the practical realisation of vector control. It measures stator currents, transforms them to the rotor flux frame, and regulates two DC-like quantities. Three layers make up the controller: a slow speed loop, a flux reference, and fast current loops. Figure 3.76 shows them together with the indirect orientation; Figure 3.77 shows the same as a flowchart executed once per PWM period.

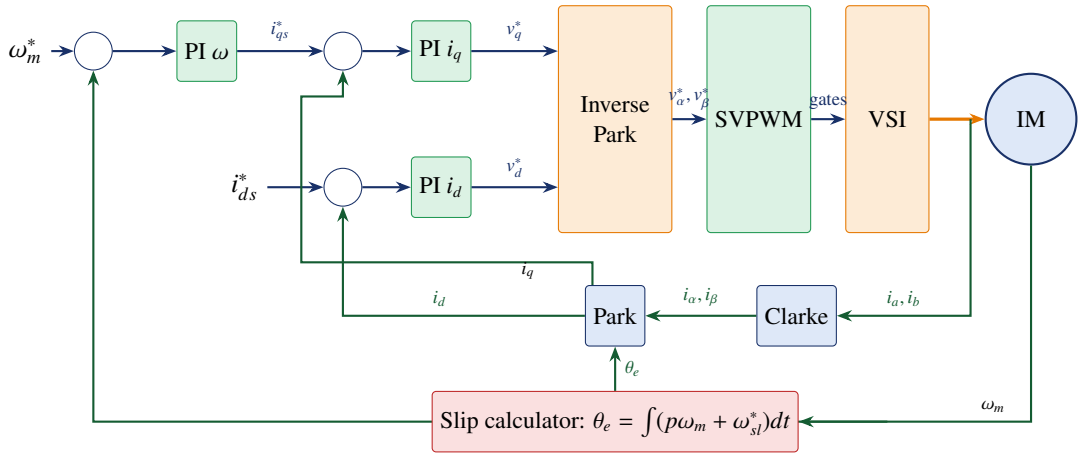


Figure 3.76: Complete induction motor FOC.

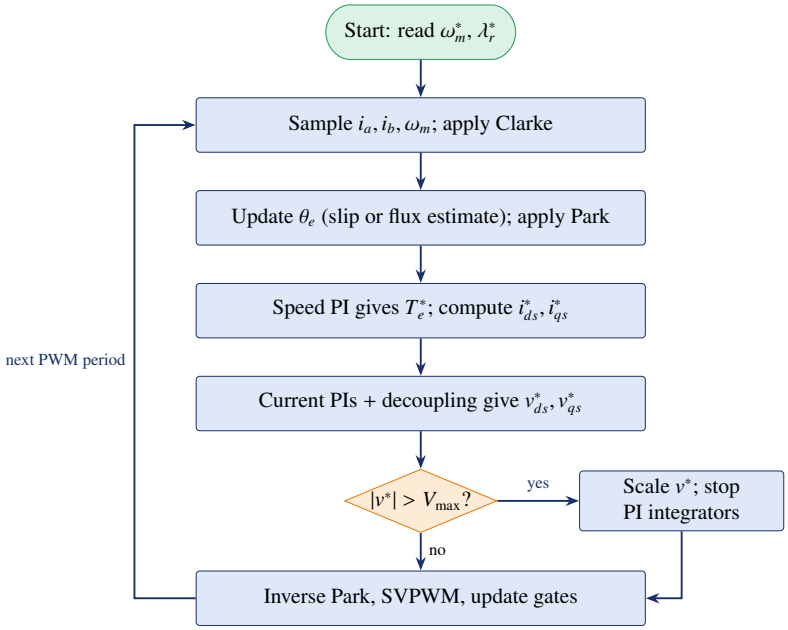


Figure 3.77: Flowchart of induction motor FOC.

3.10.2 d-q Machine Equations in the Synchronous Frame

In a frame rotating at ω_e with the conventions of Table 3.1, and with stator-referred rotor quantities and a short-circuited rotor, the machine equations, developed in Section 3.11, are

$$v_{ds} = R_s i_{ds} + \frac{d\lambda_{ds}}{dt} - \omega_e \lambda_{qs}, \tag{3.83}$$

$$v_{qs} = R_s i_{qs} + \frac{d\lambda_{qs}}{dt} + \omega_e \lambda_{ds}, \tag{3.84}$$

$$0 = R_r i_{dr} + \frac{d\lambda_{dr}}{dt} - \omega_{sl} \lambda_{qr}, \tag{3.85}$$

$$0 = R_r i_{qr} + \frac{d\lambda_{qr}}{dt} + \omega_{sl} \lambda_{dr}, \tag{3.86}$$

where $\omega_{sl} = \omega_e - \omega_r$ and the flux linkages are

$$\lambda_{ds} = L_s i_{ds} + L_m i_{dr}, \qquad \lambda_{qs} = L_s i_{qs} + L_m i_{qr}, \tag{3.87}$$

$$\lambda_{dr} = L_r i_{dr} + L_m i_{ds}, \qquad \lambda_{qr} = L_r i_{qr} + L_m i_{qs}. \tag{3.88}$$

The torque, in terms of rotor flux and stator current, is

$$T_e = \frac{3}{2} p \frac{L_m}{L_r} \left(\lambda_{dr} i_{qs} - \lambda_{qr} i_{ds} \right). \tag{3.89}$$

3.10.3 Rotor Flux Equations Under Orientation

Rotor flux orientation sets $\lambda_{qr} = 0$ and $\lambda_{dr} = \lambda_r$, with $d\lambda_{qr}/dt = 0$. From (3.88),

$$i_{qr} = -\frac{L_m}{L_r} i_{qs}, \qquad i_{dr} = \frac{\lambda_r - L_m i_{ds}}{L_r}. \tag{3.90}$$

Substituting i_{dr} into (3.85) gives the flux dynamics $\tau_r \, d\lambda_r/dt + \lambda_r = L_m i_{ds}$, which is (3.73). So λ_r follows i_{ds} through a first-order lag, and $i_{dr} = 0$ in steady state: the rotor carries no d-axis current, so i_{ds} is the *magnetizing current*, $i_{mr} = \lambda_r / L_m$. Substituting i_{qr} into (3.86) gives the slip relation (3.75). Finally, setting $\lambda_{qr} = 0$ in (3.89) gives the torque expression (3.74).

3.10.4 Stator Current Decomposition

The stator current vector is thus decomposed into the *flux-producing* current i_{ds} and the *torque-producing* current i_{qs}, subject to the inverter limit

$$i_{ds}^2 + i_{qs}^2 \le I_{\max}^2. \tag{3.91}$$

Because i_{qr} exactly opposes the q-axis MMF of i_{qs} in (3.90), the q-axis current produces no net flux change. This is the machine's equivalent of DC armature reaction compensation, and it is the root of the decoupling.

3.10.5 Decoupled Voltage Equations

Eliminate rotor current from (3.87) using (3.90):

$$\lambda_{ds} = \sigma L_s i_{ds} + \frac{L_m}{L_r} \lambda_r, \qquad \lambda_{qs} = \sigma L_s i_{qs}. \tag{3.92}$$

Substituting in (3.83) and (3.84) gives

$$v_{ds} = R_s i_{ds} + \sigma L_s \frac{di_{ds}}{dt} + \frac{L_m}{L_r} \frac{d\lambda_r}{dt} - \omega_e \sigma L_s i_{qs}, \tag{3.93}$$

$$v_{qs} = R_s i_{qs} + \sigma L_s \frac{di_{qs}}{dt} + \omega_e \sigma L_s i_{ds} + \omega_e \frac{L_m}{L_r} \lambda_r. \tag{3.94}$$

The speed-dependent terms cross-couple the axes. The controller adds them as feed-forward, so the PI regulators see only the simple R-L plant, as shown in Figure 3.78:

$$v_{ds}^{*} = v_d' - \omega_e \sigma L_s i_{qs}, \qquad v_{qs}^{*} = v_q' + \omega_e \left(\sigma L_s i_{ds} + \frac{L_m}{L_r} \lambda_r \right). \tag{3.95}$$

At steady flux, the second bracket equals $L_s i_{ds}$. Allowing for rotor current transients, the effective resistance seen by the current loops is $R_{\text{eq}} = R_s + (L_m/L_r)^2 R_r$.

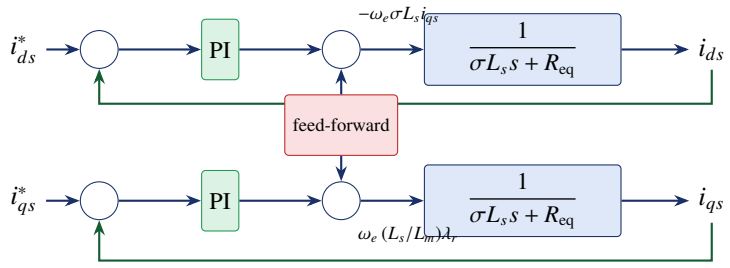


Figure 3.78: d-axis and q-axis current controllers with feed-forward decoupling.

3.10.6 PI Current Controllers

Each current loop has the form

$$v'_d = K_{pi}(i^*_{ds} - i_{ds}) + K_{ii} \int (i^*_{ds} - i_{ds})\, dt, \tag{3.96}$$

and similarly for v'_q. Placing the PI zero on the plant pole (pole-zero cancellation) with target bandwidth ω_b gives the gains $K_{pi} = \omega_b \sigma L_s$ and $K_{ii} = \omega_b R_{\text{eq}}$, and the loop becomes a first-order system of time constant $1/\omega_b$. A typical bandwidth is about a tenth of the sampling frequency, in rad/s. When the inverter saturates, integrator action must be frozen (anti-windup).

3.10.7 Speed Controller and Current Reference Generation

Figure 3.79 shows the speed loop. Its output is torque,

$$T^*_e = K_{p\omega}(\omega^*_m - \omega_m) + K_{i\omega} \int (\omega^*_m - \omega_m)\, dt. \tag{3.97}$$

In an EV, the outer speed loop is frequently absent, for the driver commands torque directly; cruise control and creep modes then form the speed loop. Torque becomes current through (3.80). The flux loop in Figure 3.80 regulates λ_r through i^*_{ds}. Above base speed, the flux reference falls with speed,

$$\lambda^*_r = \begin{cases} \lambda_{r,\text{rated}}, & |\omega_e| \le \omega_b, \\ \lambda_{r,\text{rated}}\, \omega_b / |\omega_e|, & |\omega_e| > \omega_b, \end{cases} \tag{3.98}$$

where ω_b here denotes base electrical speed.

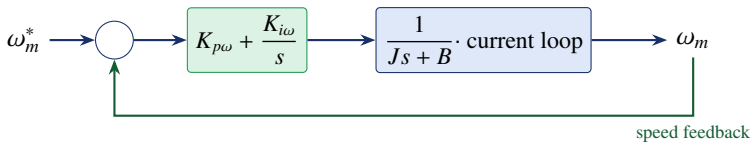


Figure 3.79: Speed control loop.

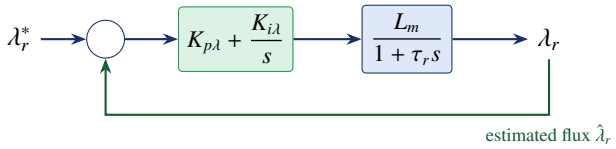


Figure 3.80: Flux control loop.

3.10.8 Slip and Flux Angle Calculation

Figure 3.81 shows how the angle is produced. Slip comes from (3.75), using the torque current and the flux estimate. Then ω_e and θ_e follow from (3.82). In discrete time, $\theta_e[k+1] = \theta_e[k] + \omega_e[k]T_c$, wrapped to 2π.

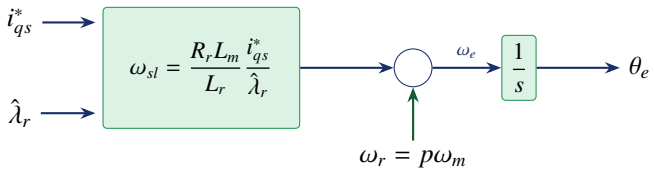


Figure 3.81: Slip frequency calculation and flux angle generation.

3.10.9 PWM Inverter Interface, Current Measurement and Speed Feedback

Figure 3.82 shows the signal path around the inverter. Voltage references are converted to PWM duties, gate drivers switch the VSI, and the motor current is sampled. Sampling is synchronised to the PWM centre, where the ripple crosses its mean, so that averaging is automatic. Speed comes from an encoder or resolver. Delay in the loop, which is about one and a half sampling periods, restricts the current bandwidth.

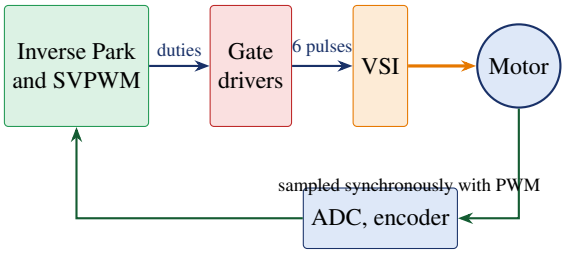


Figure 3.82: PWM inverter and motor interface.

Algorithms 10 to 12 are the routines in order of execution. The outer routine calls the current reference routine, which limits the torque component only, because the flux must be protected first.

Example 18: Induction motor FOC current reference calculation

Problem. The machine of Example 17, with $L_s = 4.2$ mH and $R_s = 0.015\ \Omega$, must produce

Algorithm 10 Complete induction motor FOC

Input: ω_m^* or T_e^*, $i_a, i_b, \omega_m, V_{\mathrm{dc}}$
Output: SVPWM duties d_a, d_b, d_c

Update flux reference by (3.98)
Update $\hat{\lambda}_r$: $\hat{\lambda}_r \leftarrow \hat{\lambda}_r + (T_c/\tau_r)(L_m i_{ds} - \hat{\lambda}_r)$
$T_e^* \leftarrow$ (3.97) if in speed mode
$i_{ds}^*, i_{qs}^* \leftarrow \textsc{CurrentReference}(T_e^*, \lambda_r^*, \hat{\lambda}_r)$
$\omega_{sl} \leftarrow R_r L_m i_{qs}^* / (L_r \hat{\lambda}_r)$; $\theta_e \leftarrow \theta_e + (p\omega_m + \omega_{sl})T_c$
$(i_{ds}, i_{qs}) \leftarrow \mathrm{Park}(\mathrm{Clarke}(i_a, i_b), \theta_e)$
$v_d', v_q' \leftarrow$ PI on current errors
$v_{ds}^*, v_{qs}^* \leftarrow$ (3.95)
if $\sqrt{v_{ds}^{*2} + v_{qs}^{*2}} > V_{\mathrm{dc}}/\sqrt{3}$ **then**
    Scale voltage vector; hold integrators
end if
$v_\alpha^*, v_\beta^* \leftarrow$ inverse Park; compute d_a, d_b, d_c

Algorithm 11 Speed and torque control of the IM drive

Input: Mode (speed or torque), ω_m^*, T_{driver}^*, ω_m, limits $T_{\max}(\omega_m)$
Output: T_e^*

if mode = speed **then**
    $e \leftarrow \omega_m^* - \omega_m$; $u \leftarrow K_{p\omega} e + I$
    $T_e^* \leftarrow \mathrm{sat}(u, \pm T_{\max})$
    if $u = T_e^*$ **then**
        $I \leftarrow I + K_{i\omega} e T_s$
    end if ▷ Anti-windup
else
    $T_e^* \leftarrow \mathrm{sat}(T_{\mathrm{driver}}^*, \pm T_{\max})$
end if

Algorithm 12 Current reference generation for the IM drive

Input: $T_e^*, \lambda_r^*, \hat{\lambda}_r, I_{\max}$
Output: i_{ds}^*, i_{qs}^*

$i_{ds}^* \leftarrow \lambda_r^*/L_m$ (or flux-PI output)
$i_{qs}^* \leftarrow 2L_r T_e^*/(3pL_m\hat{\lambda}_r)$
$i_{q,\max} \leftarrow \sqrt{I_{\max}^2 - i_{ds}^{*2}}$
$i_{qs}^* \leftarrow \mathrm{sat}(i_{qs}^*, \pm i_{q,\max})$

150 N m at $\lambda_r^* = 0.2$ Wb and $\omega_m = 300$ rad/s. Find the current references, peak and rms current, and the stator voltage in steady state.

Given. $T_e^* = 150$ N m, $p = 2$, $L_m = 4$ mH, $L_r = L_s = 4.2$ mH, $\sigma = 0.093$.

Required. i_{ds}^*, i_{qs}^*, I_s, v_{ds}, v_{qs}.

Governing equations. (3.79), (3.80), (3.93), (3.94).

Solution. $i_{ds}^* = 0.2/0.004 = 50$ A.
$i_{qs}^* = 2(0.0042)(150)/[3(2)(0.004)(0.2)] = 262.5$ A.
$\left|\vec{i}_s\right| = \sqrt{50^2 + 262.5^2} = 267.2$ A peak, $= 189$ A rms.
From Example 17, $\omega_e = 615$ rad/s. $\sigma L_s = 0.390$ mH.
$v_{ds} = 0.015(50) - 615(0.390 \times 10^{-3})(262.5) = -62.3$ V.
$v_{qs} = 0.015(262.5) + 615(0.0042)(50) = 133.1$ V, and $\left|\vec{v}_s\right| = 147$ V.

Result. $i_{ds}^* = 50$ A, $i_{qs}^* = 262.5$ A, $v_{ds} = -62.3$ V, $v_{qs} = 133.1$ V.

Physical interpretation. The current vector is close to the q axis at high torque. The voltage leads the current by a modest angle, which gives good power factor at this high load.

Engineering significance. With a 400 V link, the limit is $400/\sqrt{3} = 231$ V, so voltage margin remains at this speed. As speed rises, the margin vanishes and (3.98) takes over.

3.10.10 FOC Operation During Motoring and Regenerative Braking

In motoring, $T_e^* > 0$, so $i_{qs}^* > 0$ and $\omega_{sl} > 0$: the flux leads the rotor. In braking at forward speed, $T_e^* < 0$, so $i_{qs}^* < 0$ and $\omega_{sl} < 0$. The frame then runs *slower* than the rotor, slip is negative and power reverses, with no change in control structure. This seamless transition is one of the main benefits of FOC.

3.10.11 Dynamic Response

Figures 3.83 to 3.86 show representative responses, obtained from a simplified simulation with the torque loop modelled as a 2 ms lag, $J = 0.25$ kg m^2 and a torque limit of 200 N m. A speed step at $t = 0$ makes the speed PI saturate at maximum torque, and speed rises as a ramp. At $t = 0.8$ s, a load step causes a small dip, which the integrator removes. At $t = 1.4$ s, a reduced speed command gives negative torque, and the drive regenerates. Torque follows i_{qs} almost instantly, because flux is held constant by i_{ds}. In contrast, flux build-up after excitation takes several τ_r (Figure 3.86). This is why the machine is pre-fluxed before launch, and why i_{qs}, not i_{ds}, is the fast control channel.

3.10.12 Parameter Sensitivity

Detuning is the main practical issue. Thermal rise of R_r alters slip gain; saturation changes L_m and L_r with flux level, particularly in field weakening; and skin effect changes rotor parameters at high slip frequency. Under detuning, the actual torque constant differs from the commanded

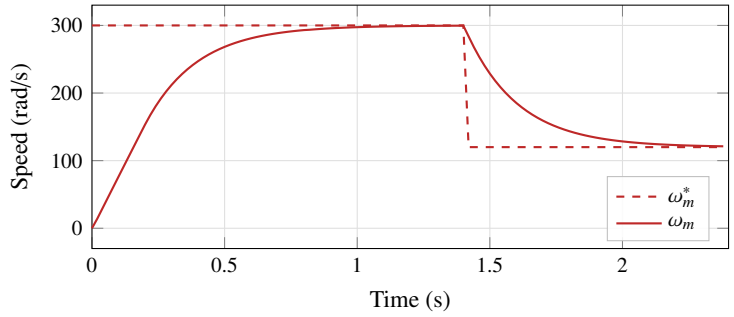


Figure 3.83: Speed response of FOC drive (representative simulation).

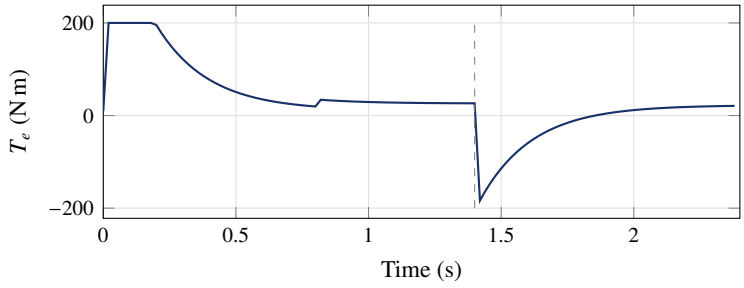


Figure 3.84: Torque response of FOC drive.

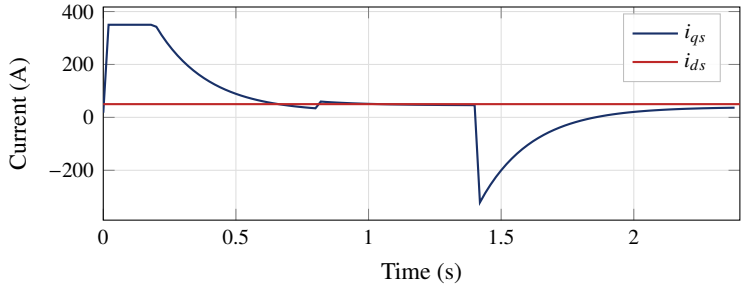


Figure 3.85: Current response: i_{qs} follows torque and i_{ds} remains constant.

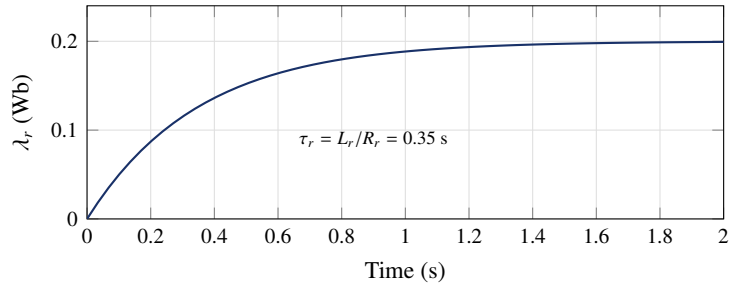


Figure 3.86: Rotor flux build-up with a step in i_{ds}.

one, and torque is no longer linear in i_{qs}^{*}. Hence automotive drives combine stored parameter maps, rotor temperature models and on-line adaptation.

3.11 Machine Model with Leakage Inductance

3.11.1 Introduction to Induction Machine Modelling

The equivalent circuit of Section 3.8 is valid only for steady state. Control design needs a dynamic model. We start from phase windings, form the coupled inductance equations, and reduce them to the dq model. The assumptions are: sinusoidal air-gap MMF, a uniform gap, linear magnetics, and balanced windings.

3.11.2 Windings and Phase-Variable Model

Figure 3.87 shows three stator windings and three equivalent rotor windings at electrical angle $\theta_r = \int \omega_r \, dt$ from the stator. In the abc reference frame, the voltage equations are

$$\mathbf{v}_s = R_s \mathbf{i}_s + \frac{d\boldsymbol{\lambda}_s}{dt}, \qquad \mathbf{v}_r = R_r \mathbf{i}_r + \frac{d\boldsymbol{\lambda}_r}{dt}, \tag{3.99}$$

with $\mathbf{v}_s = [v_{as} \; v_{bs} \; v_{cs}]^T$, etc. At this stage, R_r and the rotor variables are *actual* rotor values, and $\mathbf{v}_r = \mathbf{0}$ for a cage.

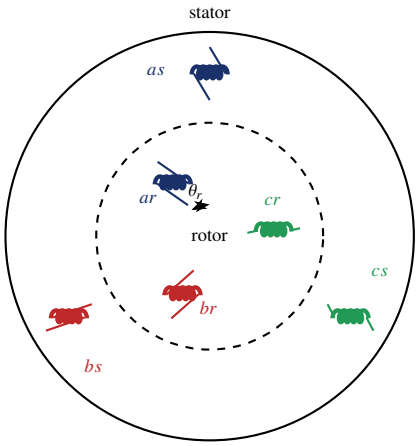


Figure 3.87: Induction motor winding arrangement.

3.11.3 Self, Mutual and Leakage Inductance

Flux from a coil divides into two parts, as depicted in Figure 3.88. Mutual flux crosses the air gap and links the other side. Leakage flux closes around the slots and end windings, linking only its own coil. Consequently, a stator phase has self-inductance $L_{ls} + L_{ms}$, where L_{ls} is *stator leakage inductance* and L_{ms} is the magnetizing part. Stator phases 120° apart share $\cos 120^\circ = -\frac{1}{2}$ of the mutual flux, so their mutual inductance is $-L_{ms}/2$. The stator-rotor mutual inductance varies as the cosine of the angle between coils, with peak value L_{sr}. The *rotor leakage inductance* L_{lr} behaves likewise on the rotor.

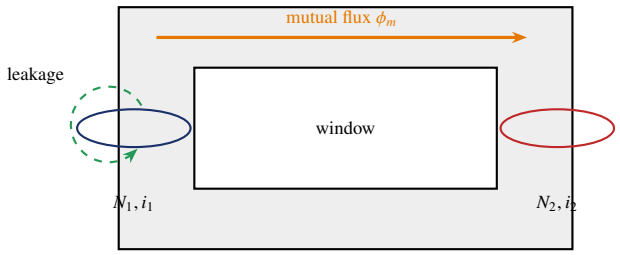


Figure 3.88: Mutual inductance representation: mutual and leakage flux paths.

The flux linkage equations of the phase-variable model are

$$\begin{bmatrix} \boldsymbol{\lambda}_s \\ \boldsymbol{\lambda}_r \end{bmatrix} = \begin{bmatrix} \mathbf{L}_{ss} & \mathbf{L}_{sr}(\theta_r) \\ \mathbf{L}_{sr}^{T}(\theta_r) & \mathbf{L}_{rr} \end{bmatrix} \begin{bmatrix} \mathbf{i}_s \\ \mathbf{i}_r \end{bmatrix}, \tag{3.100}$$

$$\mathbf{L}_{ss} = \begin{bmatrix} L_{ls}+L_{ms} & -L_{ms}/2 & -L_{ms}/2 \\ -L_{ms}/2 & L_{ls}+L_{ms} & -L_{ms}/2 \\ -L_{ms}/2 & -L_{ms}/2 & L_{ls}+L_{ms} \end{bmatrix}, \tag{3.101}$$

$$\mathbf{L}_{sr} = L_{sr} \begin{bmatrix} \cos\theta_r & \cos(\theta_r+120^\circ) & \cos(\theta_r-120^\circ) \\ \cos(\theta_r-120^\circ) & \cos\theta_r & \cos(\theta_r+120^\circ) \\ \cos(\theta_r+120^\circ) & \cos(\theta_r-120^\circ) & \cos\theta_r \end{bmatrix}. \tag{3.102}$$

$\mathbf{L}_{rr}$ has the same form as $\mathbf{L}_{ss}$, with rotor values. The difficulty lies in (3.102): the inductance matrix depends on rotor position, so the voltage equations have time-varying coefficients. Figure 3.89 shows the *abc* model in block form, with position feedback from the mechanical equation into the flux calculation.

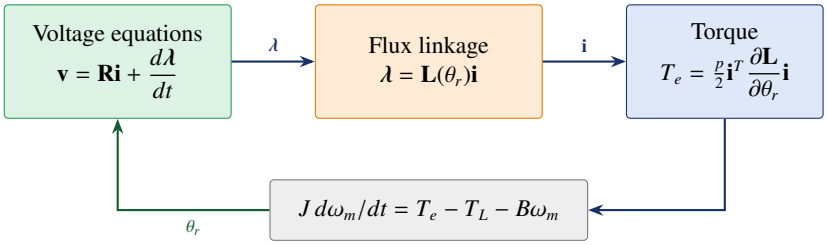


Figure 3.89: abc equivalent machine model.

Torque follows from the co-energy, $T_e = \frac{p}{2}\,\mathbf{i}^T(\partial\mathbf{L}/\partial\theta_r)\,\mathbf{i}$, where $\mathbf{i}$ collects all six currents and θ_r is the electrical angle.

3.11.4 Stator-Referred Rotor Quantities

Let $a = N_s/N_r$ be the effective turns ratio. The rotor is referred to the stator by

$$\mathbf{i}_r' = \frac{\mathbf{i}_r}{a}, \quad \mathbf{v}_r' = a\mathbf{v}_r, \quad \boldsymbol{\lambda}_r' = a\boldsymbol{\lambda}_r, \quad R_r' = a^2 R_r, \quad L_{lr}' = a^2 L_{lr}, \quad aL_{sr} = L_{ms}. \tag{3.103}$$

Primes mark referred values. Only these are measurable from the stator terminals of a cage machine, so actual cage parameters are rarely known. In the dq sections, rotor quantities are all referred, and the prime is omitted for brevity, as in (3.85): $R_r \equiv R'_r$, $L_r \equiv L'_r$.

3.11.5 Inductance Relationships and the Leakage Coefficient

In dq form, a phase's own magnetizing inductance gains the contributions of the other two phases:

$$L_m = \frac{3}{2} L_{ms}, \qquad L_s = L_{ls} + L_m, \qquad L_r = L'_{lr} + L_m. \tag{3.104}$$

This explains the factor by which L_m exceeds single-phase L_{ms}. The leakage coefficient measures imperfect coupling,

$$\sigma = 1 - \frac{L_m^2}{L_s L_r}, \tag{3.105}$$

so $\sigma = 0$ for perfect coupling. The product σL_s is the *transient inductance*: it is roughly $L_{ls} + L'_{lr}$, and it governs how quickly the stator current can change. Figure 3.90 shows the T-network of leakage and magnetizing inductance, and Figure 3.91 shows how currents build up the three flux vectors.

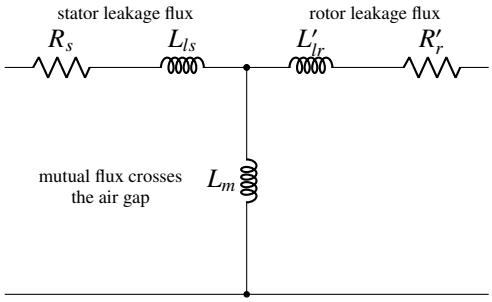


Figure 3.90: Leakage inductance representation (T-equivalent).

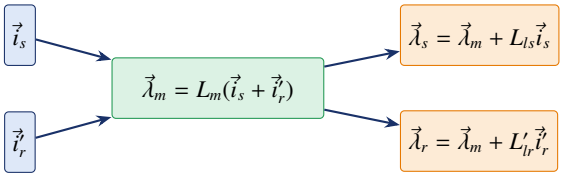


Figure 3.91: Flux linkage diagram.

3.11.6 Stationary and Synchronous Frame Models

Apply the transformation $\vec{x}_{dq} = \vec{x}e^{-j\theta}$ to (3.99). Since $\vec{\lambda} = \vec{\lambda}_{dq}e^{j\theta}$, the derivative produces an extra speed term:

$$\frac{d\vec{\lambda}}{dt} \rightarrow \frac{d\vec{\lambda}_{dq}}{dt} + j\omega\vec{\lambda}_{dq}. \tag{3.106}$$

For a frame at speed ω, we thus have

$$\vec{v}_s = R_s\vec{i}_s + \frac{d\vec{\lambda}_s}{dt} + j\omega\vec{\lambda}_s, \tag{3.107}$$

$$\vec{v}_r = R_r\vec{i}_r + \frac{d\vec{\lambda}_r}{dt} + j(\omega - \omega_r)\vec{\lambda}_r. \tag{3.108}$$

Setting $\omega = 0$ gives the stationary $\alpha\beta$ model, and $\omega = \omega_e$ gives the synchronous model of (3.83)–(3.86). Position has disappeared from the inductance terms, leaving

$$\vec{\lambda}_s = L_s\vec{i}_s + L_m\vec{i}_r, \qquad \vec{\lambda}_r = L_r\vec{i}_r + L_m\vec{i}_s. \tag{3.109}$$

Hence the equivalent inductance matrix for each axis is the constant

$$\mathbf{L}_{dq} = \begin{bmatrix} L_s & L_m \\ L_m & L_r \end{bmatrix}, \qquad \det \mathbf{L}_{dq} = \sigma L_s L_r. \tag{3.110}$$

Figure 3.92 shows these equations as circuits, in which the red sources are the speed voltages which couple the axes.

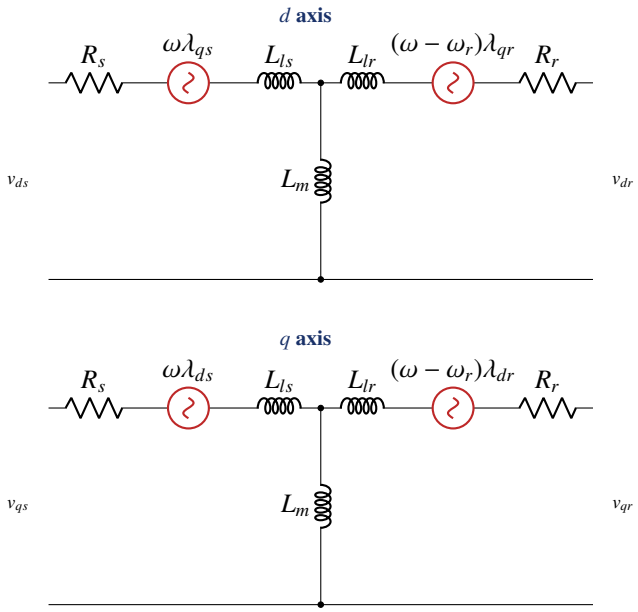


Figure 3.92: d-q machine model.

The torque can be written in several equivalent forms using (3.109) and (3.72):

$$T_e = \frac{3}{2}p\,\mathrm{Im}\{\vec{\lambda}_s^*\vec{i}_s\} = \frac{3}{2}p(\lambda_{ds}i_{qs} - \lambda_{qs}i_{ds}) = \frac{3}{2}pL_m(i_{qs}i_{dr} - i_{ds}i_{qr}). \tag{3.111}$$

The first form with the factor 3/2 reflects the amplitude-invariant transform, and p reflects conversion from electrical to mechanical angle.

3.11.7 Machine State-Space Representation

Current control benefits from stator current and rotor flux states. Let $\mathbf{x} = [i_{ds}\ i_{qs}\ \lambda_{dr}\ \lambda_{qr}]^T$, $k_r = L_m/L_r$ and $R_\sigma = R_s + k_r^2 R_r$. Eliminating i_{dr}, i_{qr} from (3.83)–(3.88) gives

$$\frac{d}{dt}\begin{bmatrix} i_{ds} \\ i_{qs} \\ \lambda_{dr} \\ \lambda_{qr} \end{bmatrix} = \begin{bmatrix} -\dfrac{R_\sigma}{\sigma L_s} & \omega_e & \dfrac{k_r}{\sigma L_s \tau_r} & \dfrac{k_r \omega_r}{\sigma L_s} \\ -\omega_e & -\dfrac{R_\sigma}{\sigma L_s} & -\dfrac{k_r \omega_r}{\sigma L_s} & \dfrac{k_r}{\sigma L_s \tau_r} \\ \dfrac{L_m}{\tau_r} & 0 & -\dfrac{1}{\tau_r} & \omega_{sl} \\ 0 & \dfrac{L_m}{\tau_r} & -\omega_{sl} & -\dfrac{1}{\tau_r} \end{bmatrix} \mathbf{x} + \frac{1}{\sigma L_s}\begin{bmatrix} v_{ds} \\ v_{qs} \\ 0 \\ 0 \end{bmatrix}. \tag{3.112}$$

With the mechanical equation (3.33) and torque (3.89), this gives a fifth-order nonlinear model, nonlinear because it contains products of speed and state. It is the form normally entered in MATLAB/Simulink through integrator blocks or an S-function, and in Simscape libraries by parameters.

3.11.8 Effect of Leakage Inductance

- **Transient response.** Fast current changes are limited by σL_s, not L_s. Stator and rotor time constants of the *R-L* type are therefore much shorter than τ_r.
- **Current dynamics.** Larger leakage lessens PWM ripple, $\Delta i \propto V_{\mathrm{dc}}/(\sigma L_s f_s)$, but demands more voltage for a given di/dt.
- **Torque production.** Leakage reduces breakdown torque, since in (3.65) it appears in the denominator. It also caps the constant power speed range.
- **Control performance.** Decoupling terms in (3.95) use σL_s, and the current-loop gain is set by it. An error in leakage shifts the estimated flux angle in (3.77).

Example 19: Flux linkage and torque from machine parameters

Problem. Using the operating point of Example 18, evaluate leakage inductances, σ, all flux linkages and the torque.

Given. $L_s = L_r = 4.2$ mH, $L_m = 4$ mH, $p = 2$, $i_{ds} = 50$ A, $i_{qs} = 262.5$ A, rotor-flux-oriented.

Required. L_{ls}, L'_{lr}, σ, rotor currents, flux linkages, T_e.

Governing equations. (3.104), (3.105), (3.90), (3.87), (3.88), (3.111).

Solution. $L_{ls} = L'_{lr} = 0.2$ mH. $\sigma = 1 - 16/17.64 = 0.093$.
$i_{dr} = 0$ in steady state, $i_{qr} = -(4/4.2)(262.5) = -250$ A.
$\lambda_{ds} = 0.0042(50) = 0.21$ Wb. $\lambda_{qs} = 0.0042(262.5) + 0.004(-250) = 0.1025$ Wb.

$\lambda_{dr} = 0.004(50) = 0.20$ Wb. $\lambda_{qr} = 0.0042(-250) + 0.004(262.5) = 0$.
$T_e = 1.5(2)[0.21(262.5) - 0.1025(50)] = 150$ N m.

Result. $\sigma = 0.093$; $\lambda_{ds} = 0.21$, $\lambda_{qs} = 0.1025$, $\lambda_{dr} = 0.20$, $\lambda_{qr} = 0$ Wb; $T_e = 150$ N m.

Physical interpretation. $\lambda_{qr} = 0$ confirms orientation, whereas stator flux is displaced from the d axis by leakage flux due to the torque current.

Engineering significance. The torque matches the reference of Example 18, which is a useful consistency check on a simulation model.

3.12 Permanent Magnet Synchronous Machine for EV Powertrain

3.12.1 Introduction to PMSM

A permanent magnet synchronous machine (PMSM) replaces the induction motor's induced rotor field by magnets. No rotor current is needed to establish flux, so there is almost no rotor copper loss, and the rotor turns at exactly synchronous speed. These qualities give the PMSM the highest torque density and efficiency among mainstream traction machines, and it is now the most widely used EV motor [15, 18].

3.12.2 PMSM Construction

Figure 3.93 shows a cross-section. The stator is the same as that of an induction motor, with laminated core and three-phase winding, as in Figure 3.94. The rotor has an iron core carrying permanent magnets of alternating polarity, arranged as in Figure 3.95. Each pair of adjacent N and S magnets spans 360° electrical.

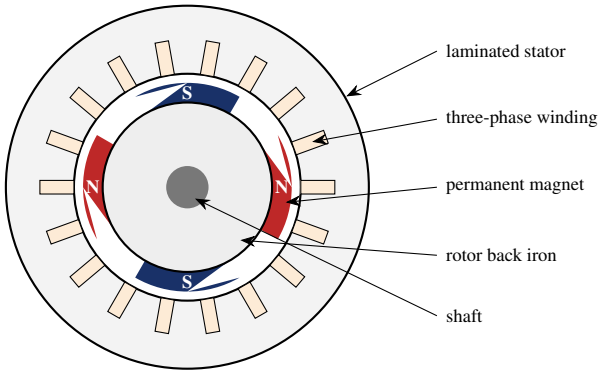


Figure 3.93: PMSM construction (surface-magnet rotor shown).

3.12.2.1 Surface-Mounted PMSM

In the surface-mounted machine (SPMSM) of Figure 3.96a, magnets are bonded to the rotor surface. Magnets have a permeability close to that of air, so the effective air gap is large and

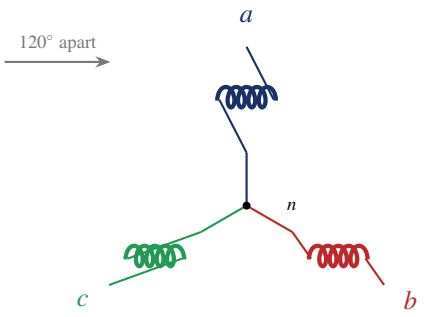


Figure 3.94: PMSM stator three-phase winding.

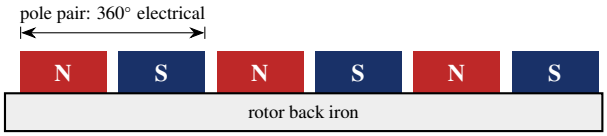
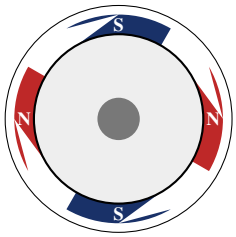


Figure 3.95: Rotor permanent magnet arrangement (developed view).

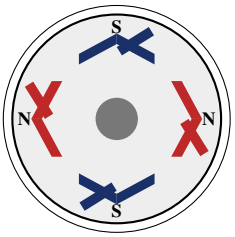
uniform. The rotor needs a sleeve to retain the magnets at high speed.

3.12.2.2 Interior PMSM

In the interior machine (IPMSM) of Figure 3.96b, magnets are buried in rotor slots, often in a V shape [25]. The rotor iron holds them mechanically and shields them somewhat from stator fields, and the iron between magnets creates a saliency which gives extra torque. Most EV PMSMs are interior machines.



(a) Surface-mounted PMSM.



(b) Interior PMSM.

Figure 3.96: PMSM rotor types.

3.12.3 Permanent Magnet Materials

- **NdFeB (neodymium-iron-boron).** Highest energy product and remanence of about 1.2–1.4 T. It loses strength with temperature, so heavy rare earths (dysprosium, terbium) are added for high-temperature grades.
- **SmCo (samarium-cobalt).** Lower remanence, but excellent temperature stability. It is brittle and costly.
- **Ferrite.** Cheap, with no rare earths, but a remanence of about 0.4 T that gives lower torque density. Researchers use it in magnet-assisted reluctance designs.

3.12.4 Magnetic Field, Back EMF and Synchronous Operation

The magnets produce a flux linkage in each stator phase that varies with rotor position. For sinusoidal distribution, phase a links $\lambda_{a,\mathrm{pm}} = \lambda_m \cos\theta_e$, with λ_m the peak flux linkage (Wb) and $\theta_e = p\theta_m$. By Faraday's law, the back EMF of phase a is

$$e_a = \frac{d\lambda_{a,\mathrm{pm}}}{dt} = -\omega_e \lambda_m \sin\theta_e, \tag{3.113}$$

with peak value $E_p = \omega_e \lambda_m$ and rms value $E = \omega_e \lambda_m / \sqrt{2}$. The back EMF grows linearly with speed and, unlike in an induction motor, is present whenever the rotor turns, even with no excitation.

Stator currents at frequency f_e create a rotating MMF at $n_s = 60 f_e / p$. The rotor locks to that field. Therefore

$$\omega_m = \frac{\omega_e}{p}, \qquad n = \frac{60 f_e}{p}. \tag{3.114}$$

Torque arises from the attempt of the magnet field to align with the stator field. It is maximum when the two are at 90° electrical, and FOC holds them there. Figure 3.97 shows the resulting per-phase equivalent, a voltage behind an impedance; note that it is exact only in the non-salient case.

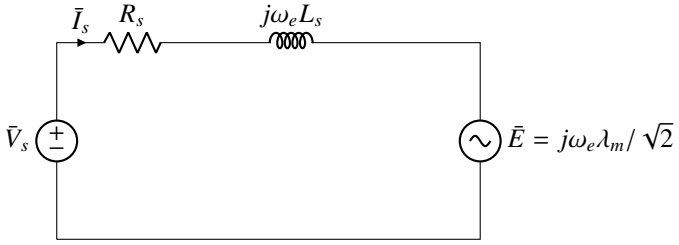


Figure 3.97: PMSM equivalent representation in steady state (non-salient, rms phasors).

3.12.5 PMSM EV Drive

Figure 3.98 shows the drive system. The controller differs from that of an induction motor in one main respect: rotor position is required for orientation, so the resolver signal is essential.

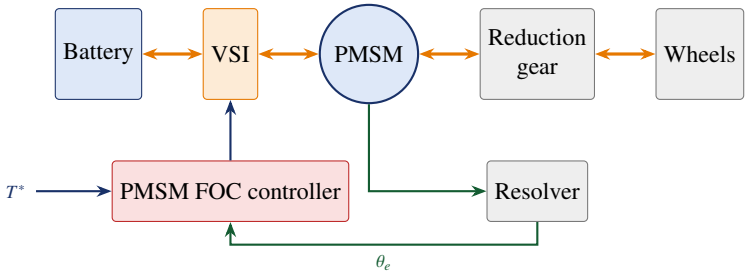


Figure 3.98: PMSM EV drive architecture.

3.12.6 EV Traction Requirements, Advantages and Limitations

Advantages:

- **High power density.** Magnets supply flux at no cost in copper, so the machine is compact.
- **High efficiency.** Rotor loss is small, and peak efficiency often reaches around 97%.
- **Regenerative braking.** Back EMF is available at all speeds, so generation requires no magnetising power.
- **Good thermal path.** Most losses are in the stator, near the cooling jacket.

Limitations:

- **Thermal limitations.** Remanence falls with temperature, by about 0.1%/K for NdFeB, which reduces torque.
- **Demagnetization.** Excessive negative d-axis MMF at high temperature can permanently damage magnets. Control limits and thermal protection must prevent this.
- **Field weakening.** The magnet flux cannot be turned off, so weakening requires continuous stator current. At high speed, an inverter fault can result in uncontrolled generation into the DC bus, because the back EMF can exceed the link voltage.
- **Cost and supply.** Rare-earth prices and supply are a strategic concern.

Tables 3.15 and 3.16 compare the machines.

Table 3.15: Induction motor versus PMSM

Property	Induction motor	PMSM
Rotor excitation	Induced, needs stator current	Magnets
Rotor speed	Slip below synchronous	Synchronous
Rotor loss	Significant	Small
Power density	Moderate	High
Part-load efficiency	Lower	Higher
Field weakening	Simple, reduce i_{ds}	Needs negative i_d
Open-circuit drag	None when unexcited	Magnet-induced iron loss
Rare-earth use	None	Usually
Controller parameter risk	τ_r	λ_m varies with temperature

The two tables show why dual-motor cars have sometimes used a PMSM on one axle for efficiency and an IM on the other for low idle drag. No machine wins on every criterion, and selection depends on the drive cycle and cost target.

Table 3.16: EV motor selection considerations

Criterion	Implication
Efficiency over drive cycle	Favours IPMSM for primary axle
Peak power per kg	Favours PMSM; IM needs higher speed
Cost and rare-earth risk	Favours IM or magnet-free designs
Spin losses on secondary axle	Favours IM, or PMSM with disconnect
High-speed range	Requires IPMSM or IM with sufficient flux-weakening
Thermal environment	Magnet grade and rotor cooling are key
Fault tolerance	IM safer at high speed with loss of control

3.13 Non-Salient and Salient Drives

3.13.1 The Concept of Saliency

Saliency means that the magnetic reluctance seen by the stator depends on rotor position. It is measured along two axes fixed to the rotor. The d axis lies on the magnet axis, and the q axis leads it by 90° electrical, midway between magnets. Figure 3.99 shows these axes with a general current vector, whose angle β is measured from the q axis towards the negative d axis, so that

$$i_d = -I_s \sin\beta, \qquad i_q = I_s \cos\beta. \tag{3.115}$$

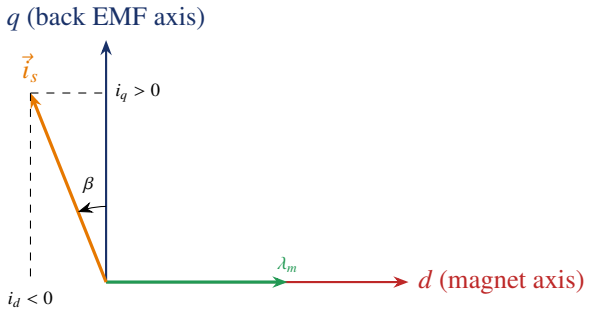


Figure 3.99: d-q axis representation of the PMSM current vector.

3.13.2 Non-Salient PMSM

In a surface-magnet machine (Figure 3.100a), the rotor looks magnetically round: both axes pass through magnet material or an equal effective gap. Therefore $L_d = L_q = L_s$, and these are low because the effective gap is large.

3.13.2.1 Torque Production and Current Control

The derivation given in Section 3.14 leads to

$$T_e = \frac{3}{2} p \left[\lambda_m i_q + (L_d - L_q) i_d i_q \right], \tag{3.116}$$

where p denotes pole *pairs*, and the $3/2$ belongs to the amplitude-invariant Park convention of Table 3.1. With a power-invariant transform, the factor is 1 and currents are $\sqrt{3/2}$ times larger, so the physics does not change. For a non-salient machine, $L_d = L_q$, giving the simplified linear equation

$$T_e = \frac{3}{2} p \lambda_m i_q. \tag{3.117}$$

Torque depends only on i_q, so setting $i_d = 0$ gives maximum torque per ampere. In steady operation below base speed, the current vector therefore stays on the q axis, and the torque-speed characteristic is flat until the voltage limit, which is reached at base speed.

3.13.3 Salient PMSM

In an interior machine (Figure 3.100b), flux along the d axis has to cross the magnets, while flux along the q axis stays in the iron between them, as in Figure 3.101. As the permeability of magnet material is about that of air,

$$L_q > L_d. \tag{3.118}$$

This *inverse saliency* differs from a wound-rotor salient-pole synchronous machine, in which $L_d > L_q$.



(a) Non-salient rotor. **(b)** Salient rotor.

Figure 3.100: Non-salient and salient PMSM rotors with d and q axes.

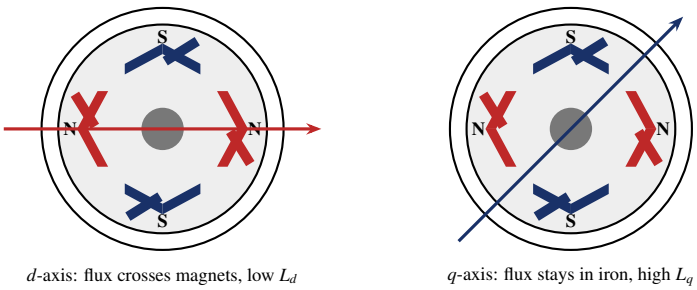


Figure 3.101: Flux distribution along the d and q axes of an interior rotor.

3.13.3.1 Magnet Torque and Reluctance Torque

Two components appear in (3.116).

- **Permanent magnet torque**, $T_{\text{pm}} = \frac{3}{2} p \lambda_m i_q$, arises from the magnet field and the q-axis current.
- **Reluctance torque**, $T_{\text{rel}} = \frac{3}{2} p (L_d - L_q) i_d i_q$, arises from the rotor's tendency to align its low-reluctance path with the stator field.

$L_d - L_q$ is negative, so reluctance torque is positive when $i_d < 0$ and $i_q > 0$. Thus negative d-axis current *adds* torque. The roles of the quantities are: i_q is the main torque current; i_d adjusts flux and, in a salient machine, torque; L_d sets the flux-weakening ability; and L_q sets the reluctance contribution and the voltage drop due to torque current. In terms of the current angle of (3.115), the combined electromagnetic torque is

$$T_e = \frac{3}{2} p \left[\lambda_m I_s \cos\beta + \tfrac{1}{2} (L_q - L_d) I_s^2 \sin 2\beta \right]. \tag{3.119}$$

Figure 3.102 plots these components for the IPMSM of Example 20. Magnet torque peaks at $\beta = 0$, and reluctance torque peaks at 45°. Their sum peaks in between, at the MTPA angle, here 28.8°.

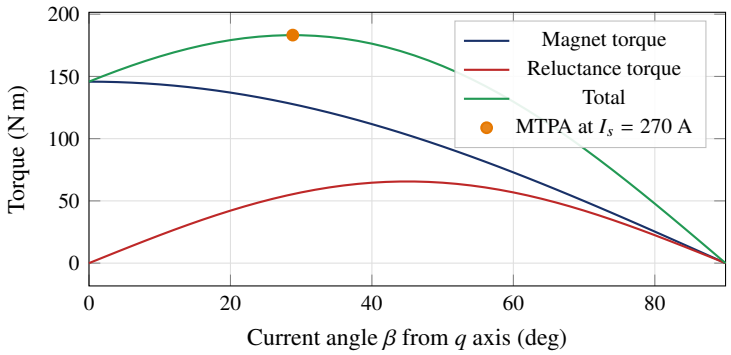


Figure 3.102: Torque components of the example IPMSM at $I_s = 270$ A.

3.13.3.2 Maximum Torque per Ampere

For a given current magnitude, the angle for maximum torque comes from $\partial T_e / \partial \beta = 0$ applied to (3.119), which gives $\lambda_m \sin\beta = (L_q - L_d) I_s \cos 2\beta$. Rewriting with $\Delta L = L_q - L_d$ and solving for i_d,

$$i_d = \frac{\lambda_m - \sqrt{\lambda_m^2 + 8 \Delta L^2 I_s^2}}{4 \Delta L}. \tag{3.120}$$

It may equally be written using i_q as $i_d = \lambda_m / (2\Delta L) - \sqrt{\lambda_m^2 / (4\Delta L^2) + i_q^2}$. This is the MTPA trajectory.

3.13.3.3 Torque-Speed Characteristics and Field Weakening

Figure 3.103 compares computed capability envelopes of the two example machines. The operating envelope has a flat constant torque region up to base speed, followed by a falling region in which the stator current is rotated towards the negative d axis to hold terminal voltage at the limit, as in Figure 3.104. Saliency lets an IPMSM get more torque from the same current and a smaller magnet.

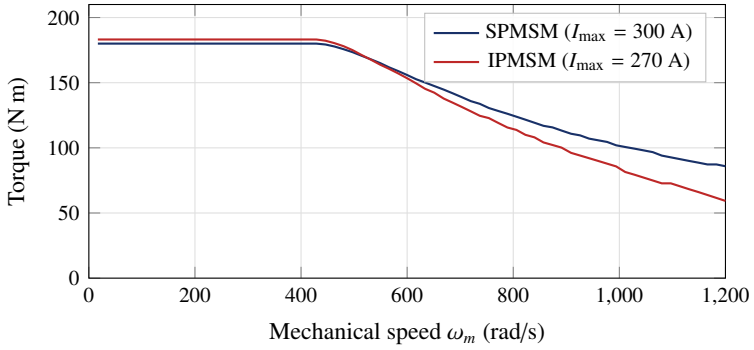


Figure 3.103: Torque-speed characteristics of the example SPMSM and IPMSM (computed, V_{max} = 230 V).

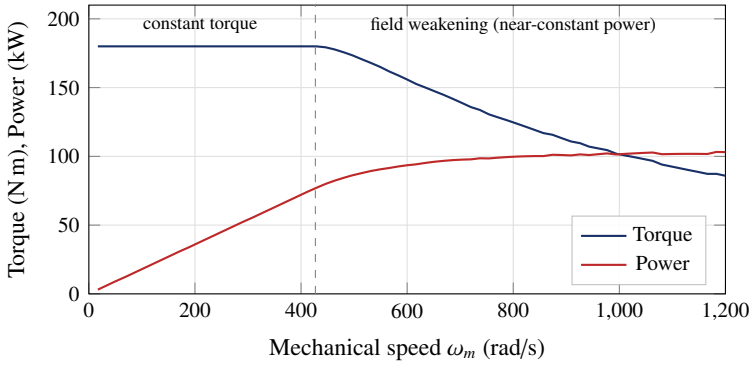


Figure 3.104: Constant torque and field weakening regions (example SPMSM).

Tables 3.17 and 3.18 give the comparison.

Table 3.17: SPMSM versus IPMSM: construction and application

Feature	SPMSM	IPMSM
Magnet location	Rotor surface	Buried in iron
Retention	Sleeve or banding	Rotor iron bridges
Maximum speed	Limited by sleeve	High
Magnet quantity	Higher for same torque	Lower
Rotor manufacture	Simple	More complex
Typical use	Servo, in-wheel	Mainstream EV traction

The tables explain why the IPMSM dominates despite more complex control: saliency reduces magnet use, and the extra control effort is software cost.

Table 3.18: Non-salient versus salient PMSM: electromagnetic behaviour

Feature	Non-salient	Salient
Inductances	$L_d = L_q$	$L_q > L_d$
Torque	$\frac{3}{2}p\lambda_m i_q$	$\frac{3}{2}p[\lambda_m i_q + (L_d - L_q) i_d i_q]$
Reluctance torque	None	Present
Best current below base speed	$i_d = 0$	MTPA with $i_d < 0$
Torque-current relation	Linear	Nonlinear
Field-weakening range	Limited by low L	Wider
Control complexity	Low	Higher, needs tables

3.14 Generic Model of PMSM

3.14.1 Generic PMSM Representation

The model assumes sinusoidal back EMF and winding distribution, no saturation, negligible iron loss and no damper windings. It uses motor convention and the amplitude-invariant transform of Table 3.1. Figure 3.105 shows the three-phase machine, with rotor position θ_e measured from the phase a axis to the magnet north axis.

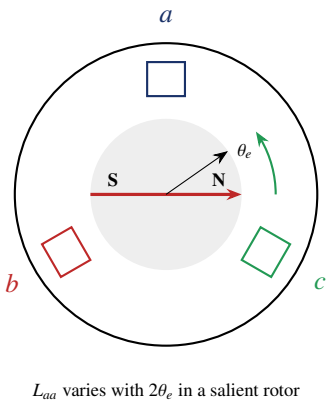


Figure 3.105: PMSM abc representation.

3.14.2 Three-Phase abc Model

Each phase voltage is resistive drop plus rate of change of flux linkage:

$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = R_s \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \frac{d}{dt} \begin{bmatrix} \lambda_a \\ \lambda_b \\ \lambda_c \end{bmatrix}. \tag{3.121}$$

Flux linkage has inductive and magnet parts,

$$\boldsymbol{\lambda}_{abc} = \mathbf{L}(\theta_e)\,\mathbf{i}_{abc} + \lambda_m \begin{bmatrix} \cos\theta_e \\ \cos(\theta_e - 120^\circ) \\ \cos(\theta_e + 120^\circ) \end{bmatrix}. \tag{3.122}$$

In a salient machine, the self and mutual inductances vary at twice the electrical angle, for example

$$L_{aa} = L_{ls} + L_A + L_B \cos 2\theta_e, \qquad L_{ab} = -\tfrac{1}{2}L_A + L_B \cos(2\theta_e - 120^\circ). \tag{3.123}$$

It may be shown that $L_d = L_{ls} + \frac{3}{2}(L_A + L_B)$ and $L_q = L_{ls} + \frac{3}{2}(L_A - L_B)$, so $L_B < 0$ for an IPMSM, and $L_B = 0$ recovers a position-independent SPMSM. Because $\mathbf{L}$ depends on θ_e, the equations are not directly suited to controller design.

3.14.3 Clarke and Park Transformations

Applying (3.68) to (3.121), and then (3.70) with $\theta = \theta_e$, gives the rotor frame shown in Figure 3.106. In complex form, $\vec{x}_{dq} = \vec{x}_{\alpha\beta} e^{-j\theta_e}$. The resulting flux linkages have no angle dependence.

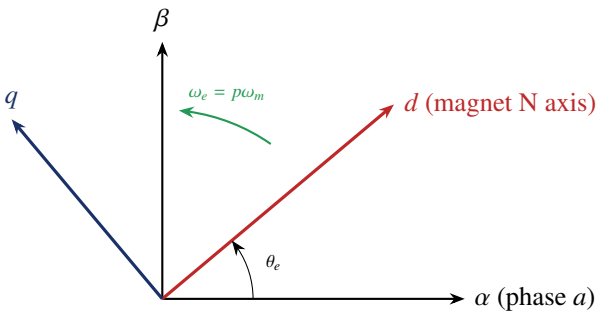


Figure 3.106: PMSM d-q reference frame.

3.14.4 d-q Flux Linkage and Voltage Equations

With d on the magnet axis, the magnet links only the d axis. Hence

$$\lambda_d = L_d i_d + \lambda_m, \tag{3.124}$$

$$\lambda_q = L_q i_q, \tag{3.125}$$

as Figure 3.107 illustrates. To obtain the voltages, write (3.121) as a space vector and substitute $\vec{\lambda} = \vec{\lambda}_{dq} e^{j\theta_e}$:

$$\vec{v}_{dq} = R_s \vec{i}_{dq} + \frac{d\vec{\lambda}_{dq}}{dt} + j\omega_e \vec{\lambda}_{dq}. \tag{3.126}$$

Separating real and imaginary parts verifies the starting equations given in the syllabus:

$$v_d = R_s i_d + \frac{d\lambda_d}{dt} - \omega_e \lambda_q, \tag{3.127}$$

$$v_q = R_s i_q + \frac{d\lambda_q}{dt} + \omega_e \lambda_d. \tag{3.128}$$

The terms have a clear physical meaning:

- $R_s i_d$ and $R_s i_q$ are resistive drops.
- $d\lambda_d/dt$ and $d\lambda_q/dt$ are transformer voltages, which appear only during transients.
- $-\omega_e \lambda_q$ and $+\omega_e \lambda_d$ are speed (rotational) voltages, caused by flux on one axis moving past the other axis.
- $\omega_e \lambda_m$, contained in the q-axis speed voltage, is the back EMF, which lies along the q axis.

Substituting (3.124) and (3.125) with constant inductances,

$$v_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q, \tag{3.129}$$

$$v_q = R_s i_q + L_q \frac{di_q}{dt} + \omega_e L_d i_d + \omega_e \lambda_m. \tag{3.130}$$

These are the circuits of Figure 3.108. A surface-mounted model has $L_d = L_q = L_s$. An interior model keeps them distinct.

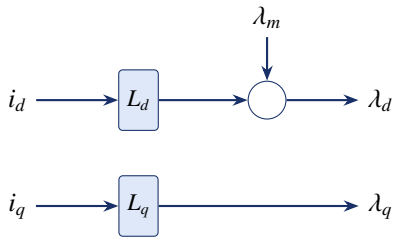


Figure 3.107: PMSM flux linkage model.

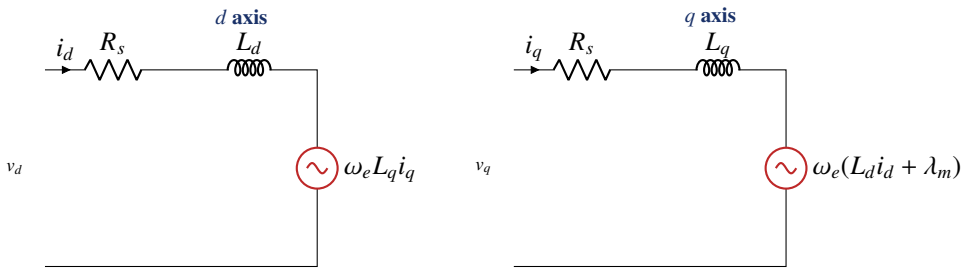


Figure 3.108: PMSM voltage model.

3.14.5 Electromagnetic Torque Equation

Multiply (3.127) by i_d and (3.128) by i_q, add them, and apply (3.72):

$$p_{\text{in}} = \underbrace{\tfrac{3}{2}R_s(i_d^2 + i_q^2)}_{\text{copper loss}} + \underbrace{\tfrac{3}{2}\left(i_d\frac{d\lambda_d}{dt} + i_q\frac{d\lambda_q}{dt}\right)}_{\text{field energy rate}} + \underbrace{\tfrac{3}{2}\omega_e(\lambda_d i_q - \lambda_q i_d)}_{\text{mechanical power}}. \tag{3.131}$$

The last term must equal $T_e\omega_m$. With $\omega_e = p\omega_m$, we have

$$T_e = \frac{3}{2}p(\lambda_d i_q - \lambda_q i_d) = \frac{3}{2}p\left[\lambda_m i_q + (L_d - L_q)i_d i_q\right], \tag{3.132}$$

confirming (3.116). The result would have a factor $3P/4$ if written with the pole number P.

3.14.6 Mechanical Dynamic Equation

From (3.33), with position and electrical speed,

$$J\frac{d\omega_m}{dt} = T_e - T_L - B\omega_m, \qquad \frac{d\theta_m}{dt} = \omega_m, \qquad \theta_e = p\theta_m, \qquad \omega_e = p\omega_m. \tag{3.133}$$

3.14.7 State-Space Model

Choosing states $\mathbf{x} = [i_d\ i_q\ \omega_m\ \theta_e]^T$ and inputs v_d, v_q, T_L, rearranging (3.129), (3.130) and (3.133) gives

$$\frac{d}{dt}\begin{bmatrix} i_d \\ i_q \\ \omega_m \\ \theta_e \end{bmatrix} = \begin{bmatrix} \dfrac{1}{L_d}\left(v_d - R_s i_d + \omega_e L_q i_q\right) \\ \dfrac{1}{L_q}\left(v_q - R_s i_q - \omega_e L_d i_d - \omega_e\lambda_m\right) \\ \dfrac{1}{J}\left(\tfrac{3}{2}p[\lambda_m i_q + (L_d - L_q)i_d i_q] - T_L - B\omega_m\right) \\ p\,\omega_m \end{bmatrix}. \tag{3.134}$$

Figure 3.109 gives the block structure: electrical states yield torque, torque drives the mechanical state, and speed feeds back into the electrical equations through the speed voltages. The products $\omega_e i_d$, $\omega_e i_q$ and $i_d i_q$ make the model nonlinear. Algorithm 13 shows a simple discrete implementation.

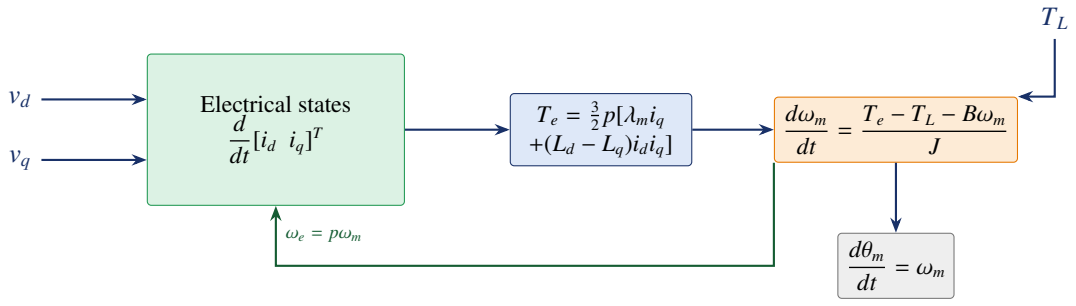


Figure 3.109: PMSM state-space architecture.

Algorithm 13 PMSM mathematical model implementation

Input: Parameters $R_s, L_d, L_q, \lambda_m, p, J, B$; step h; inputs v_a, v_b, v_c, T_L
Output: $i_{abc}, T_e, \omega_m, \theta_e$

```
Initialise i_d, i_q, ω_m, θ_e ← 0
for each time step do
    (v_d, v_q) ← Park(Clarke(v_a, v_b, v_c), θ_e)
    ω_e ← p ω_m
    Evaluate the derivative ẋ of (3.134)
    x ← x + h ẋ (or a higher-order solver)
    Wrap θ_e; compute T_e using (3.132)
    (i_a, i_b, i_c) ← inverse Clarke(inverse Park(i_d, i_q, θ_e))
end for
```

3.15 Steady-State Analysis of PMSM

3.15.1 Steady-State Assumptions

Assume constant speed and sinusoidal balanced currents. In the rotor frame all quantities are then constant, and d/dt terms vanish. Capital letters denote these DC values, which are peak phase quantities under the adopted scaling. Rms values are $1/\sqrt{2}$ of these.

3.15.2 Steady-State Equations

Setting derivatives to zero in (3.129) and (3.130),

$$V_d = R_s I_d - \omega_e L_q I_q, \tag{3.135}$$

$$V_q = R_s I_q + \omega_e L_d I_d + \omega_e \lambda_m. \tag{3.136}$$

Flux linkages are $\Lambda_d = L_d I_d + \lambda_m$ and $\Lambda_q = L_q I_q$. Back EMF is $E_p = \omega_e \lambda_m$ along the q axis. Torque and power are

$$T_e = \frac{3}{2} p \left[\lambda_m I_q + (L_d - L_q) I_d I_q \right], \qquad P_m = T_e \omega_m = T_e \frac{\omega_e}{p}. \tag{3.137}$$

Electrical power follows from (3.72), $P_e = \frac{3}{2}(V_d I_d + V_q I_q) = P_m + \frac{3}{2} R_s I_s^2$, ignoring iron loss.

Example 20: PMSM d-q voltage and torque calculation

Problem. An IPMSM runs at 3000 r/min with $I_d = -100$ A and $I_q = 250$ A. Find its synchronous frequency, back EMF, voltages, torque and power, and check against a 400 V link.

Given (assumed). $p = 4$, $R_s = 0.02\ \Omega$, $L_d = 0.2$ mH, $L_q = 0.5$ mH, $\lambda_m = 0.09$ Wb.

Required. $f_e, E_p, V_d, V_q, |V|, T_e, P_m$.

Governing equations. (3.114), (3.135), (3.136), (3.137).

Algorithm 14 PMSM steady-state analysis

Input: n, I_d, I_q, parameters, V_{dc}, $I_{\max}$
Output: $V_d, V_q, |V|, T_e, P_m$, feasibility
1: $\omega_m \leftarrow 2\pi n/60$; $\omega_e \leftarrow p\omega_m$; $f_e \leftarrow \omega_e/2\pi$
2: $E_p \leftarrow \omega_e \lambda_m$
3: $V_d, V_q \leftarrow$ (3.135), (3.136); $|V| \leftarrow \sqrt{V_d^2 + V_q^2}$
4: $T_e, P_m \leftarrow$ (3.137); $P_e \leftarrow \frac{3}{2}(V_d I_d + V_q I_q)$
5: **if** $\sqrt{I_d^2 + I_q^2} > I_{\max}$ **or** $|V| > V_{\mathrm{dc}}/\sqrt{3}$ **then**
6: Mark point infeasible
7: **end if**

Solution. $f_e = 3000 \times 4/60 = 200$ Hz. $\omega_m = 314.2$ rad/s, $\omega_e = 1256.6$ rad/s.
$E_p = 1256.6 \times 0.09 = 113.1$ V peak (80 V rms).
$V_d = 0.02(-100) - 1256.6(0.0005)(250) = -159.1$ V.
$V_q = 0.02(250) + 1256.6[0.0002(-100) + 0.09] = 93.0$ V.
$|V| = 184.3$ V, below $V_{\max} = 400/\sqrt{3} = 230.9$ V.
$T_{\mathrm{pm}} = 1.5(4)(0.09)(250) = 135$ N m. $T_{\mathrm{rel}} = 1.5(4)(-0.0003)(-100)(250) = 45$ N m.
$T_e = 180$ N m. $P_m = 180 \times 314.2 = 56.5$ kW.
Check: $P_e = 1.5(V_d I_d + V_q I_q) = 58.7$ kW; copper loss $1.5(0.02)(269.3)^2 = 2.18$ kW.

Result. $f_e = 200$ Hz, $E_p = 113.1$ V, $V_d = -159.1$ V, $V_q = 93.0$ V, $T_e = 180$ N m, $P_m = 56.5$ kW.

Physical interpretation. A quarter of the torque is reluctance torque, gained without extra magnet. The d-axis voltage is large because of the voltage drop of i_q across L_q.

Engineering significance. Power balance closes: 56.5 + 2.18 = 58.7 kW. Such a check is the quickest way to validate a model.

Figure 3.110 sketches the same steady state as a phasor diagram on the rotor axes, with stator resistance neglected. The back EMF lies on the q axis, negative I_d subtracts from it, and $\omega_e L_q I_q$ acts in the $-d$ direction. Their sum is the terminal voltage, which is why V_d was negative and large in Example 20.

3.15.3 Voltage and Current Limits

The inverter limits the voltage vector and thermal rating limits the current:

$$V_d^2 + V_q^2 \le V_{\max}^2, \qquad V_{\max} \approx \frac{V_{\mathrm{dc}}}{\sqrt{3}} \ \text{(SVPWM)}, \tag{3.138}$$

$$I_d^2 + I_q^2 \le I_{\max}^2. \tag{3.139}$$

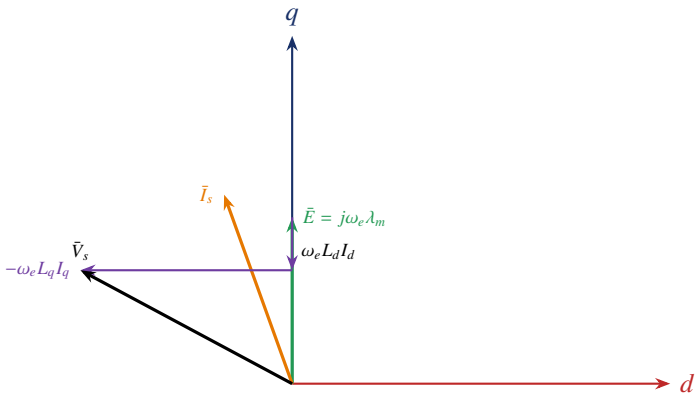
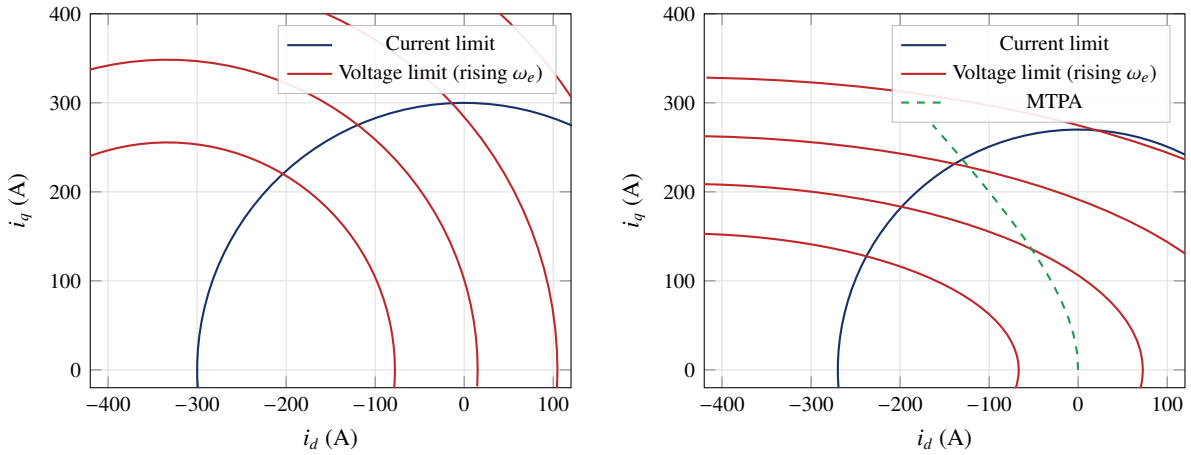


Figure 3.110: PMSM steady-state phasor diagram for motoring with $I_d < 0$ (not to scale).

Neglecting R_s and substituting (3.135) and (3.136),

$$(L_d I_d + \lambda_m)^2 + (L_q I_q)^2 \leq \left(\frac{V_{\max}}{\omega_e}\right)^2. \tag{3.140}$$

In the i_d-i_q plane, this is an ellipse centred at $(-\lambda_m/L_d, 0)$ that shrinks as speed rises. It is a circle for an SPMSM. The allowed operating points lie within *both* the current circle and the voltage ellipse, as Figures 3.111a and 3.111b show.



(a) SPMSM.

(b) IPMSM with MTPA path.

Figure 3.111: Current limit and voltage limit curves at four speeds for the example machines.

3.15.4 Constant Torque Region and Base Speed

Below base speed, the voltage limit lies outside the current operating point, and the drive runs at the MTPA point of $I_{\max}$ (or $i_d = 0$ for an SPMSM). Base speed is where the voltage limit first reaches this point. For an SPMSM with $i_d = 0$, (3.140) gives

$$\omega_{e,b} = \frac{V_{\max}}{\sqrt{\lambda_m^2 + (L_s I_{\max})^2}}. \tag{3.141}$$

3.15.5 Field Weakening and the Constant Power Region

Beyond base speed, $|V|$ would exceed the limit, so negative i_d is applied to cancel part of the magnet flux in (3.140). Solving the equality with the current limit for an SPMSM gives

$$I_d = \frac{(V_{\max}/\omega_e)^2 - \lambda_m^2 - L_s^2 I_{\max}^2}{2L_s\lambda_m}, \qquad I_q = \sqrt{I_{\max}^2 - I_d^2}. \tag{3.142}$$

The field weakening relationship shows that I_d grows in magnitude with speed. The quantity $I_{\mathrm{ch}} = \lambda_m/L_d$, called the *characteristic current*, sets the speed range. If $I_{\mathrm{ch}} \le I_{\max}$, the centre of the ellipse lies inside the current circle, so torque remains available up to very high speed. If $I_{\mathrm{ch}} > I_{\max}$, the maximum speed is finite,

$$\omega_{e,\max} = \frac{V_{\max}}{\lambda_m - L_d I_{\max}}. \tag{3.143}$$

3.15.6 Maximum Torque per Voltage

At high speed in a machine with $I_{\mathrm{ch}} < I_{\max}$, the current circle no longer limits. The largest torque on a given ellipse is at the point nearest the ellipse centre along a specified curve, called maximum torque per voltage (MTPV), or maximum torque per flux. For an SPMSM, this is simply $I_d = -\lambda_m/L_s$, the ellipse centre, with I_q from $L_s I_q = V_{\max}/\omega_e$. For an IPMSM, the d-axis current is offset by saliency, and it is usually read from a table [26]. The example machines have I_{ch} near or above $I_{\max}$, so MTPV is not reached within their speed range.

3.15.7 Operating Characteristics and Saliency Effects

The curves were computed by searching over current vectors at each speed, subject to (3.138) and (3.139) with R_s neglected. The torque-speed curve is shown in Figure 3.103, and the regions in Figure 3.104. Figure 3.112 gives current against speed: $|I_s|$ stays at $I_{\max}$, i_d goes more negative above base speed, and i_q falls accordingly. Figure 3.113 shows how i_d differs between machines. Below base speed, an SPMSM needs $i_d = 0$, whereas an IPMSM has a negative MTPA value. Figure 3.114 plots q-axis current against torque: it is linear for an SPMSM, while the IPMSM needs less i_q at high torque. Figure 3.115 shows the resulting power. With the example values, both machines hold nearly constant power over a broad speed range.

Example 21: PMSM operating point under a voltage constraint

Problem. An SPMSM runs on its current limit from a 400 V link. Find the base speed, base torque, and the operating point at $\omega_e = 2000$ rad/s.

Given (assumed). $p = 4$, $L_s = 0.3$ mH, $\lambda_m = 0.1$ Wb, $I_{\max} = 300$ A, $V_{\max} = 230$ V; R_s neglected.

Required. $\omega_{e,b}$, T_b, I_d, I_q, T_e, P_m.

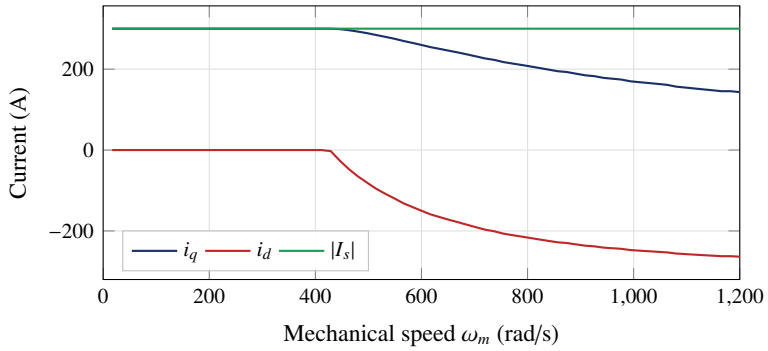


Figure 3.112: Current-speed curves of the example SPMSM at the torque limit.

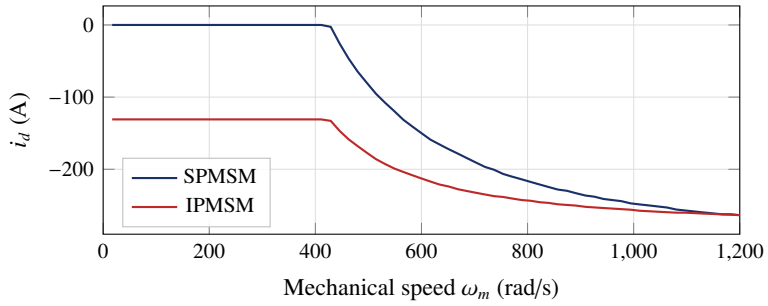


Figure 3.113: d-axis current versus speed.

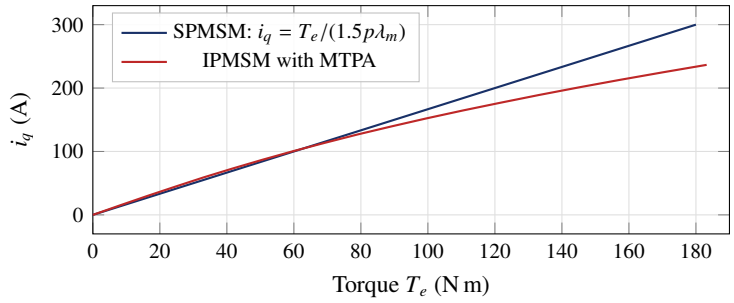


Figure 3.114: q-axis current versus torque.

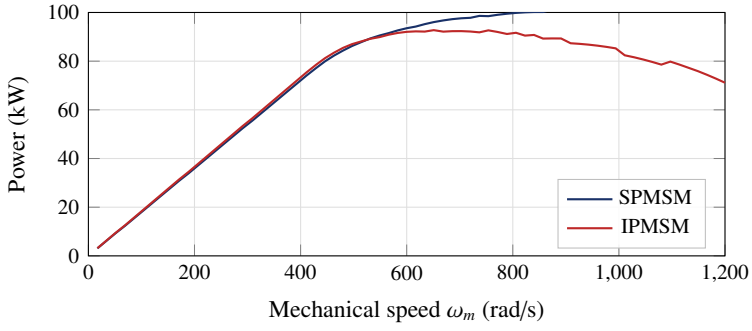


Figure 3.115: SPMSM versus IPMSM power characteristics.

Governing equations. (3.141), (3.142), (3.117).

Solution. $L_s I_{\max} = 0.09$ Wb. $\omega_{e,b} = 230/\sqrt{0.1^2 + 0.09^2} = 1710$ rad/s, so $\omega_{m,b} = 427.4$ rad/s (4081 r/min).
$T_b = 1.5(4)(0.1)(300) = 180$ N m, $P_b = 76.9$ kW.
At $\omega_e = 2000$: $V_{\max}/\omega_e = 0.115$ Wb.
$I_d = (0.013225 - 0.01 - 0.0081)/(2 \times 0.0003 \times 0.1) = -81.3$ A.
$I_q = \sqrt{300^2 - 81.3^2} = 288.8$ A.
$T_e = 0.6(288.8) = 173.3$ N m, $\omega_m = 500$ rad/s, $P_m = 86.6$ kW.

Result. $n_b = 4081$ r/min, $T_b = 180$ N m; at 4775 r/min, $I_d = -81.3$ A, $T_e = 173.3$ N m.

Physical interpretation. Moving part of the current on to the d axis slightly lowers torque but allows operation above base speed. Power actually rises, since the voltage vector has rotated closer to the current vector.

Engineering significance. $I_{\text{ch}} = 333$ A is just above $I_{\max}$. Eq. (3.143) then gives a very large $\omega_{e,\max}$ of 23 000 rad/s, so the speed range is limited in practice by mechanical and iron-loss constraints rather than voltage.

3.16 Field Oriented Control of PMSM

3.16.1 Introduction to PMSM FOC

FOC of a PMSM follows the logic of Section 3.10, but it is simpler in one respect: the rotor flux angle is the rotor position, so no slip calculation is needed. Figure 3.116 shows the complete architecture. Inputs are ω_m^* (or a torque command), rotor position and phase currents. Output is the set of six gate signals. The outer speed loop gives the torque or q-axis command, the d-axis command comes from MTPA and field-weakening logic, the inner PIs give voltage commands, and the inverse Park and SVPWM blocks produce gating.

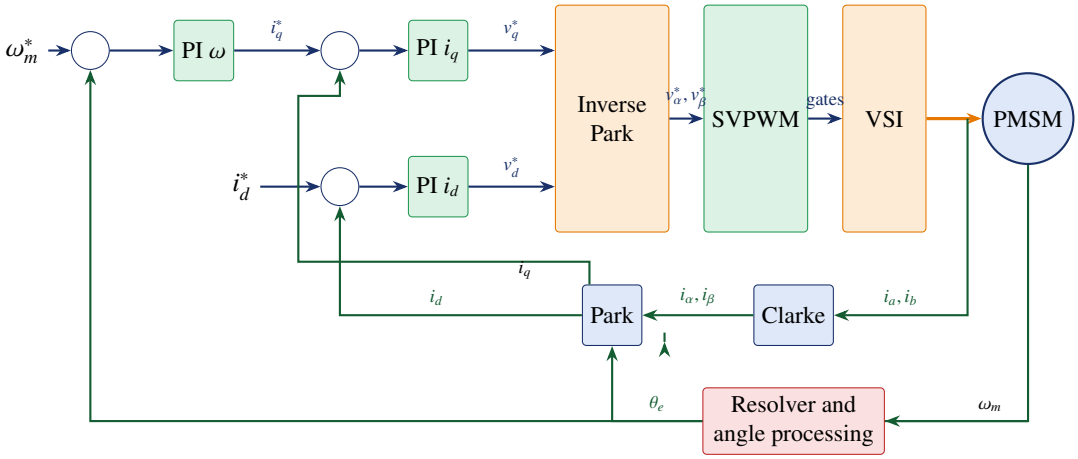


Figure 3.116: Complete PMSM FOC architecture.

3.16.2 Principle of Rotor Flux Orientation

The Park angle is $\theta_e = p\theta_m + \theta_0$, where θ_0 is a calibrated offset between the sensor and the magnet north axis. With this, the d axis lies on the magnet by construction, i_q is the torque current, and i_d controls the net d-axis flux. Current decoupling then means that, in the rotor frame, the two currents can be set independently by separate loops.

3.16.3 Rotor Position Feedback

Figure 3.117 shows the position chain. A **resolver**, a rotary transformer, gives sine and cosine signals modulated by an excitation carrier. It is robust against heat and vibration, and hence the automotive standard. A resolver-to-digital converter (RDC) or software tracking observer extracts angle and speed. An **encoder** gives higher resolution by digital pulses but is less tolerant of the environment. An angle error $\Delta\theta$ rotates the frame, producing a torque error of about $T_e(1 - \cos\Delta\theta)$ together with d-axis disturbance. Offset calibration is therefore a production step.

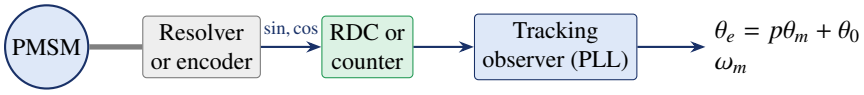


Figure 3.117: Rotor position feedback.

3.16.3.1 Sensorless PMSM Control Overview

Without a position sensor, the angle is estimated. At medium and high speed, a model-based observer reconstructs the back EMF from (3.127)–(3.128), and a phase-locked loop tracks its angle, as shown in Figure 3.118. At low speed, back EMF vanishes, so an IPMSM can be probed instead with high-frequency voltage injection, which exploits the position dependence of inductance. In traction, sensorless estimates mainly serve as back-up for a failed resolver.

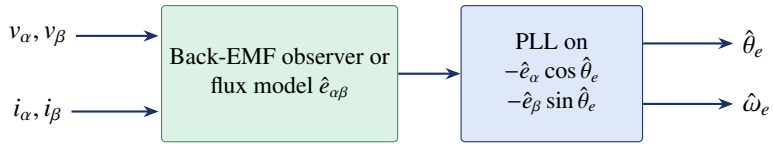


Figure 3.118: Rotor position estimation.

3.16.4 Current Controller Equations and Decoupling

Rearranging (3.129) and (3.130) shows that each axis is an R-L plant with disturbance terms:

$$L_d \frac{di_d}{dt} + R_s i_d = v_d + \omega_e L_q i_q, \tag{3.144}$$

$$L_q \frac{di_q}{dt} + R_s i_q = v_q - \omega_e L_d i_d - \omega_e \lambda_m. \tag{3.145}$$

Figures 3.119 and 3.120 show the current loops, in which PI controllers act on the errors:

$$v'_d = K_{pd} e_d + K_{id} \int e_d \, dt, \qquad v'_q = K_{pq} e_q + K_{iq} \int e_q \, dt, \tag{3.146}$$

where $e_d = i^*_d - i_d$ and $e_q = i^*_q - i_q$. Pole-zero cancellation for a bandwidth ω_{cc} gives $K_{pd} = \omega_{cc} L_d$, $K_{pq} = \omega_{cc} L_q$ and $K_{id} = K_{iq} = \omega_{cc} R_s$. The decoupling terms of Figure 3.121 cancel the disturbances:

$$v^*_d = v'_d - \omega_e L_q i_q, \qquad v^*_q = v'_q + \omega_e (L_d i_d + \lambda_m). \tag{3.147}$$

Thus each PI sees only $1/(Ls + R_s)$, and the q-axis term includes the back EMF feed-forward that keeps the integrator from having to supply it.

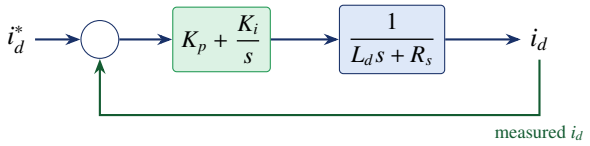


Figure 3.119: d-axis current controller.

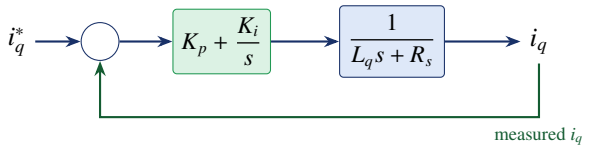


Figure 3.120: q-axis current controller.

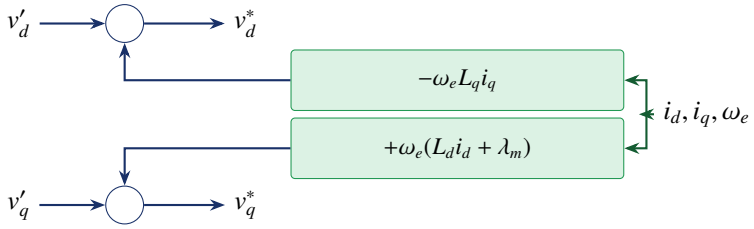


Figure 3.121: Decoupling compensation.

3.16.5 Speed Control Loop

Figure 3.122 shows the outer loop. With $i_d = 0$ or small, and the current loop treated as ideal, the plant from i^*_q to speed is $1.5 p \lambda_m / (Js + B)$, and the PI gives

$$i^*_q = K_{p\omega} e_\omega + K_{i\omega} \int e_\omega \, dt, \qquad e_\omega = \omega^*_m - \omega_m, \tag{3.148}$$

with the torque form $T^*_e = 1.5 p \lambda_m i^*_q$. The speed bandwidth is kept at least several times below the current bandwidth, so that the loops separate.

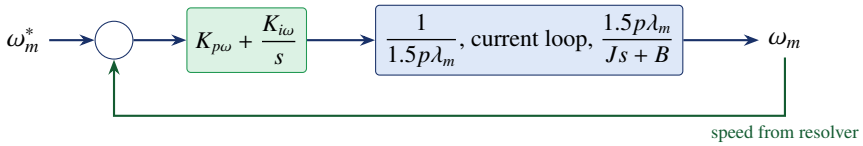


Figure 3.122: PMSM speed control loop.

3.16.6 Current Reference Generation

3.16.6.1 SPMSM

For a surface machine below base speed,

$$i_d^* = 0, \qquad i_q^* = \frac{2T_e^*}{3p\lambda_m}. \tag{3.149}$$

3.16.6.2 IPMSM

For an interior machine, MTPA with (3.120) applies. Since torque is nonlinear in current, i_q^* and i_d^* for a demanded T_e^* are normally pre-computed and stored in a table, as in Figure 3.123. On a dynamometer map, the tables also include saturation of L_d, L_q and λ_m. Algorithm 15 gives a calculation using iteration, and Algorithm 17 merges it with field weakening.

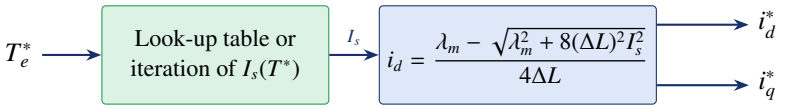


Figure 3.123: MTPA control block.

Algorithm 15 MTPA control for IPMSM

Input: T_e^*, λ_m, L_d, L_q, p, $I_{\max}$, tolerance ϵ
Output: $i_{d,\mathrm{MTPA}}^*$, i_q^*

$\Delta L \leftarrow L_q - L_d$; $I_s \leftarrow \left|T_e^*\right| / (1.5p\lambda_m)$
repeat
  $i_d \leftarrow [\lambda_m - \sqrt{\lambda_m^2 + 8\Delta L^2 I_s^2}]/(4\Delta L)$
  $i_q \leftarrow \mathrm{sign}(T_e^*)\sqrt{I_s^2 - i_d^2}$
  $T \leftarrow 1.5p[\lambda_m i_q - \Delta L i_d i_q]$
  $I_s \leftarrow I_s \cdot \left|T_e^*/T\right|^{1/2}$
until $\left|T - T_e^*\right| < \epsilon$ **or** $I_s \geq I_{\max}$
$i_{d,\mathrm{MTPA}}^* \leftarrow i_d$; $i_q^* \leftarrow i_q$

3.16.7 Field Weakening Control

Voltage-feedback field weakening is widely used since it does not depend strongly on parameters. In Figure 3.124, the magnitude of the voltage command is compared with a limit slightly below $V_{\mathrm{dc}}/\sqrt{3}$, and any excess drives a PI that makes i_d^* more negative. The current circle then

limits i_q^*. A fast torque reversal therefore cannot saturate the inverter. Sufficient voltage margin is left for current control, otherwise the loops lose authority. Note that i_d is also bounded to avoid demagnetization.

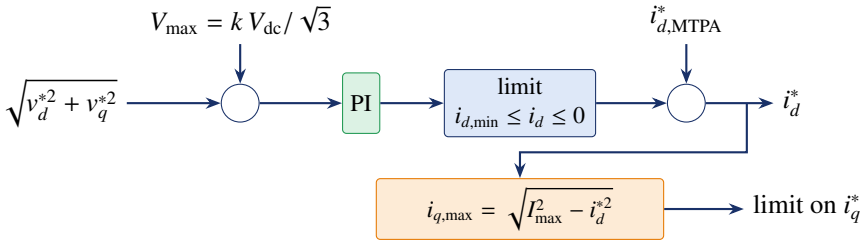


Figure 3.124: Field weakening control.

Algorithm 16 PMSM field weakening control

Input: v_d^*, v_q^*, V_{dc}, $i^*_{d,\mathrm{MTPA}}$, $I_{\max}$, $i_{d,\min}$, margin $k < 1$
Output: i_d^*, $i_{q,\max}$

$V_{\max} \leftarrow kV_{\mathrm{dc}}/\sqrt{3}$; $V \leftarrow \sqrt{v_d^{*2} + v_q^{*2}}$
$\Delta i_d \leftarrow \mathrm{PI}(V_{\max} - V)$, limited to $\Delta i_d \le 0$
$i_d^* \leftarrow \max(i^*_{d,\mathrm{MTPA}} + \Delta i_d,\ i_{d,\min})$
$i_{q,\max} \leftarrow \sqrt{I_{\max}^2 - i_d^{*2}}$
if $i_d^* = i_{d,\min}$ and $V > V_{\max}$ **then**
    Reduce torque command
end if

Algorithm 17 PMSM d-q current reference generation

Input: T_e^*, machine type, ω_e, V_{dc}, limits
Output: i_d^*, i_q^*

if machine is SPMSM **then**
    $i_{d0} \leftarrow 0$; $i_{q0} \leftarrow 2T_e^*/(3p\lambda_m)$
else
    $(i_{d0}, i_{q0}) \leftarrow$ MTPA table or Algorithm 15
end if
$(i_d^*, i_{q,\max}) \leftarrow$ field-weakening routine (Algorithm 16) with $i^*_{d,\mathrm{MTPA}} = i_{d0}$
Recompute i_q^* from (3.116) with the new i_d^*
$i_q^* \leftarrow \mathrm{sat}(i_q^*, \pm i_{q,\max})$

Example 22: PMSM FOC current reference and operating point calculation

Problem. The IPMSM of Example 20 runs under MTPA at $I_s = 270$ A and 2500 r/min. Find the current references, torque, MTPA angle, voltage references and power, and compare with $i_d = 0$.

Given. $p = 4$, $R_s = 0.02\ \Omega$, $L_d = 0.2$ mH, $L_q = 0.5$ mH, $\lambda_m = 0.09$ Wb, $\Delta L = 0.3$ mH.

Required. i_d^*, i_q^*, β, T_e, v_d^*, v_q^*, P_m.

Governing equations. (3.120), (3.116), (3.135), (3.136).

Solution. $8\Delta L^2 I_s^2 = 8(9\times10^{-8})(72\,900) = 0.0525$. Then
$i_d^* = (0.09 - \sqrt{0.0081 + 0.0525})/0.0012 = -130.1$ A.
$i_q^* = \sqrt{270^2 - 130.1^2} = 236.6$ A, $\beta = \arctan(130.1/236.6) = 28.8°$.
$T_e = 6[0.09(236.6) + 0.0003(130.1)(236.6)] = 183.2$ N m.
With $i_d = 0$, $i_q = 270$ A: $T_e = 145.8$ N m.
$\omega_e = 1047.2$ rad/s. $v_d^* = -2.6 - 123.9 = -126.5$ V. $v_q^* = 4.7 + 67.0 = 71.7$ V.
$|v| = 145.4$ V. $P_m = 183.2 \times 261.8 = 47.95$ kW.

Result. $i_d^* = -130.1$ A, $i_q^* = 236.6$ A, $T_e = 183.2$ N m, $|v| = 145.4$ V, $P_m = 48.0$ kW.

Physical interpretation. MTPA gains 25.6% more torque from the same current, because reluctance torque outweighs the loss of magnet torque.

Engineering significance. The voltage is well within the 230.9 V limit, so this point lies in the constant torque region and no field weakening is needed.

3.16.8 Control of SPMSM and IPMSM

The two machines share one structure, but differ in the reference block. An SPMSM uses $i_d = 0$ and a linear torque constant, with negative i_d only for field weakening. An IPMSM uses MTPA tables below base speed and a coordinated move along the voltage ellipse above it, with $L_d \neq L_q$ in decoupling. The PI gains also differ between axes. Table 3.19 compares FOC of the two machine families.

Table 3.19: Induction motor FOC versus PMSM FOC

Feature	IM FOC	PMSM FOC
Flux source	i_{ds}	Magnets
Frame angle	Estimated or slip-derived	Measured position
Slip	Required, $\omega_{sl} \neq 0$	None
Normal d-axis current	Positive magnetising	Zero or negative
Torque equation	$\frac{3}{2}p\frac{L_m}{L_r}\lambda_r i_{qs}$	$\frac{3}{2}p[\lambda_m i_q + (L_d - L_q)i_d i_q]$
Key sensitivity	τ_r	λ_m, position offset
Field weakening	Reduce λ_r^*	Negative i_d
Start-up	Pre-flux required	Initial angle required

The table highlights that the control skeleton is identical. Differences arise from where flux comes from: the IM must create its own, while the PMSM must locate the magnets and, at high speed, oppose them.

3.16.9 PWM Generation and SVPWM

The VSI has eight switching states, of which six give active voltage vectors and two give zero, as in Figure 3.125. Using (3.67) with the pole voltages, the active vectors have length $2V_{\mathrm{dc}}/3$ and are 60° apart. SVPWM produces the reference $\vec{v}^{*}$ on average by applying the two adjacent active vectors and the zero vectors in each period T_s [24]. Let γ be the angle of $\vec{v}^{*}$ within its sector. Equating volt-seconds, $\vec{v}^{*}T_s = \vec{V}_k T_1 + \vec{V}_{k+1} T_2$, gives

$$T_1 = \frac{\sqrt{3}\,T_s\left|\vec{v}^{*}\right|}{V_{\mathrm{dc}}}\sin(60^\circ - \gamma), \qquad T_2 = \frac{\sqrt{3}\,T_s\left|\vec{v}^{*}\right|}{V_{\mathrm{dc}}}\sin\gamma, \qquad T_0 = T_s - T_1 - T_2. \tag{3.150}$$

The linear limit is the largest circle inside the hexagon, $\left|\vec{v}^{*}\right| \le V_{\mathrm{dc}}/\sqrt{3}$. Voltage references come from the inverse Park transform with the decoupled outputs:

$$v_\alpha^{*} = v_d^{*}\cos\theta_e - v_q^{*}\sin\theta_e, \qquad v_\beta^{*} = v_d^{*}\sin\theta_e + v_q^{*}\cos\theta_e. \tag{3.151}$$

In practice, θ_e is advanced by $1.5\,\omega_e T_c$ to compensate for computation and PWM delay. Figure 3.126 shows the SVPWM-based drive path, and Figure 3.127 shows the PMSM and inverter interface.

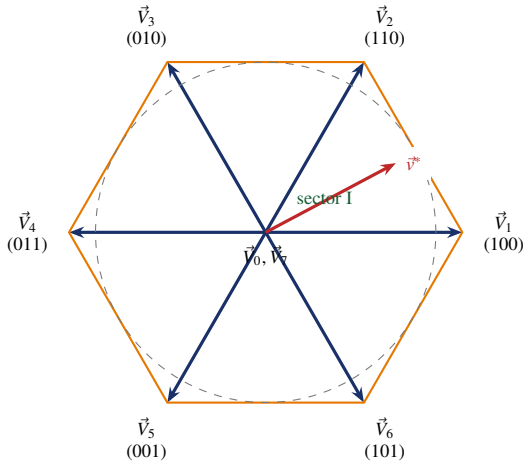


Figure 3.125: Voltage vectors of a two-level VSI and the SVPWM reference.

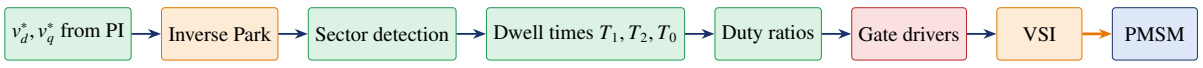


Figure 3.126: SVPWM-based PMSM drive signal path.

3.16.10 Complete Algorithms

Algorithm 18 is the fast interrupt routine, and Figure 3.128 shows the same logic as a flowchart. Algorithm 19 is the slower outer task. Note the rates: the current loop is typically run at the PWM frequency, around 10 kHz, and the speed loop an order of magnitude slower.

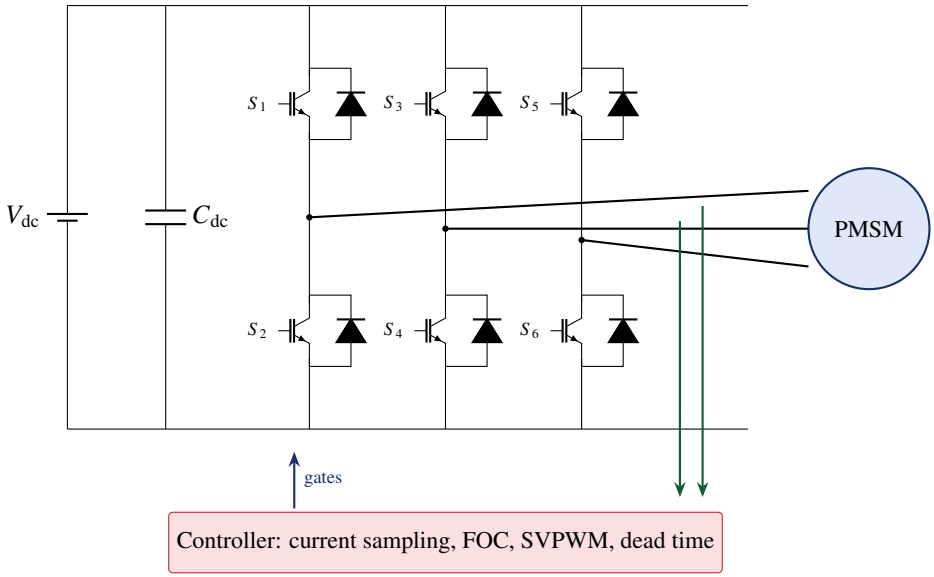


Figure 3.127: PMSM and three-phase inverter interface.

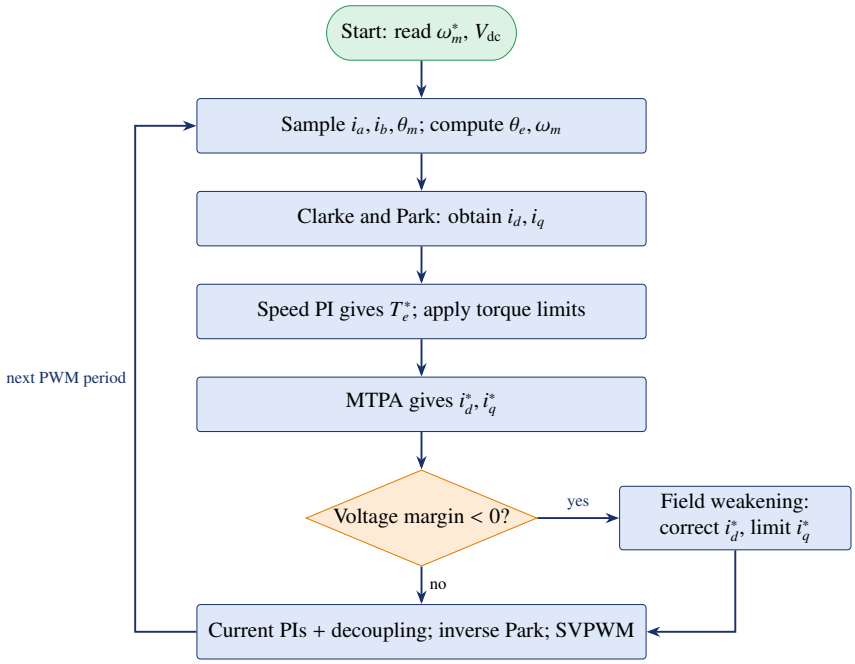


Figure 3.128: Flowchart of PMSM field oriented control.

Algorithm 18 PMSM field-oriented control

Input: T_e^* or ω_m^*, i_a, i_b, θ_m, V_{dc}
Output: Duties d_a, d_b, d_c for $S_1 \ldots S_6$

Sample at PWM centre; $\theta_e \leftarrow p\theta_m + \theta_0$; $\omega_e \leftarrow p\,\omega_m$ from observer
$(i_\alpha, i_\beta) \leftarrow \mathrm{Clarke}(i_a, i_b)$; $(i_d, i_q) \leftarrow \mathrm{Park}(i_\alpha, i_\beta, \theta_e)$
$(i_d^*, i_q^*) \leftarrow$ Algorithm 17
$e_d \leftarrow i_d^* - i_d$; $e_q \leftarrow i_q^* - i_q$
$v_d' \leftarrow \mathrm{PI}_d(e_d)$; $v_q' \leftarrow \mathrm{PI}_q(e_q)$
$v_d^* \leftarrow v_d' - \omega_e L_q i_q$; $v_q^* \leftarrow v_q' + \omega_e(L_d i_d + \lambda_m)$
if $\sqrt{v_d^{*2} + v_q^{*2}} > V_{\mathrm{dc}}/\sqrt{3}$ **then**
    Scale (v_d^*, v_q^*) with q-axis priority; hold integrators
end if
$(v_\alpha^*, v_\beta^*) \leftarrow$ inverse Park with $\theta_e + 1.5\omega_e T_c$
Determine sector and γ; compute T_1, T_2, T_0 using (3.150)
Load duties into PWM timers

Algorithm 19 PMSM speed and torque control

Input: Driver torque request or ω_m^*, ω_m, BMS limits, temperatures
Output: T_e^*

if speed mode (creep, cruise) **then**
    $T_{\mathrm{req}} \leftarrow \mathrm{PI}_\omega(\omega_m^* - \omega_m)$
else
    $T_{\mathrm{req}} \leftarrow$ pedal map value
end if
Apply the envelope $T_{\max}(\omega_m, V_{\mathrm{dc}})$
Apply power limits as in Algorithm 4
if winding or magnet temperature high **then**
    Derate T_{req}
end if
$T_e^* \leftarrow \mathrm{rate\text{-}limit}(T_{\mathrm{req}})$ ▷ Protects gears against backlash shock

3.16.11 Motoring, Regeneration and Dynamic Performance

Motoring is $i_q^* > 0$ at forward speed, in which case the voltage and current give $P_e > 0$. For regeneration, $i_q^* < 0$. By (3.137), torque changes sign, and the DC current reverses through the inverter diodes and switches, with no change of control structure. For an IPMSM in regeneration, MTPA keeps $i_d < 0$ and β moves into the third quadrant of the current plane, since reluctance torque follows the sign of $i_d i_q$.

Figures 3.129 to 3.134 show representative waveforms from an idealised profile for the SPMSM of Example 21, with ideal torque tracking and $i_d = 0$. From 0 to 0.6 s, torque is 145 N m and speed ramps up (motoring). A low cruising torque follows. At 1.2 s, negative torque of −120 N m brings regenerative braking, during which i_q reverses and mechanical power is negative. Phase currents, in Figure 3.132, are sinusoidal, and their frequency rises in step with speed, while i_d, i_q remain constant, which is what lets PI controllers work. In Figure 3.133, λ_d is constant at the magnet value because $i_d = 0$, and $|\lambda_q| = L_q |i_q|$ follows torque.

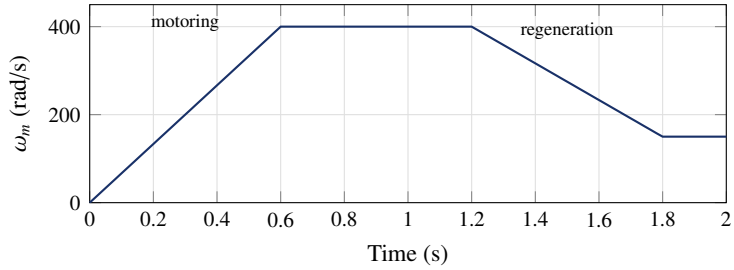


Figure 3.129: Dynamic speed response: motoring followed by regenerative braking.

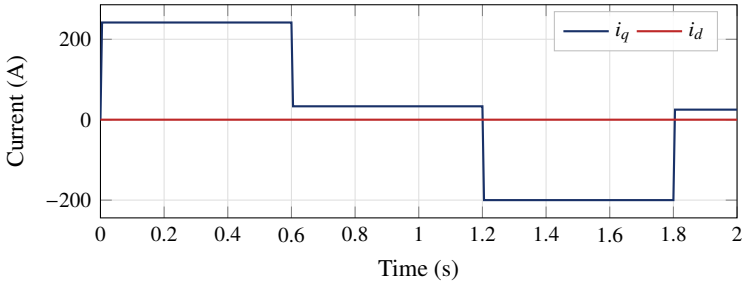


Figure 3.130: d-axis and q-axis currents.

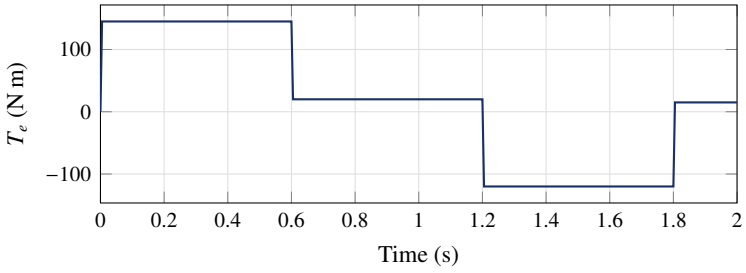


Figure 3.131: Electromagnetic torque.

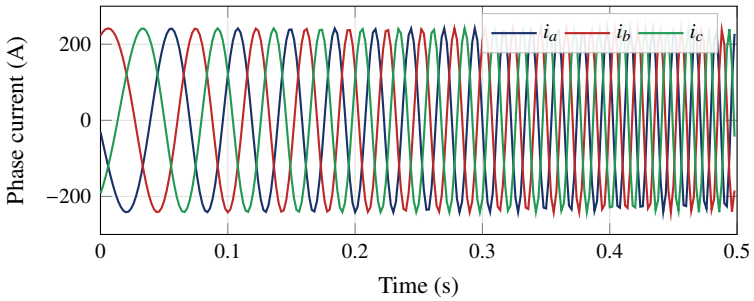


Figure 3.132: Three-phase stator currents during acceleration.

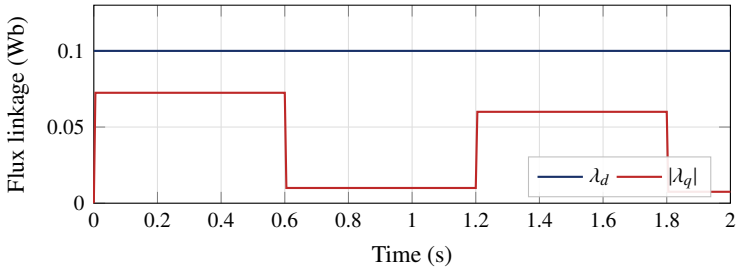


Figure 3.133: PMSM d-q flux linkages.

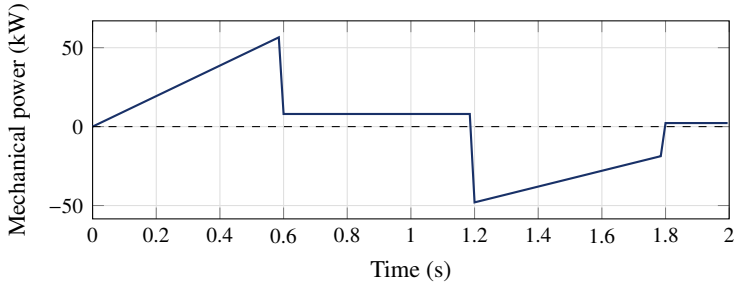


Figure 3.134: Mechanical power during motoring and regeneration.

A well-tuned PMSM drive gives a torque rise time of 1–2 ms, far quicker than the mechanical time constants of a vehicle, so the torque response is effectively instantaneous from the driver's viewpoint. Performance is constrained not by the controller but by DC-link voltage (particularly at low state of charge), by thermal limits, and by torsional oscillation in the driveline, for which the rate limiter of Algorithm 19 and active damping are used. In MATLAB/Simulink, the same structure can be assembled with Simscape Electrical and Motor Control Blockset components [27], using the equations and parameter values of this section.

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